

# Democracy and Its Unequal Effect on Capital Inflows\*

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**Abstract:** We study the implications of democratic regime change for the magnitude of capital inflows, a possible conduit for the large ‘democratic dividend’ for long-run economic prosperity found in the recent empirical literature. Our focus is on systemic heterogeneity in the causal effect of democracy across countries: we hypothesise and empirically demonstrate that benefits from democratisation (increased capital inflows) are considerably lower in countries with unfavourable geography (defined by climate and disease environment). Throughout our analysis we consider and dismiss ‘deep determinants’ of comparative development related to culture, history, and legal origin as alternative explanations for the patterns of unequal effects of democratic regime change. We also study episodes of excessive capital inflows, which are known triggers of financial crises, and find that their propensity is in the majority reduced by democratic regime change in ‘good’ geography countries, but in the majority worsened in ‘poor’ geography countries. While we cannot provide causal evidence, additional analysis enables us to suggest that the strong geographic patterns we observe are really driven by colonial history, and less likely due to factors like culture or legal origin.

**Keywords:** capital inflows, economic development, democratic regime change, bonanzas, surges, geography, deep determinants of prosperity, difference-in-differences, interactive fixed effects, heterogeneous treatment effects

**JEL codes:** F21, F34, O10, O43, P16

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# 1 Introduction

*“[Geography] tells an unpleasant truth, namely, that nature like life is unpleasant, unequal in its favours; further, that nature’s unfairness is not easily remedied.”*

(Landes, 1999, 4/5)

In this paper we study the relationship between democratic regime change and capital inflows. We look at the outcomes from a positive (increased inflows) and negative (excessive inflows) perspective and hypothesise that ‘nature’ (geography), related to characteristics of climate and historical disease environment, plays an important role in this context.<sup>1</sup> Democratic regime change, establishing a bundle of economic, political and legal institutions (Acemoglu et al., 2019), clearly represents the sort of reforms that “curtail the power of entrenched economic interests and liberate the economy’s productive potential” (Obstfeld, 2009, 63), while at the same time making “economies safe for international asset trade” (ibid). Democratic regime change should reduce economies’ objective or perceived political and economic risk factors, and hence attract higher foreign direct investment and other financial inflows (Li and Resnick, 2003; Papaioannou, 2009; Asiedu and Lien, 2011). Yet, we argue, deep-seated structural factors prevail: Geography determines economic ‘structure’, e.g. the complexity and diversity of the export basket (Malik and Temple, 2009), or the potential for and speed of structural transformation away from agriculture (Eberhardt and Vollrath, 2018), and this, in turn, determines investment opportunities (opportunities for economic returns). ‘Poor geography’ is associated with a lack of investment opportunities, (perceived) poor returns, and hence low capital inflows. Using descriptive analysis we highlight that countries with ‘poor geography’ (a) have relatively higher export concentration (in terms of goods/products) and hence are more exposed to global commodity market fluctuations, resulting in greater aggregate commodity price volatility; (b) are characterised by a productive system of lower economic complexity; and (c) suffer from higher trade costs. Different geography may translate into very different economic benefits from democratic regime change.

There still is somewhat of a taboo surrounding the suggestion that prosperity today may be determined by the ‘fate’ of geography (Landes, 1999, 4, refers to this as geography’s “sulphurous odour of heresy”). While immutable characteristics may determine development *to some extent*, from a policymaker’s perspective, it is important to understand what role institutional change can play over a much shorter time-horizon *despite geography*:<sup>2</sup> what are the patterns and underlying mechanisms of cross-country differences in the democratic dividend for economic prosperity.

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<sup>1</sup>We do not use the term ‘natural endowment’ employed in some of the existing literature since this too readily leads to association with ‘natural resources’ in the form of minerals, oil, etc., which are explicitly *not* part of our concept of geography. Our proxies for geography are, in terms of climate: (i) some land area in the tropical climate zone; (ii) no land area in the temperate climate zone; and (iii) low absolute latitude. In terms of disease environment: (i) the share of population at risk of malaria in 1965, (ii) malaria ecology, and (iii) the historical prevalence of 7 endemic diseases. Continuous indicators are dichotomised at the full sample median (including countries which democratised and those that were democracies or autocracies throughout the sample period).

<sup>2</sup>See also arguments in Nunn (2020) in this regard.

In our analysis of capital inflows (total inflows or FDI) for a large panel of countries (1975-2015) we use a novel methodology from the heterogeneous treatment effects literature (Xu, 2017; Chan and Kwok, 2022) employing a common factor structure to capture unobserved time-varying heterogeneity (Pesaran, 2006; Bai, 2009; Gobillon and Magnac, 2016). A simple empirical setup enables us to isolate the differential effect of democracy *by geography*: we construct separate sets of treatment and control samples by geography ('good', 'poor') and estimate average treatment effects on the treated (ATET) for democratic regime change using a difference-in-differences estimator (the principal component DID, PCDDID, of Chan and Kwok, 2022).<sup>3</sup> Why don't we just estimate the heterogeneous PCDDID for *all* countries and then compute averages by geography? This would adopt a control sample made up of countries with good and poor geography, undermining the clean counterfactual setup of our above strategy: 'poor' geography control countries for 'poor' geography treated countries, and 'good' geography control countries for 'good' geography treated countries. Setting a deliberately high bar for our definition of democratic regime change avoids the concern that democracy might 'mean different things' in countries with 'good' versus 'poor' geography: we employ V-Dem data (Coppedge et al., 2021) for 'liberal democracy' and also the indicator devised by Acemoglu et al. (2019) which captures similar institutional building blocks related to electoral democracy and the rule of law (see Appendix Figure A-1 for a visualisation).<sup>4</sup> We also make sure that countries in the poor vs good geography samples spend a similar number of years in democracy and do not have differential rates of reverting to autocracy. If the effect of geography on regime change is thus taken out of the equation, we can separately identify the causal effect of democracy on capital flows in 'good' and 'poor' geography countries, respectively, and aided by our graphical presentation of the results can compare the economic magnitudes. This analysis answers the question of whether geography is unequal in its favours for the effect of democratic regime change on capital inflows.

We also study excessive financial inflows for 1980-2015, building on the early warning system approach in the banking crisis literature (Bussière and Fratzscher, 2006; Schularick and Taylor, 2012) and investigate the *within-country effect* of four widely-accepted macroeconomic 'triggers' of capital flow bonanzas and surges (Ghosh et al., 2014; Caballero, 2016). If democracy has stronger effects for capital inflows in 'good' geography countries, maybe the potential for such costly episodes is also elevated. Our estimator is again country-specific and we adapt our treatment effects methodology to account for unobserved time-varying heterogeneity.

Naturally, geography is not the only deep determinant of economic prosperity banded about in the literature (La Porta et al., 1998; Stulz and Williamson, 2003; Rajan and Zingales, 2003; Nunn, 2009; Gorodnichenko and Roland, 2017), hence we demonstrate that alternative explana-

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<sup>3</sup>Under reasonable assumptions our estimator accounts for selection into regime change and we are able to test a (weaker) form of the parallel trend test required for identification.

<sup>4</sup>In additional analysis we take advantage of the hierarchical nature of the V-Dem indices to investigate whether institutions related to electoral democracy (polyarchy) have different implications from those associated with the rule of law and executive constraints (liberal component).

tions related to legal origin (French civil law origins provide lower legal protection for investors), history (colonial experience, extractive colonisation) and culture (individualism emphasizing personal freedom and achievement; linguistic similarity across countries enabling communication and exchange fostering innovation and modern growth) fail to provide manifest differences between groups of countries like the patterns in the case of geography. We cannot make causal claims for the deep determinants but the systematic patterns induced by geography are uncanny.

Our benchmark results ignoring deep determinants establish that democratic regime change (in this summary we use the liberal democracy definition) has a positive effect on total capital inflows over GDP on the order of 1 to 2 percentage points (depending on the set of controls included) for a sample mean of 3.0% (during the autocratic period of all countries that subsequently experienced democratic regime change).<sup>5</sup> For FDI inflows the effects amount to 0.6 to 1.4 percentage points, for a mean FDI/GDP of 1.4%. These indicate sizeable economic effects of democratisation on capital inflows. These averages hide substantial heterogeneity by geography: countries with 'good' geography experience 1 to 4 percentage points higher total inflow/GDP following democratic regime change for a mean of 3.8%, whereas those with 'poor' geography experience a 0 to 2 percentage point increase for a mean of 2.6%. For FDI the effects are a 0.5-2.5pp increase (mean 1.7%) and a 0-1.5pp increase (mean 1.2%) in 'good' and 'poor' geography countries, respectively — this summary ignores statistical significance, which is given for virtually all effects in the 'good' geography samples but in fewer 'poor' geography samples. When we explore alternative deep determinants related to culture, history and legal origin, treatment effects estimates are of near-uniform magnitude across the two samples ('good', 'poor'). For the analysis of bonanzas and surges we can broadly confirm that democratic regime change is in the majority associated with *reduced* vulnerability in the 'good' geography sample, whereas it is in the majority associated with *increased* vulnerability in the 'bad' geography sample.

Revealing patterns of differential democratic regime change effects can offer important insights for academics and policy makers, but cannot provide the reasons why we observe them. There are many instances of geography influencing history (Nunn, 2009), most prominently so in the context of the 'extraction' of slaves in Africa (Nunn, 2008) and the patterns of different forms of colonisation (Acemoglu et al., 2001), which are further intimately linked to the spread of legal origin (La Porta et al., 1998). Similarly, culture can be influenced by geography, where unfavourable nature can create barriers to the diffusion of ideologies, beliefs, ideas, or the means of communication (Gorodnichenko and Roland, 2017), effectively limiting the spread of certain 'culture' or preserving cultural isolation. Our aim in additional analysis is to provide some evidence that the strong geographical patterns we reveal may really be attributable to cultural, historical, or

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<sup>5</sup>Using this benchmark somewhat inflates the percentage increases since regime change took place later in our sample with capital inflows also (broadly) rising over time. If we simply take the mean capital inflow/GDP and FDI/GDP in the treated countries regardless of democratic regime, the percentage increases are 20-45% and 25-55%, respectively. By geography: capital inflows 20-70% and 0-50% for 'good' and 'bad' geography samples, FDI 16-80% and 0-65%.

legal factors *shaped* by nature, and to a lesser extent due to nature itself. We answer a very simple question: what are the patterns of capital flow increases following democratic regime change between ‘good’ and ‘poor’ geography samples if we focus on different deep determinants within these groups? Among the ‘good’ geography sample, how did countries with French legal origin fare, and are their average treatment effects similar or very different from countries with French legal origin which are endowed with ‘poor’ geography? Adopting this strategy for a large number of proxies for culture, legal origin and history, we find that proxies related to colonial experience appear to consistently eradicate the substantial differences in the ‘democratic dividend’ we observe when comparing good and bad geography countries. Countries with favourable geography but a history of colonisation experience only similar, in the case of FDI flows even lower, capital inflows after regime change as those countries with the same history but poor geographic endowment.

**Background** The past fifty years represent a new era of financial globalisation during which international capital flows have increased dramatically and are regarded as drivers of economic growth (due to increased investment, a productivity boost or reduced misallocation: [Prasad et al., 2007](#); [Larrain and Stumpner, 2017](#); [Erten et al., 2021](#)) but also as potential triggers of financial crises (due to capital inflow ‘bonanzas’: [Kaminsky and Reinhart, 1999](#); [Kaminsky, 2019](#); [Reinhart and Reinhart, 2009](#); [Ghosh et al., 2014](#); [Caballero, 2016](#)). Globally, most capital controls impeding capital mobility were removed by the early 1990s in a process which started with Germany and the US after the collapse of the Bretton Woods System in 1973, followed by Japan, the UK and Latin America later that decade and the rest of Europe in the 1980s ([Aizenman et al., 2013](#); [Kaminsky, 2019](#)). Some of the existing literature on excessive capital inflows suggests that these are primarily caused by cyclical push factors ([Reinhart et al., 2017](#)), with “domestic macroeconomic characteristics... generally less important” ([Forbes and Warnock, 2012](#), 235). Dissenting voices do not question that global factors are driving global surges to some extent but argue that the incidence and magnitude of a surge for an individual country are largely dependent on domestic factors ([Ghosh et al., 2014](#)) and prime among these the quality of institutions ([Fratzscher, 2012](#)). Most recently, [Cerutti et al. \(2019\)](#) suggest that the vast majority of variation in capital flow patterns cannot be explained by global factors.

For a long time, many academics, policymakers and development practitioners doubted the economic dividends of democratic regime change: enabling the populace to remove an incumbent government through the power of the electoral process (one of the fundamental definitions of democracy) would drive up (government) consumption and, via the threat of tax increases to finance these redistribution efforts, reduce the rate of investment, to the detriment of economic growth (e.g. [Baum and Lake, 2003](#), 334f). Doubters would further point to the stellar growth rates in autocratic regimes such as China or Singapore to question whether democracy is *necessary* for economic prosperity. While that may not be the case, beyond cherry-picking success stories it is widely recognised that growth outcomes vary substantially across autocracies ([Persson and](#)

Tabellini, 2009; Knutsen, 2012), and the strong *average* improvement in economic development in democratising countries established more recently (Acemoglu et al., 2019) provided convincing evidence for a positive and large causal effect of democratic regime change. Our definition of democracy (liberal democracy) represents a bundle of institutions, covering both electoral democracy (polyarchy) as well as aspects of individual liberties (e.g. freedom of movement), equality before the law and secure property rights. Linking capital flows to democracy rather than individual institutions (e.g. Papaioannou, 2009, distinguishes and tests seven features of institutional quality) brings our work in line with the new ‘democracy causes growth’ literature and highlights a specific channel through which democratic regime change can lead to greater economic prosperity.

**Empirical Strategy** We adopt a treatment effects framework (Papaioannou and Siourounis, 2008; Acemoglu et al., 2019), but use an implementation which adds common factors estimated from control sample regressions to the country-specific treatment regression model: the Chan and Kwok (2022) Principal Component Difference-in-Differences (PCDID) estimator. Like any other Difference-in-Differences estimator the PCDID studies treated countries before and after treatment, but there are no control country observations included in our treatment regression: these are instead captured in the form of estimated common factors. The intuition is as follows: our country-specific specification of capital inflows as a function of a democracy dummy, an intercept, and some control variables crucially omits a great deal of unobserved heterogeneity, time-varying determinants of capital flows which are also affecting democratic regime change as well as the controls — country-specific productivity or absorptive capacity may be good examples (Eberhardt and Presbitero, 2015; De Visscher et al., 2020). Factor models construct proxies for these omitted factors, either by Principal Component Analysis from regression residuals (Bai, 2009) or by use of cross-section averages of all observed variables in the model (Pesaran, 2006). These proxies are then entered into the estimation equation: like a country fixed effect in a pooled panel model solves the problem of unobserved time-invariant determinants correlated with the outcome (capital flows) and the independent variables (democracy, controls), these ‘interactive fixed effects’ solve the problem of unobserved time-varying determinants correlated with outcome and independent variables. This is the setup in standard heterogeneous panel models. In the difference-in-differences context, there is a tweak: here, the common factors are estimated from the residuals of *control country regressions* (capital flows regressed on an intercept and the controls, country by country, in the sample of never-democratisers), and then included in the country-specific treatment effects regression as additional controls with country-specific parameters. Instead of a standard parallel trend test, the empirical specification has to satisfy the Alpha test for ‘weak parallel trends’ (Chan and Kwok, 2022): in essence, this checks that the ‘information’ about unobserved heterogeneity the PCDID extracts from the control sample is equally ‘relevant’ in the treatment sample.

Our investigation of excessive capital inflows and the effects of democracy employs a country-specific linear probability model for capital flow ‘surges’ or ‘bonanzas’, which we augment with

factors estimated from the control samples of never-democratisers: this enables us to account for the unobserved heterogeneity in the determinants of excessive capital flows and the endogeneity of regime change. We interact (in separate regressions) each of known ‘trigger variables’ for excessive capital inflows with a democracy dummy and hence can estimate the within-country difference of these triggers in autocracy versus democracy. This setup builds on the common factor structure in a generalised linear model ([Boneva and Linton, 2017](#)) and the PCDID to devise a factor-augmented early warning system model. We then report the patterns of the within-country effect of an increase in the trigger variable during autocracy and its relative effect during democracy.

**Related Literature** We contribute to three separate strands of literature. First, our work is related to the empirical literature on democracy and long-run growth, which only recently established a positive and large causal relationship ([Madsen et al., 2015](#); [Acemoglu et al., 2019](#); [Eberhardt, 2022](#); [Boese-Schlösser and Eberhardt, 2024](#)). Two important challenges to a better understanding of *how* democracy causes growth remain: (a) existing studies assume that the democracy-growth relationship is common across countries, which makes it difficult to derive policy implications for individual countries ([Durlauf, 2020](#)); and (b) the direct transmission mechanisms by which democracy leads to growth have not been studied systematically. Our paper explores geography as an important factor governing the patterns of heterogeneous democracy effects and capital inflows — both as a positive driver of development and a source of increased financial vulnerability — as the conduit for the effect of democratic regime change on economic prosperity.

Second, we contribute to two literatures on the determinants of capital inflows and of excessive capital flow episodes, studying the domestic ‘pull factors’. Capital inflows are widely suggested to have a positive impact on growth ([Alfaro et al., 2004](#); [Durham, 2004](#); [Prasad et al., 2007](#); [Erten et al., 2021](#)). But, where capital does or does not flow and why has, of course, occupied the profession for a long time (the ‘Lucas Paradox’, [Lucas, 1990](#)). Existing work has suggested that ‘institutions’ are important determinants of capital inflows ([Feldstein, 1999](#); [Li and Resnick, 2003](#); [Alfaro et al., 2008](#); [Papaioannou, 2009](#); [Asiedu and Lien, 2011](#)) and hence a partial solution to these puzzles. Like in the literature on financial development (e.g. [Law and Singh, 2014](#); [Arcand et al., 2015](#)), recent empirical work on financial flows seems to emphasise the dangers of ‘too much of a good thing’ much more than the benefits of capital flows *per se*: excessive capital inflows are a primary candidate for increased financial vulnerability ([Kaminsky and Reinhart, 1999](#); [Lopez-Mejia, 1999](#); [Reinhart and Reinhart, 2009](#); [Ghosh et al., 2014](#); [Caballero, 2016](#); [Erten et al., 2021](#)), a topic which has seen a wealth of contributions following the Global Financial Crisis ([Gourinchas and Obstfeld, 2012](#); [Schularick and Taylor, 2012](#)). We contribute to these two literatures by highlighting the differential effect of institutional change on capital inflows and bonanzas/surges *by geography*.

Third, we contribute to an older cross-country empirical literature on the deep determinants of comparative economic development. 2021 marked the twentieth anniversary of the publication

of ‘*The colonial origins of comparative development*’ (Acemoglu et al., 2001). Though not the first empirical contribution on the link between institutions and growth (e.g. Hall and Jones, 1999), it is arguably the paper which firmly established the quality of institutions as the most significant ‘deep determinant’ of long-run economic development. In the years after its publication empirical battles were fought over the supremacy of institutions over geography and trade openness (e.g. Dollar and Kraay, 2003; Easterly and Levine, 2003; Rodrik et al., 2004) as well as over the precise definition of institutional quality which did (or did not) cause development over the long-run (Glaeser et al., 2004). Related work has shifted attention to the study of culture (Stulz and Williamson, 2003; Gorodnichenko and Roland, 2017; Ang, 2019), history (Nunn, 2009) or legal origins (La Porta et al., 1998, 2008; Monnet and Velde, 2021), at times focusing on the determinants of financial development rather than economic prosperity more broadly, including work on the political economy of financial development (Rajan and Zingales, 2003). Most of this work is based on regressions in the cross-section and defines ‘institutions’ as time-invariant. We contribute to this literature by considering democracy as a time-varying bundle of institutions (in line with recent seminal contributions on democracy and growth), and studying the differential effects of democratic regime change across different sets of country groups defined by immutable characteristics proxying for geography, history, legal origin or culture.

The remainder of this paper is structured as follows: in the next section, we provide some background on and descriptive analysis of the ‘deep determinants’ of economic development. In Section 3 we study the causal effect of democratic regime on capital flows, on undesirable capital flow surges and bonanzas. Section 4 first provides descriptive analysis to highlight the ‘structural’ disadvantage of countries with unfavourable geography along a number of factors determining trade and production. We then explore whether the strong correlations between capital inflows and geography may not hide indirect cultural or historical factors. Section 5 concludes.

## 2 Deep Determinants of Comparative Development

In this section, we provide an overview of some of the existing literature on deep determinants and economic prosperity, covering empirical literature which investigates per capita income, financial development or capital flows. We then provide details on the range of proxies we use in the paper to study geography, culture, history and legal origin. We offer a first glimpse of the deep determinants of capital flows in the context of democratic regime change in descriptive analysis but provide evidence against simple mechanical explanations for these results related to regime fragility and length of time spent in democracy.

### 2.1 Geography, Legal Origin, Colonial History, and Culture

**Why geography?** Arguments supporting a link between geography (climate, disease environment) and contemporary economic development are frequently centred on their impact on land,

labour and production technology (Diamond, 1998; Bloom and Sachs, 1998; Gallup et al., 1998), illustrated by the suggestion that in tropical climates people are “enervated by the slightest physical or mental exertion” (a Bangladeshi diplomat cited in Landes, 1999, 15), which makes for a “slow rhythm [of work] with long and frequent pauses” (ibid.: 16);<sup>6</sup> or that (modern) innovations in production technology favour agriculture in temperate versus tropical countries (Diamond, 1998). Yet, these arguments are difficult to uphold given the ‘reversal of fortune’ (Acemoglu et al., 2002) whereby if climate had such a profound impact then countries which were rich in 1500 should still be rich today (but frequently are not). These authors further convincingly dismiss related explanations that agricultural technology reversed the early advantage of tropical over temperate agriculture. We therefore need to identify distinctly more ‘modern’ features of growth and development as likely reasons for a democracy-geography-growth link.

Standard gravity arguments for the flow of traded *goods* between countries (Bergstrand, 1985; Anderson and Van Wincoop, 2003) find similar effects of distance and remoteness for capital flows (Portes et al., 2001; Bergstrand and Egger, 2007; Head and Ries, 2008; Lane and Milesi-Ferretti, 2008; Papaioannou, 2009; Pellegrino et al., 2021), suggesting that “the geography of information is the main determinant of the pattern of international transactions” (Portes and Rey, 2005, 269). This speaks to distance as an important factor. Besides geographic predisposition to trade and capital flows, not just in terms of remoteness but also distance from the equator (Frankel and Romer, 1999), nature affects the structure of exports, which can leave countries prone to external (terms of trade) shocks (Malik and Temple, 2009). We provide evidence in Section 4 that trade costs, concentration of exports, complexity of production strongly correlates with our proxies for climate and disease environment.

**Why legal origins?** A sizeable literature has investigated the economic consequences of legal origin, in particular for financial development (e.g. La Porta et al., 1998; Beck et al., 2003). The conceptual arguments for such a link, that legal protection for outside investors (and hence financial development) is stronger in countries with origins in (British) common law, rather than (French) civil law, are well-known (La Porta et al., 2008), though not without controversy: while modern (post-WWII) financial development seems to follow the suggested patterns, history provides many instances of a ‘reversal’ in the correlation (Monnet and Velde, 2021), thus undermining a structural link. We consider legal origin since arguments for investor protection seem equally relevant in the context of capital flows, and since the legal system represents a ‘meta-institution’ (Koyama, 2022) and hence merits wider investigation beyond financial development.

**Why history?** One strand of the empirical literature on ‘the long arm of history’ relates to the ‘triangular’ global trade system from the 15th century onwards, which connected manufactured goods from European colonisers, raw materials from the Americas, and slave labour from

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<sup>6</sup>It is important to emphasise that they speak of local and non-local individuals being affected in this way: there is no suggestion that the people residing in tropical locales inherently exert a lower work effort and productivity.

Sub-Saharan Africa (Nunn, 2009). These exploitative connections have been causally linked to prosperity in Europe, underdevelopment in the Americas, and African stagnation. While in our setup of democratic regime change ‘shocks’ it is prohibitive to study the slave trade corner of this triangle,<sup>7</sup> broader notions of colonial experience (during the ‘Columbian exchange’ and the ‘Scramble for Africa’) and within this more extractive colonisation (Acemoglu et al., 2001) can provide insights into the divergent effects of historical contact with Europe (Nunn, 2020).

**Why culture?** The origins of a proposed link between culture — typically defined as a shared set of values, beliefs and norms of behaviour — and economic prosperity are usually found in Max Weber’s *protestant work ethic*. While earlier empirical work made a link to religion (Landes, 1999; Stulz and Williamson, 2003), it was the study of Gorodnichenko and Roland (2017) which systematically approached the distinction between individualism (said to be fostering personal freedom, achievement, and hence innovation) and collectivism (emphasising embeddedness, group loyalty and discouraging ‘standing out’) — a distinction suggested to be *the* primary dimension of cultural differences (Heine, 2007).<sup>8</sup> Adopting a range of instrumentation strategies Gorodnichenko and Roland (2017) demonstrate a causal link between individualism and income per worker.

Language “makes information operational” (Ginsburgh and Weber, 2020, 348) and provides a ‘social technology’ we can use to construct divisions of our sample into those with more similar and others with more dissimilar common language, as a crude proxy for culture: “The various aspects of culture are hard to describe and for the sake of simplicity, language may be, and is often, used as a proxy for culture and/or ethnicity” (ibid, 363). The specific definitions of language we focus on relate to ‘intercommunication distances’, which have primarily found application in the study of bilateral trade flows (e.g. Melitz and Toubal, 2014) — commonly finding that language distance reduces trade — or of lexicographical bias in firm-level exports (Cheng et al., 2020). Although the presence of a *lingua franca* enables communication, it is the notion of common ethnicity and trust captured by intercommunication distance we adopt (Ginsburgh and Weber, 2020) which makes such indices attractive proxies for culture.

**Proxies** Throughout our analysis, we adopt a range of proxies for geography ( $\times 6$ ), history ( $\times 2$ ) and culture ( $\times 3$ ), as well as data on legal origin to capture these different ‘deep determinants’.<sup>9</sup>

For the disease environment aspect of geography, we use two datasets for ‘malaria ecology’: (i) from McCord et al. (2017, malpct) the percentage share of population at risk of malaria in 1965 and (ii) from Kiszewski et al. (2004, ME) malaria ‘ecology’, an “ecologically-based variable that is predictive of malaria risk” (Sachs, 2003, 7). We further adopt (iii) data on the historical

<sup>7</sup>The analysis of slavery is limited to 52 African countries, which in samples split by median slave export numbers would amount to only 8 countries with democratic regime change (definition: Liberal democracy), respectively.

<sup>8</sup>Existing research, reviewed in Spolaore and Wacziarg (2013), has focused on the (intergenerational) transmission of culture and also its effect on contract enforcement, fertility choice, regulation, etc.

<sup>9</sup>For some of these proxies, a range of sources are available, we selected those which maximised the sample size.

prevalence of parasitic and infectious diseases from [Murray and Schaller \(2010, hdp\\_7\)](#) — the variable considering seven diseases has the best coverage. For the climate-related aspects of geography we adopt (i) a dummy variable for zero land area in the temperate climate zone constructed from [Spolaore and Wacziarg \(2013, kgatep\)](#), and (ii) a dummy variable for ‘some’ land area in the tropical climate zone constructed from [Nunn and Puga \(2012, tropical\)](#), from where we (iii) also construct absolute average latitude (using lat).

For French legal origin we use a dummy from [La Porta et al. \(2008, legor\\_fr\)](#).<sup>10</sup>

For aspects of history we adopt the ‘Colonial Transformation’ dataset from [Ziltener and Künzler \(2013\)](#) for 85 countries with colonial experience. Our first indicator is simply whether a country experienced colonialism or not. The second indicator is limited to the 85 countries in the data with colonial experience and uses information on plantations, gold/silver mining and general mining during the colonial period to establish whether colonial institutions were ‘extractive’ (any of the three measures having a value of ‘2’ indicating ‘extensive’).

For culture, we use (i) data from [Gorodnichenko and Roland \(2017, distE\\_UK\)](#) relating to a measure of distance (from the UK, one of the world’s most individualist countries) in terms of frequencies of blood types. For the language aspect of culture, we use (ii) data from [Gurevich et al. \(2021\)](#) who compiled the domestic and international common language (DICL) database. Our proxies capture the probability that two individuals picked at random from each pair of countries speak the same native language (cni) and a population-weighted proximity measure based on ‘linguistic trees’ which categorise languages (lp). These are dyadic data for country pairs, and we compute the country-specific averages for the average common native language (cni) index and average language similarity (lp) for country  $i$  across all other countries  $j$ .<sup>11</sup>

For all of the above: where we do not already indicate that the proxies are dummy variables we dichotomise continuous or categorical variables at the cross-country median (the proxies selected are all time-invariant or in case of malpct for a single year).<sup>12</sup> Throughout our analysis and in the presentation of results a geography dummy value of 0 is for ‘good geography’ and 1 for ‘poor geography’: e.g. for absolute latitude the dummy is 1 for countries below (low latitude) and 0 for countries above the median (higher latitude). Similarly for cultural and historical proxies. We provide some maps indicating the distribution of deep determinants across countries in Appendix Figure A-3. The mean (median) pairwise correlation coefficient within the geography proxies is 0.60 (0.58) and within culture proxies 0.44 (0.47). The mean (median) correlation coefficient between geography and culture proxies is 0.32 (0.33), between geography and French legal origin

<sup>10</sup>Additional countries covered in [Nunn and Puga \(2012\)](#) are not part of our sample and these authors also do not seem to have carried over the changes in [La Porta et al. \(2008\)](#) vis-à-vis [La Porta et al. \(1998\)](#).

<sup>11</sup>Following the ideas of Max Weber, dummies for ‘no European descendants’ or ‘no European settlers’ at the turn of the 20th century would have been an insightful alternative proxy for culture. However, this setup fails to create a sufficient set of control countries in the ‘non-zero’ group (only 5 for the ANRR and 8 for the liberal democracy specifications, respectively, for 31 and 32 democratisers).

<sup>12</sup>This is for the full sample which includes always-democracies, never-democracies and democratisers.

0.03 (0.02) and between French legal origin and culture -0.18 (-0.16) — the latter two comparisons are for merely six and three correlation coefficients, respectively.<sup>13</sup> See Appendix Table A-3 for all pairwise correlations.

## 2.2 The uneven effect of democracy

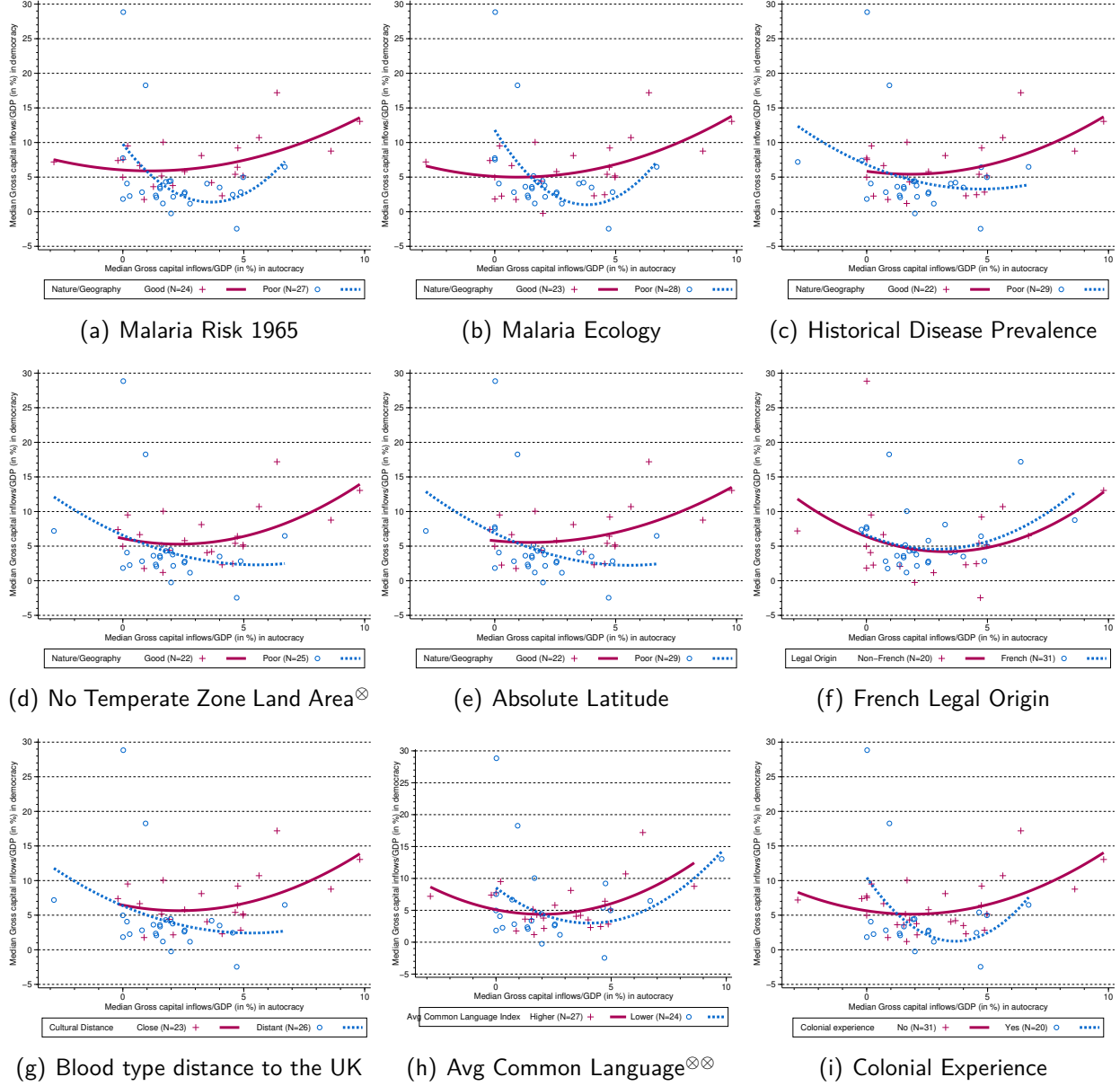
In Figure 1 we provide first descriptive evidence that democracy has an uneven effect on capital inflows, and that geography appears to be one good candidate (albeit not the sole candidate) to explain the observed patterns. Each plot is for a proxy for deep determinants relating to geography, legal origin, history or culture. We plot the country-specific median capital inflow/GDP value during democracy (on the  $y$ -axis) against its median value during autocracy (on the  $x$ -axis) for two sets of countries: those with good geography (or more individualistic/proximate culture, or non-French legal origin or no colonial experience) using dark pink markers and solid quadratic regression lines, and for countries with poor geography (or more collectivist/distant culture, or French legal origin or colonial history) using blue markers and dashed quadratic regression lines.<sup>14</sup>

The resulting patterns are quite similar across the geography proxies in panels (a) to (e): for similar levels of capital inflows during autocracy, regime change in ‘good’ geography countries on average leads to higher capital inflows than in ‘poor’ geography countries. Take the Malaria Risk proxy in panel (a): most blue markers are in the area between 0% and 5% (both in terms of the  $x$ - and  $y$ -axis), whereas many dark pink markers between 0% and 5% on the  $x$ -axis (in autocracy) have post-regime change median inflows in excess of 5%. Equivalently, beyond a pre-regime change inflow in excess of 1%, the fitted regression lines for ‘good’ geography countries is to the North of the ‘bad’ geography one and rising. The divergence between ‘good’ and ‘bad’ geography samples is particularly marked for the climate-related measures in panels (d) and (e). In contrast, using the same strategy but distinguishing countries by (French) legal origin in panel (f) yields virtually no differences between the two sets of country results. The proxy for culture based on blood type distance to the UK in panel (g), however, shows a similar deviation to the above geography proxies. The measure for common language (analysis of the alternative language proxy is omitted as results are virtually identical) once again indicates no substantive deviation between the two country groups. Finally, sample split by colonial experience in panel (i) is once again aligned with the geographic splits, with lower regime change effects for colonised countries.

<sup>13</sup>These are pairwise correlations for single cross-sections of our treatment and control samples using the liberal democracy definition (between 99 and 109 countries). The correlations are moderately lower than equivalent pairwise correlations for the sample of all (137-147) countries, including ‘always democracies’.

<sup>14</sup>We omit the plots for tropical land area and average language similarity, which respectively are qualitatively identical to the temperate land zone and average common language index versions presented.

**Figure 1:** Patterns of Capital Inflows before/after regime change by Deep Determinant



**Notes:** We present scatters and quadratic regression lines for the relationship between median total capital inflows/GDP before ( $x$ -axis) and after democratic regime change ( $y$ -axis), distinguishing ‘deep determinants’ in each plot as proxies for geography, Legal Origin or culture.  $\otimes$  and  $\otimes\otimes$  indicate that we omit the plots for tropical land area and average language similarity, which respectively are qualitatively identical to the temperate land zone and average common language index versions presented. To aid presentation, we omit the observation for Panama (a ‘bad’ geography, French legal origin and close/proximate cultural distance country) in all plots, which during autocracy (democracy) experienced a range of capital inflows/GDP between -295% and +206% (-23% and +70%) with a median of 22% (20%).

## 3 Democracy and Capital Inflows

### 3.1 Data, Methodology and Presentation

**Data and Transformations** We focus on two indicators for democratic regime change which combine elements of electoral democracy and aspects related to the rule of law and executive constraints: first, we adopt the binary indicator of democratic regime change from [Acemoglu et al. \(2019, ANRR, ending in 2010\)](#). This represents a union, or sorts, of a positive Polity IV polity2 index and a Freedom House index (FHI) coded as ‘free’ or ‘partially free’ to “purge spurious changes in each” (50) — panel (b) of Appendix Figure [A-1](#) provides a visualisation of the institutions covered by these indices. ANRR further build on the practice of [Papaioannou and Siourounis \(2008\)](#) and consider each case of democratisation in their data against the historical narrative. Finally, in contrast to the practice in much of the earlier work, they do not retrospectively re-code short episodes of democracy. Second, we take the V-Dem definition of ‘liberal democracy’ combining the principle of electoral democracy (polyarchy, following the work by [Dahl, 1971](#)) with executive constraints and the rule of law (summarised as the ‘liberal component’ in the V-Dem data, [Coppedge et al., 2021](#)) — the latter two institutional factors are seen as the “truly distinctive” feature of liberal democracy ([Mukand and Rodrik, 2020, 765](#)). This measure for liberal democracy<sup>15</sup> is an index between 0 and 1, we adopt the cross-country mean for this index as our cut-off for democracy.<sup>16</sup> In additional analysis, enabled by the hierarchical structure of the V-Dem indices (see panel (a) of Appendix Figure [A-1](#) for a visualisation), we ask whether results differ according to the two building blocks of liberal democracy, adopting the sample mean of the polyarchy index and the liberal component as respective cutoffs. This distinction is of interest as political scientists have favoured electoral democracy as the minimal definition whereas economists have typically highlighted the institutional qualities of property rights and executive constraints (see [Glaeser et al., 2004](#); [Rodrik et al., 2004](#), for an earlier debate on whether ‘institutions rule’).

We study two measures of capital inflows from the IMF Financial Flow Analysis (FFA) database: (1) total capital inflows, excluding the official sector,<sup>17</sup> and (2) FDI inflows.<sup>18</sup> These measures are expressed in percent of GDP although we also employ per capita series in robustness checks. We adopt *gross* capital inflows:<sup>19</sup> net capital flow dynamics may be driven by inflows or outflows and the factors underlying these may be different ([Rothenberg and Warnock, 2011](#);

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<sup>15</sup>The similarity in names is unfortunate, but it is important to stress that we do *not* employ the [Lührmann et al. \(2018\)](#) ROW ‘liberal democracy’ definition.

<sup>16</sup>In robustness checks, we adopt the mean plus 1/4 or 1/2 standard deviation of the respective V-Dem index.

<sup>17</sup>Total Non-Official Capital Inflows, defined as `icapf1 - iothfg`: Total inflows less other inflows to official sector. Total inflows are made up of `ifdi + ipf + idrvtv + iothf`: FDI inflows, portfolio inflows, derivative inflows and other inflows. The resulting flow is expressed in percent of GDP (`icapflp_gdp`).

<sup>18</sup>FDI inflows, expressed in percent of GDP (`ifdi_gdp`).

<sup>19</sup>Appendix Figure [B-10](#) illustrates the results (by geography and alternative deep determinants) using net capital flows. While the patterns are similar to those in the analysis of gross flows, virtually none of the ATET estimates are statistically significant and the vast majority of specifications fail the weak parallel trend test — see Appendix Tables [B-9](#) and [B-10](#).

Byrne and Fiess, 2016), an insight which came to prominent attention in the Global Financial Crisis when gross inflows increased dramatically while net flows remained ‘subdued’ (Kaminsky, 2019; see also Forbes and Warnock, 2012 and Broner et al., 2013 ).<sup>20</sup> Appendix Figure A-2 charts the median evolution of capital inflows over the past 40 years.

In robustness checks, we include additional controls for export/trade (constructed from IMF DOTS) as well as population growth and per capita GDP growth (from the updated ‘Maddison’ database, Bolt and van Zanden, 2020).

**Sample** Studying the details of the sample makeup in Appendix Table A-1 it is very clear that our analysis here largely *excludes* advanced economies: 33 High-income economies were always democracies (Liberal Democracy definition), only nine experienced democratic regime change (out of a treated sample of 51 countries) and only six are in the control sample (out of a control sample of 58 countries).<sup>21</sup>

**Principal Component DID** We estimate country regressions for treated countries only but augment each country-regression with common factors estimated from the residuals of the same regression model *in the control sample* via Principal Component Analysis (following Chan and Kwok, 2022).<sup>22</sup> The basic intuition of this approach is that the unobserved time-varying heterogeneity driving outcomes (capital flows) and determinants (democratic regime change, controls) in the treated sample of countries (which did democratize at one point) can be proxied by information collected in the control sample (countries which never democratized). If we ignored unobserved time-varying heterogeneity in our treatment regression, then it would suffer from omitted variable bias. Using estimated ‘placeholders’ for this heterogeneity, we can (under reasonable and testable assumptions) identify a causal treatment effect. Consider a standard fixed effects regression: adding country fixed effects solves the problem that time-invariant heterogeneity could be correlated with the dependent and independent variables, hence biasing any estimates for the latter. The PCDD is part of a suite of empirical estimators exploiting ‘interactive fixed effects’ (Bai, 2009; Gobillon and Magnac, 2016; Xu, 2017): adding estimated common factors in the treatment regression and allowing each factor to have a country-specific coefficient solves the problem that treatment could be endogenous *and* that treated and control countries may be on different ‘trajectories’ before the treatment already (non-parallel trends). Like any DID estimator, there is some variant of a parallel trend assumption that needs to be satisfied: for the PCDD, the requirement is that the ‘information’ captured by the factors in the control sample is ‘relevant’ for the treated

<sup>20</sup>The FFA series start in 1970, however, we do not use the first five years of data: our empirical setup would imply that a mere 1 or 2 control group countries were available for 1970-74 and as a result, virtually all ‘weak parallel trend’ Alpha tests (see below) reject.

<sup>21</sup>The nine treated countries are Croatia, Hungary, Uruguay, Panama, South Korea, Poland, Chile, Spain, and Portugal. The six control countries are Hong Kong, Kuwait, Oman, Saudi Arabia, Singapore, and the Seychelles.

<sup>22</sup>This dimensionality-reducing approach is very popular in the forecasting literature (Stock and Watson, 2002) but has also been employed to proxy productivity in cross-country analysis (e.g. Eberhardt et al., 2013; Eberhardt and Presbitero, 2015; De Visscher et al., 2020).

sample — the factor coefficients should on average be equal between treated and control samples, which we can investigate using the [Chan and Kwok \(2022\)](#) Alpha test. We discuss our empirical strategy in more formal terms in the following.

Using potential outcomes, the observed outcome of treatment  $D_{it}$  for panel unit  $i$  at heterogeneous time  $T_{0i}$  can be written as

$$y_{it} = D_{it}y_{it}(0) + (1 - D_{it})y_{it}(1) = \Delta_{it}\mathbf{1}_{\{i \in E\}}\mathbf{1}_{\{t > T_{0i}\}} + y_{it}(0) \quad (1)$$

$$\text{with } y_{it}(0) = \varsigma_i + \beta'_i x_{it} + \mu'_i f_t + \tilde{\epsilon}_{it}, \quad (2)$$

where the two indicator variables  $\mathbf{1}_{\{\cdot\}}$  are for the treated panel unit and time period, respectively,  $\Delta_{it}$  is the time-varying heterogeneous treatment effect,  $x$  is a vector of control variables with associated country-specific parameters  $\beta_i$ ,<sup>23</sup>  $\mu'_i f_t$  represents a set of unobserved common factors  $f_t$  (which can be nonstationary) with country-specific factor loadings  $\mu_i$ , and  $\tilde{\epsilon}_{it}$  is the error term.

The treatment effect is assumed to decompose into  $\Delta_{it} = \bar{\Delta}_i + \tilde{\Delta}_{it}$ , with  $E(\tilde{\Delta}_{it}|t > T_{0i}) = 0$   $\forall i \in E$  since  $\tilde{\Delta}_{it}$  is the demeaned, time-varying idiosyncratic component of  $\Delta_{it}$ ; we refer to  $\bar{\Delta}_i$  as ITET, the treatment effect of unit  $i$  averaged over the treatment period — this is our parameter of interest. The reduced-form model is

$$y_{it} = \bar{\Delta}_i\mathbf{1}_{\{i \in E\}}\mathbf{1}_{\{t > T_{0i}\}} + \varsigma_i + \beta'_i x_{it} + \mu'_i f_t + \epsilon_{it}, \quad (3)$$

with  $\epsilon_{it} = \tilde{\epsilon}_{it} + \tilde{\Delta}_{it}\mathbf{1}_{\{i \in E\}}\mathbf{1}_{\{t > T_{0i}\}}$ . Given the treatment effect decomposition  $\epsilon_{it}$  has zero mean but may be heteroskedastic and/or weakly dependent.

The factor structure has a long tradition in the panel time series literature to capture strong cross-section dependence ([Pesaran, 2006](#); [Bai, 2009](#)), a form of unobserved, time-varying heterogeneity. Strong correlation across panel members is distinct from weaker forms of dependence, such as spatial correlation, and if ignored can lead to serious (omitted variable) bias in the estimated coefficients on observable variables ([Andrews, 2005](#)). Here, the combination of common factors and heterogeneous parameters also allows for potentially non-parallel trends across panel units, most importantly between treated and control units. The above setup can further accommodate endogeneity of treatment  $D_{it}$  in the form of *inter alia* correlation between treated units and factor loadings, the timing of treatment and factor loadings, or between observed covariates and timing or units of treatment.

The estimation of the country-specific treatment effect (ITET)  $\bar{\Delta}_i$  proceeds in two steps: first, using Principal Component Analysis (PCA), we estimate proxies of the unobserved common factors from data in the control group (details below); second, country-specific least squares regressions of treated countries are augmented with these factor proxies as additional covariates.

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<sup>23</sup>We assume  $\beta_i = \bar{\beta} + \tilde{\beta}_i$  with  $E(\tilde{\beta}_i) = 0$  ([Pesaran, 2006](#)).  $x$  can be a function of  $f$ .

The main identifying assumptions are that all unobserved determinants of capital inflows are captured by the factors, a standard assumption in the panel time series literature (Pesaran, 2006; Bai, 2009) and related causal panel models (Athey and Imbens, 2022). Since the factors are estimated with error, there is a potential correlation between the errors of treated and control countries, which will bias the treatment estimate. This bias asymptotically disappears if we require that  $\sqrt{T}/N_c \rightarrow 0$ , where  $T$  is the time series dimension and  $N_c$  is the number of control countries. It is further assumed that conditional on the estimated factors the control variables  $x$  are jointly insignificant predictors for the treatment — they do not constitute ‘bad controls’.<sup>24</sup> Treated countries further have to satisfy the ‘weak parallel trend’ test, which we have described above as a way of confirming that the ‘information’ (the space spanned by the estimated factors) from the control sample on average has the same effect in treatment and control sample — see discussion in the paragraph on Diagnostic Testing below.

The estimation equation for each treated country  $i \in E$  is then:

$$y_{it} = b_{0i} + \delta_i \text{Dem}_{it} + a'_i \hat{f}_t + b'_{1i} x_{it} + u_{it}, \quad (4)$$

where  $\hat{f}$  are the estimated factors obtained by PCA on the residuals  $\hat{e}$  from the heterogeneous regression of  $y_{it} = b_{0i} + b'_{1i} x_{it} + e_{it}$  in the control group sample, and  $\delta_i$  is the country-specific parameter of interest for the democratic regime change dummy  $\text{Dem}_{it}$ .  $y$  is the capital flow measure and  $x$  are additional controls we include in robustness checks (export/trade, population growth, per capita GDP growth). We estimate (4) augmented with two to six common factors, given that determining the ‘relevant’ number of factors is fraught with difficulty and ambivalence. The average treatment effect (ATET,  $\hat{\delta}^{MG}$ ) is simply the average of the country estimates  $\hat{\delta}_i$ . We follow the practice in the literature and use the robust mean group estimate (Hamilton, 1992) with the associated standard errors based on  $\Sigma^{MG} = (N - 1)^{-1} \sum_i (\hat{\delta}_i - \hat{\delta}^{MG})^2$  (Pesaran, 2006).

All of the above is laid out for a sample of  $N$  countries. In our analysis we will estimate separate models by deep determinant of development. We do not rule out that geography or culture or history or legal origin may have an effect on the propensity of countries becoming a democracy, but adopting high barriers on our definition of democratic regime change (following Acemoglu et al. (2019) and the V-Dem definition of liberal democracy) in each treatment sample of, say, ‘good’ and ‘poor’ geography, we in effect hold the correlation between the deep determinant and democratic regime change constant across samples. This allows us to study the effect of geography on the implications of democratic regime change in isolation between countries with ‘good’ and ‘poor’ geography and equivalent distinctions for alternative deep determinants.

**Diagnostic testing** The validity of standard pooled Difference-in-Differences estimators crucially relies on the parallel trend assumption: treatment and control samples cannot be on different tra-

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<sup>24</sup>We carry out Wald tests for this assumption — see discussion in the following paragraph and Appendix Tables B-3 and B-5.

jectories prior to the treatment. In the context of the PCDID, we can allow for non-parallel trends between treated and control samples by means of a common factor model with heterogeneous factor loadings, but we nevertheless need to confirm the assumption of ‘weak parallel trends’ via the Alpha test ([Chan and Kwok, 2022](#)): we conduct an auxiliary regression for the treated sample

$$y_{it} = \alpha_i + \beta_i \text{Dem}_{it} + \gamma_i \bar{e}_t + b'_{1i} x_{it} + \epsilon_{it}, \quad (5)$$

where  $\bar{e}_t$  is the cross-section average of the residual of the control sample regression  $y_{it} = b_{0i} + b'_{1i} x_{it} + e_{it}$  from which in the PCDID we extract the common factors. The null hypothesis of the Alpha test is that treatment and control samples are driven by the same set of common factors and rejection of the null suggests the PCDID model is potentially misspecified. The test is in the form of a  $t$ -test for the cross-country average coefficient of  $\gamma_i$  in equation (5) being equal to 1, implemented via the [Pesaran and Smith \(1995\)](#) Mean Group estimator.

A second concern arises if we add controls to the regression model, since these may be ‘bad controls’ in the sense of [Angrist and Pischke \(2008, 64\)](#): “Bad controls are variables that are themselves outcome variables in the notional experiment at hand.” In the present case, we assume that conditional on the estimated factors in equation (4) there is no correlation between the treatment variable  $\text{Dem}_{it}$  and the control variables  $x$ . We test this assumption by regressing the democracy dummy on estimated factors and controls in the treated sample and carrying out a Wald test for the joint insignificance of the controls. If the null is rejected we need to conclude that the controls may constitute ‘bad controls’. Implementation is via the Mean Group estimator.

**Presentation of results** The common practice in the treatment effects literature is to report the ATET,  $\hat{\delta}^{MG}$ . Given the uncertainty over how many estimated factors to include (from [Moon and Weidner, 2015](#), we know that including too many has only minimal effect on consistent estimation in OLS models like ours), the number of democracy indicators and proxies for deep determinants, and alternative specifications with additional controls the reporting of our findings will largely be confined to visual presentation. This enables us to highlight the patterns in the unequal effects of democracy on capital flows, while important diagnostic test results will be reported in an Appendix and described in broad terms in our discussion of the results.

## 3.2 Main Results

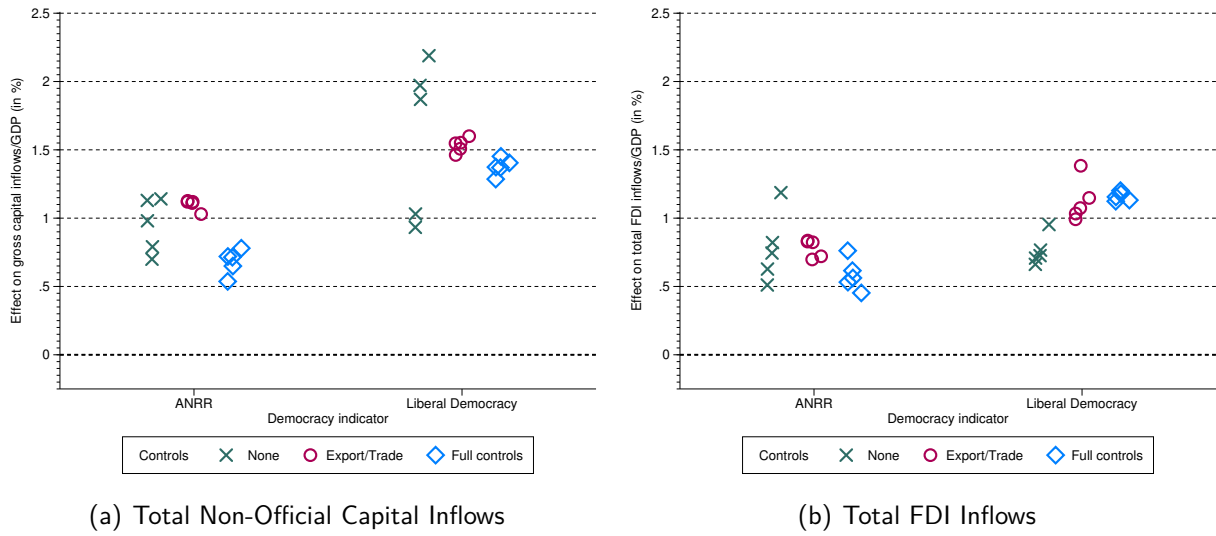
**Ignoring Deep Determinants** Before we investigate the uneven effect of geography we estimate the full sample PCDID average treatment effects on the treated (ATET) of democratic regime change on capital inflows (total, FDI) to offer a benchmark for later analysis. Depending on the definition of the democracy dummy the treated samples amount to between 51 and 69 countries, with control samples ranging from 31 to 58 countries.

**Table 1:** Democratic Regime Change and Capital Inflows (1975-2015)

|                                                                      | Total Capital Inflows |                     | FDI Inflows         |                     |
|----------------------------------------------------------------------|-----------------------|---------------------|---------------------|---------------------|
|                                                                      | (1)                   | (2)                 | (3)                 | (4)                 |
|                                                                      | ANRR                  | LibDem              | ANRR                | LibDem              |
| <b>Panel A: No control variables</b>                                 |                       |                     |                     |                     |
| Democratic Regime Change                                             | 0.773**<br>[0.388]    | 1.852***<br>[0.498] | 0.804***<br>[0.178] | 0.749***<br>[0.245] |
| Alpha Test ( $t$ )                                                   | -0.92                 | -2.08               | 0.23                | 0.31                |
| <i>Alternative factor augmentation</i>                               |                       |                     |                     |                     |
| 2 factors                                                            | 1.147***              | 2.195***            | 1.193***            | 0.959***            |
| 3 factors                                                            | 0.707*                | 1.979***            | 0.752***            | 0.733***            |
| 4 factors                                                            | 0.773**               | 1.852***            | 0.804***            | 0.749***            |
| 5 factors                                                            | 0.989**               | 1.039***            | 0.636***            | 0.716***            |
| 6 factors                                                            | 1.127**               | 0.930**             | 0.508***            | 0.659***            |
| <b>Panel B: Export/Trade as control variable</b>                     |                       |                     |                     |                     |
| Democratic Regime Change                                             | 1.102***<br>[0.374]   | 1.535***<br>[0.409] | 0.806***<br>[0.196] | 1.366***<br>[0.324] |
| Alpha Test ( $t$ )                                                   | -0.53                 | -0.48               | 0.25                | 1.21                |
| $\chi^2$ Test ( $p$ )                                                | 0.36                  | 0.30                | 0.78                | 0.28                |
| <i>Alternative factor augmentation</i>                               |                       |                     |                     |                     |
| 2 factors                                                            | 1.036***              | 1.606***            | 0.727***            | 1.154***            |
| 3 factors                                                            | 1.118***              | 1.515***            | 0.705***            | 1.081***            |
| 4 factors                                                            | 1.102***              | 1.535***            | 0.806***            | 1.366***            |
| 5 factors                                                            | 1.135***              | 1.471***            | 0.844***            | 1.042***            |
| 6 factors                                                            | 1.117***              | 1.545***            | 0.826***            | 0.989***            |
| <b>Panel C: Export/Trade, pop. growth, GDP pc growth as controls</b> |                       |                     |                     |                     |
| Democratic Regime Change                                             | 0.632*<br>[0.372]     | 1.435***<br>[0.483] | 0.545***<br>[0.190] | 1.162***<br>[0.335] |
| Alpha Test ( $t$ )                                                   | -0.48                 | -1.05               | 0.86                | 0.29                |
| $\chi^2$ Test ( $p$ )                                                | 0.00                  | 0.10                | 0.03                | 0.07                |
| <i>Alternative factor augmentation</i>                               |                       |                     |                     |                     |
| 2 factors                                                            | 0.786**               | 1.411***            | 0.459**             | 1.138***            |
| 3 factors                                                            | 0.721*                | 1.377***            | 0.623***            | 1.209***            |
| 4 factors                                                            | 0.632*                | 1.435***            | 0.545***            | 1.162***            |
| 5 factors                                                            | 0.728**               | 1.382***            | 0.540***            | 1.164***            |
| 6 factors                                                            | 0.534                 | 1.283**             | 0.759***            | 1.121***            |
| Treated Countries                                                    | 69                    | 51                  | 69                  | 51                  |
| Treated Observations                                                 | 2087                  | 1830                | 2072                | 1830                |
| Control Countries                                                    | 31                    | 58                  | 31                  | 58                  |
| Control Observations                                                 | 825                   | 1800                | 819                 | 1779                |

Notes: We present robust mean estimates from PCDID regressions of total non-official capital inflows and FDI inflows and a democracy dummy defined as indicated in each column — these estimates can be interpreted as Average Treatment Effects on the Treated (ATET). The main results and standard errors in square brackets (estimated non-parametrically following [Pesaran, 2006](#)) are for the specification augmented with *four* common factors. In a lower part of each panel, we report the ATET estimates for specification with two to six factors. We further provide details of the Alpha test for weak parallel trends ( $t$ -ratio reported) and a  $\chi^2$  test for the control variables ( $p$ -value reported) — in both cases sound diagnostics imply we would not want to reject the null. Sample details are reported in the bottom rows of the table. We use \*, \*\*, and \*\*\* to indicate statistical significance at the 10%, 5% and 1% level.

**Figure 2: Democracy and Capital Inflows**



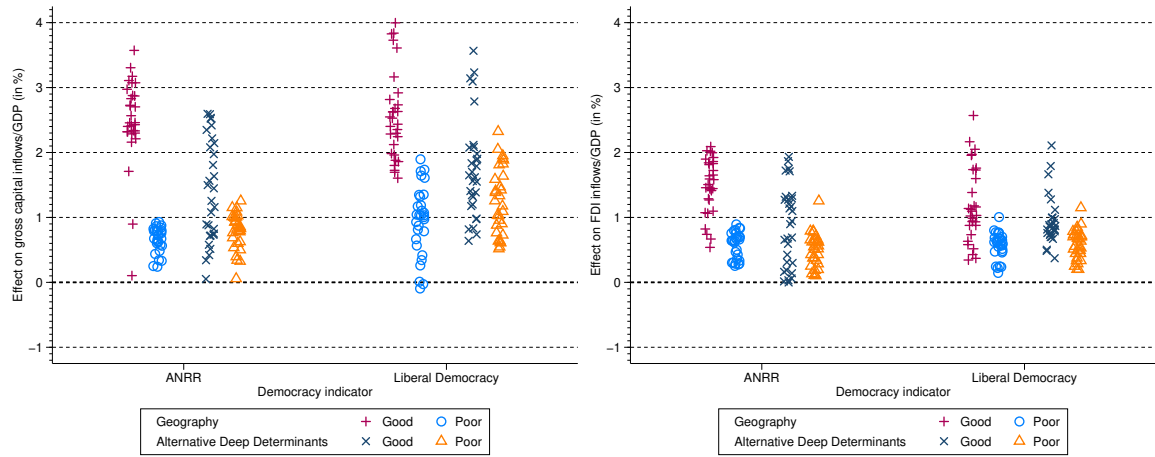
**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows (left) and FDI inflows (right) for two definitions of democratic regime change. Each result ‘cloud’ (markers are randomly perturbed to aid visualisation) features PCDID augmentations with 2 to 6 estimated factors (hence: 5 markers per ‘cloud’). For each democracy definition, we present results for a specification without any controls (x), with export/trade (o), and with all controls (◇). The plots ignore statistical significance or weak parallel trend tests — see Table 1.

Table 1 presents the results with different panels referring to the specifications with no controls, export/trade as control and, additionally, population growth and per capita GDP growth as controls. Our diagnostic tests indicate that the assumption of weak parallel trends is typically confirmed, with the notable exceptions of LibDem in model (2) of Panel A. Export/trade (Panel B) is not a ‘bad control’, while the more elaborate set of controls in Panel C does not pass this test — a pattern that will repeat itself throughout our analysis below. Figure 2 visualises all ATETs for total capital inflows (left panel) and FDI inflows (right panel).

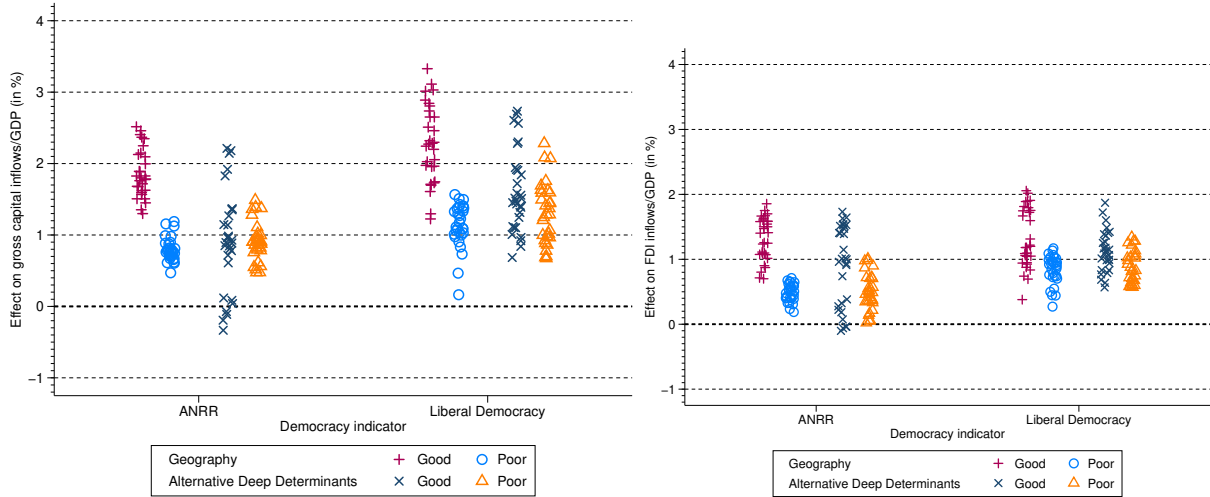
We find ample evidence for statistically significant and economically sizeable effects of democratic regime change: focusing on the specification with exports as additional control (pink circles), regime change has a causal effect of 1 (ANRR) to 1.5 (LibDem) percentage points higher gross capital inflows and 0.75 (ANRR) to 1.25 (LibDem) percentage points higher FDI inflows. These effects are economically large, given the average 3% capital flow/GDP ratio and 1.4% FDI/GDP ratio for treated countries prior to regime change. Although the existing literature primarily focused on proxies for different institutions instead of an overarching concept of democracy (Alfaro et al., 2008; Papaioannou, 2009), our findings underline the notion that a substantial improvement in institutions is associated with substantial additional capital inflows.

**Gross Capital Inflows** We present our results for total non-official capital inflows in Appendix Table B-1 — these are only the results for the specification without any additional control variables,

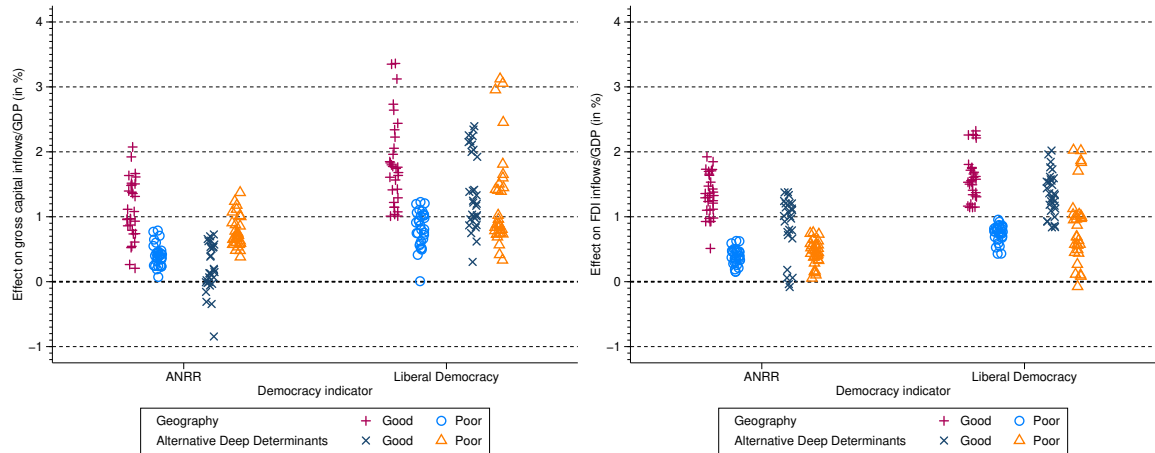
**Figure 3: Democracy, Deep Determinants and Capital Inflows**



(a) Total Non-Official Capital (left) and FDI (right) Inflows – no controls



(b) dto – export/trade as control (not a bad control)



(c) dto – full controls (bad controls)

**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by geography (+ and o for good and poor geography, respectively) and alternative deep determinants (x for non-French LO/proximate culture/no colonial experience and  $\triangle$  for French LO/distant culture/colonial experience). Each result 'cloud' (markers are randomly perturbed to aid visualisation) features 30 estimates: six proxies of good/bad geography or other deep determinants and PCDID augmentations with 2 to 6 estimated factors. The plots ignore statistical significance or weak parallel trend tests (see Appendix Tables B-3 and B-4). Further tests (Appendix Table B-3) indicate exports/trade on its own is not a bad control.

distinguished by geography. Results for the same specification distinguished by alternative deep determinants are presented in Appendix Table B-2. Treated samples typically cover 25 to 35 countries, control samples are generally more modest in size. The tables report Alpha test results for which a  $t$ -statistic in excess of 1.96 indicates the weak parallel trend assumption is violated, suggesting that the PCDID model may be misspecified.<sup>25</sup>

There are a very large number of estimates (120 in each of Tables B-1 and B-2), even though we just present one specification (without controls) and we use a visual representation of the results in Figure 3 to highlight the general patterns. The left column of plots is for total capital inflows, the right column for FDI inflows; in each column the plot in panel (a) is for the specification without additional controls, that in panel (b) with trade/GDP as additional control, and that in panel (c) with full controls. Note that the latter always fail the ‘bad controls’ test and these are just presented for completeness. Each plot is organised by the definition of democracy ( $x$ -axis) and markers signifying ‘good’ versus ‘poor’ natural environment (pink + and blue o), or ‘good’ versus ‘poor’ alternative deep determinant (navy x and orange  $\Delta$ ). Each marker indicates the effect of democratic regime change on capital inflows; each result ‘cloud’ is made up of 30 estimates, since we have five alternative PCDID augmentations with estimated factors and six proxies for geography or the other deep determinants, respectively. These plots ignore the statistical significance of the ATET estimates and further do not indicate whether individual PCDID models satisfy the weak parallel trends test — we comment on these in broad brushes below and refer to detailed results in Appendix Tables B-3 and B-4.

The left column of Figure 3 is for total capital inflows. Most of the estimates for ‘good’ geography (pink +) are statistically significant at the 10% level, for ‘poor’ (blue o) geography only around two-thirds are; and all satisfy the weak parallel trend assumption. These causal effects of democracy on total capital inflows show distinct patterns which are visually striking: in the model without controls, democratising countries with ‘good’ geography experience a 2-4 percentage point increase in capital inflows, whereas those with ‘poor’ geography see much more moderate effects, a 0-2 percentage point increase, if that. In the model with trade/GDP as additional control, these magnitudes are somewhat moderated, but the striking pattern remains: ‘good’ geography results are substantially larger than those in the ‘poor’ geography samples.<sup>26</sup> This pattern also does not change in the model with full controls. The sample mean of total capital inflows over GDP for the respective treated sample prior to democratic regime change (i.e. all years in autocracy) is between 2.5% and 3.3%, which indicates that our average treatment effects are economically large.

The distinction between results for ‘good’ (navy x) and ‘poor’ (orange  $\Delta$ ) alternative deep determinants (legal origin, culture, colonial history) is presented in the same plots. In broad terms

<sup>25</sup>In Appendix Table B-11 we report results for four more proxies related to geography: being landlocked, high UV radiation exposure, limited frost days and low suitability for agriculture – with the exception of the latter, the same patterns as discussed below prevail.

<sup>26</sup>The largest estimates for ‘good’ alternative deep determinant samples are always for colonial experience — see Appendix Figure B-1.

this distinction does not yield evidently systematically different treatment effects (particularly so for the LibDem democracy definition), whether we study democratising countries with French or other legal origins, those with or without colonial experience, or whether we compare more individualistic and more collective societies as well as alternative proxies for cultural ‘clustering’.<sup>27</sup> This is best illustrated in the model with export as additional control in panel (b), where for the ANRR definition small clusters of ‘good determinant’ estimates are respectively larger and smaller than the cluster of estimates for ‘poor determinants’, while for the Liberal Democracy definition there is very little between the two. Hence, our results for total capital inflows suggest that we see substantial and systemic differences in the effects of democratic regime change by geography but not by alternative deep determinants.

**FDI inflows** The right column of plots in Figure 3 visualises the ATET estimates for causal effects on FDI inflows in the format introduced above. For the distinction by geography the overwhelming majority of estimates are statistically significantly different from zero, and we have few specifications which reject the weak parallel trend assumption. Compared with the results for total capital inflows the differences in the result patterns are less substantial but still marked: the ATET for countries with poor geography is typically below 1 percentage point, while in countries with good geography, the effect ranges from 0.25 to 2.5 percentage points. The economic effects of democratic regime change are again substantial (in relative terms even more so than for total capital inflows), given that the mean for FDI inflows over GDP is around 1.1-1.4%.

For the distinction by alternative deep determinants<sup>28</sup> the difference in treatment effects is less marked than in the case of gross total inflows. If we again take the model with export/trade as additional control in panel (b) as preferred specification, some of the ‘good’ alternative deep determinant estimates are clearly of greater magnitude than the ‘poor’ sample counterparts. It should be noted that in both cases (for ANRR and the liberal democracy definitions) the largest ‘good deep determinant’ ATET estimates are for the subsamples analysis of countries with colonial experience — see Appendix Figure B-1.

**Mechanical explanations** Naturally, there would be concerns if ‘treated’ countries (i.e. democratisers) with one type of deep determinant (say, low absolute latitude) would have a significantly higher propensity to revert to autocracy than those of the other type (high absolute latitude): we’d be comparing ‘solid’ and ‘shaky’ democracies. The average ‘reversal’ probability in treated samples is between 1 and 2.5 percent. Carrying out comparison in means between the treated samples of the two ‘types’ we find that those with ‘favourable’ deep determinants (e.g. low absolute latitude, legal origin other than French civil law, etc.) typically have lower propensity of reversal (1.2%), but the differences between these and samples for ‘unfavourable’ deep determinants (1.6%) are

<sup>27</sup>In Appendix Figures B-2 and B-3 we separate out colonialist experience from the other deep determinants.

<sup>28</sup>Again, the vast majority of effects are statistically significantly different from zero and most models pass the weak parallel trend test.

typically not statistically significant (results available on request).<sup>29</sup>

Similarly, for the magnitudes of the treatment effects, it is important to check whether the length of time spent in democracy does not differ substantially across treatment samples of the two types, since otherwise a bigger boost to growth may simply be down to having spent more years in democracy. The average number of sample years in democracy is between 15 and 23, depending on the definition. We find that samples for countries with ‘favourable’ deep determinants have an advantage, over two-and-a-half additional years of treatment (19.8) compared with countries with ‘unfavourable’ deep determinants (17.2), though once again the difference is not typically statistically significant (results available on request).<sup>30</sup>

### 3.3 Robustness checks and extensions

**Alternative definition of capital flows** Some researchers in the capital flow surge/bonanza literature (see discussion in [Caballero, 2016](#), footnote 10) maintain that the capital flow to GDP ratio is unsuitable for analysis given potentially differential dynamics/trends of the numerator and denominator, suggesting the use of capital flow *per capita* instead. We repeat the analysis using this alternative definition of the dependent variable and present the findings in Appendix Figures [B-4](#) and [B-5](#). Alpha tests for these specifications frequently reject the weak parallel trend assumption while the analysis of bad controls follows the same patterns as above (see Appendix Table [B-5](#)).

We also re-ran the capital flow analysis employing data from the World Bank World Development Indicators to construct control variables, resulting in a substantial reduction in sample size (33% fewer observations in the treated sample, using the liberal democracy definition). Our findings in Appendix Figures [B-6](#) and [B-7](#) show qualitatively similar but weaker patterns to those described in our main results. Due to the smaller sample size the treatment and control samples frequently number only very few countries in the years up to the mid-1990s, with the result that the ‘Alpha’ test for weak parallel trends typically rejects in more than half the specifications whether we use no controls, only trade/GDP or the full set of controls (test results in Appendix Table [B-7](#)).

**Alternative Definitions of Democracy** Our dichotomised regime change indicators for liberal democracy (and in robustness checks its building blocks, polyarchy and the liberal component) are constructed by using the full sample mean as the threshold. The congruence of patterns of results with those based on the [Acemoglu et al. \(2019, ANRR\)](#) definition are an indication that we successfully capture a significant step in the institutional development of our sample countries. Nevertheless, the adopted threshold is arbitrary and to check the robustness of our findings we provide alternative versions where we take the mean plus 1/4 or 1/2 of the standard deviation,

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<sup>29</sup>The exception is the ‘zero land in the temperate zone’ proxy for geography, for which ANRR and liberal democracy definitions of democratic regime change indicate statistical significance at the 5% level.

<sup>30</sup>The exception is the ‘average common native language’ proxy for culture, for which ANRR and liberal democracy definitions of democratic regime change indicate statistical significance at the 5% level.

providing a ‘tighter’ definition of democracy. This substantially reduces the sample size of treated countries (reductions in the number of countries of 20% and 39%, respectively).<sup>31</sup> Despite this caveat, results presented in Appendix Figures B-8 and B-9 for geography and alternative deep determinants are qualitatively very similar to those we present above using the mean index cut-off.

**Building blocks of liberal democracy** Our analysis adopts two data proxies (ANRR, LibDem) for a concept of liberal democracy which encompasses (a) polyarchy (electoral democracy), and (b) the rule of law combined with executive constraints. The nature of the V-Dem data enables us separate these two aspects, with the latter referred to as the ‘liberal component’ (results available on request). Across all specifications we can observe a very clear pattern whereby treatment effects of regime change are *very similar in magnitude* across the two samples of countries with ‘good’ and ‘poor’ geography when we consider polyarchy. In contrast, the treatment effects are *substantially larger* in good geography countries in the analysis of the liberal component: the differential effects of democratic regime change appear to be driven not by aspects related to clean elections, or freedom of speech but to executive constraints and the rule of law (including property rights and individual freedoms). Distinction instead by culture, history or legal origins again offers no discernible differences in treatment effect size.

### 3.4 Excessive Capital Inflows

In this section we consider whether democratic regime change can trigger *excessive* capital inflows and whether it does so with distinct geographic patterns: our above findings of much stronger capital flow effects for countries with ‘good’ geography may be viewed in a very different light if this benefit of regime change is associated with the cost of a higher propensity for excessive capital inflows, thus undermining financial stability and potentially triggering financial crises.

**Data, Sources and Transformations** We gather a small set of primary domestic determinants of excessive capital inflows on the basis of the existing literature. We investigate current account deterioration, international reserves accumulation, real exchange rate appreciation, and drops in CPI (deflation).<sup>32</sup> The data for these determinants are taken from the World Bank WDI. The dependent variables in our Early Warning System (EWS) analysis are various indicator variables for bonanzas and/or surges, where 1 indicates the start year of a bonanza or surge and 0 is for all years without such an event. We use the gross non-official capital inflow series from the IMF FFA to construct these ‘excessive capital inflow’ indicators — see details below. All variables are expressed as changes, growth rates or ratios (these are less likely to be subject to stochastic

<sup>31</sup>Using Liberal Democracy (index cut-offs 0.41 for the mean, 0.48 for mean+1/4SD, and 0.55 for mean+1/2SD) sample size drops from 51 (22 and 29 ‘good’ and ‘poor’ geography countries, respectively) to 41 (20 and 21) and 31 (16 and 15); for Polyarchy (0.51, 0.58 and 0.65) from 59 (25, 34) to 49 (22 and 27) and 38 (19 and 19); for the Liberal Component (0.61, 0.68 and 0.75) from 57 (24, 33) to 48 (21 and 27) and 44 (21 and 23).

<sup>32</sup>Empirical results ignoring regime status or deep determinants in Appendix Figure C-4 give some indication that our methodology captures the association of these trigger variables with episodes of excessive capital inflows.

trends), and are winsorized at the top and bottom 1% of observations, to minimize the role of outliers. We use an MA(3) transformation for all independent variables to capture the cumulative build-up of crisis triggers over time (see [Eberhardt and Presbitero, 2021](#)).<sup>33</sup> Our empirical analysis faces similar difficulties as our earlier analysis with limited control sample coverage in the early years of the dataset and we therefore adopt 1980 as the start year for the analysis.

**Bonanzas and Surges – Definitions** Bonanzas are defined as large deviations from the HP-filtered trend of capital inflows, where following [Caballero \(2016\)](#) we adopt a 1 SD-threshold. Surges at time  $t$  are exceptional levels of capital inflows in the top 30th percentile of both the country-specific and the full sample distribution (following [Ghosh et al., 2014](#)). For illustrations see Appendix C. As these definitions already indicate, bonanzas will be rarer than surges.

**Factor-Augmented Early Warning System** We specify a latent variable model of excessive capital inflows  $Y_{it}^*$  as a function of their dominant predictors in the literature (illustrated below using inflation in the MA(3) transformation) for each country in a ‘treated’ sample of countries for ‘good’ and ‘poor’ geography, respectively:

$$Y_{it}^* = \alpha_i + \beta_i^A \overline{\Delta CPI}_{i,t-1/t-3} + \beta_i^B \mathbf{1}_{\{t > T_{0i}\}} \overline{\Delta CPI}_{i,t-1/t-3} \quad (6)$$

$$+ \gamma_i' \bar{x}_{i,t-1/t-3} + \mu_i' f_t + \varepsilon_{it}, \quad \forall i \in E^* \{\text{good, bad}\}$$

where  $f$  is a set of unobserved common factors with heterogeneous factor loadings  $\mu$  and additional controls are represented by  $x$ . The indicator variable  $\mathbf{1}_{\{\cdot\}}$  captures the time periods country  $i$  spent in democracy:  $\beta_i^A$  is thus the benchmark effect of inflation in autocracy, while  $\beta_i^B$  indicates how the inflation effect deviates from this benchmark during democracy. A positive  $\beta_i^A$  and negative  $\beta_i^B$  would suggest that higher inflation is associated with surges or bonanzas, but that democratic regime change reduces this effect. The EWS is estimated separately for countries with ‘good’ and ‘poor’ geography, like in our earlier capital flow analysis, and we highlight the differential effects of  $\beta_i^A$  and  $\beta_i^B$  across these two samples.

We implement this model by combining work on common factors in a generalised linear model ([Boneva and Linton, 2017](#)) with that on the PCDID ([Chan and Kwok, 2022](#)) to create a factor-augmented EWS approach.<sup>34</sup> We adopt a linear probability model for the year of a surge or bonanza,<sup>35</sup>  $Y_{it}$ , in those countries which experienced democratic regime change (‘treated sample’). The country-specific estimation equation is augmented with up to  $k$  common factors, estimated from the same equation for a set of countries which always remained autocratic (‘control sample’).

<sup>33</sup>Our notation below indicates that  $\overline{\Delta CPI}_{i,t-1/t-3}$  is the unweighted average of inflation for periods  $t - 3$  to  $t - 1$ ; in analogy for all other independent variables.

<sup>34</sup>[Boneva and Linton \(2017\)](#) extend the [Pesaran \(2006\)](#) common correlated effects estimator to the probit model but also support the linear probability model.

<sup>35</sup>If these events extend beyond a single year then we pick the first year and subsequent years are dropped from the sample as per practice in the literature on banking crisis EWS.

The common factors  $\hat{f}$  are estimated via PCA from the residuals of the same model in the control sample (albeit by construction without the interaction term).<sup>36</sup>

We estimate for treated countries  $i \in E^*$  by geography (good, poor)

$$\begin{aligned} Y_{it} = & a_i + b_i^A \overline{\Delta \text{CPI}}_{i,t-1/t-3} + b_i^B \mathbf{1}_{\{t > T_{0i}\}} \overline{\Delta \text{CPI}}_{i,t-1/t-3} \\ & \{ + c_{1i} \overline{\Delta \text{CA}/\text{GDP}}_{i,t-1/t-3} + c_{2i} \overline{\Delta \text{Reserves}/\text{GDP}}_{i,t-1/t-3} \\ & + c_{3i} \overline{\Delta \text{ER}}_{i,t-1/t-3} \} + d_i' \hat{f}_t + \varepsilon_{it}, \end{aligned} \quad (7)$$

where we spell out additional control variables for the case of inflation in detail.  $\varepsilon$  is the error term, which can be heteroskedastic and/or serially correlated. Note that equation (7) features only a single interaction term, namely for inflation: given the number of trigger variables, adding the interaction with the democracy dummy for each of these in a single specification would substantially increase the number of parameters to be estimated for each country in a moderately long panel. Alternative specifications with an interaction term for the change in the current account to GDP ratio, the change in reserves/GDP, and depreciation are constructed analogously.<sup>37</sup> Equation (7) includes *all* controls ('full specification'), we also estimate a 'simple specification' which excludes the additional controls in curly brackets. The factor augmentation captures the developments in the countries which remained autocratic throughout, while the interaction term setup allows us to investigate differential effects of dominant surge/bonanza predictors *within* individual countries when they were autocratic versus democratic.

**Results: Ignoring Geography** We consider the effect of democratic regime change on excessive capital inflows while ignoring geography. Appendix Figure C-5 has very busy and somewhat confusing plots.<sup>38</sup> For a clearer impression of the central tendency we use an M-estimator (Rousseeuw and Leroy, 1987) to compute an outlier-robust mean for the coefficients on the triggers and, separately, for the interaction effects across augmentations with two to four factors. In Table 2 there are pairs of markers where the first is for the trigger effect under autocracy, and the second for the relative trigger effect under democracy. We use  $\uparrow$  ( $\downarrow$ ) for a positive (negative) significant effect, for an insignificant effect we use 0 (10% significance level). A  $(\cdot, \uparrow)$  pair suggests a shift to democracy increased the propensity of an excessive capital inflow episode,  $(\cdot, \downarrow)$  that such a shift reduced the propensity. The former is marked in red, the latter in green to highlight the respective detrimental and beneficial effect of democratic regime change.

<sup>36</sup>The error term includes some estimation error of this process,  $\tilde{f}_t$ , which vanishes as  $\sqrt{T}/N_C \rightarrow 0$  for  $T$  the time series dimension and  $N_C$  the number of control group countries, in which case this term in square brackets is time-invariant. Note further that the estimated factors are *not* MA(3)-transformed since they are estimated from the residuals of a control sample regression in which all regressors are already MA(3)-transformed.

<sup>37</sup>Inclusion of the democracy dummy in the model leads to qualitatively very similar results (available on request).

<sup>38</sup>These provide the differential effect of a 1 SD increase in each of the four trigger variables before (hollow markers or +) and, in relative terms, after democratic regime change (filled markers or x) on the propensity of experiencing a bonanza (left plot) or surge (right plot).

**Table 2:** Schematic Presentation of Vulnerability Results — ignoring geography (1980-2015)

|                                          | (1)       |   | (2)    |   | (3)     |   | (4)      |   |
|------------------------------------------|-----------|---|--------|---|---------|---|----------|---|
|                                          | Deflation |   | CA Det |   | ER Appr |   | Reserves |   |
| Panel (a): Total Capital Inflow Bonanzas |           |   |        |   |         |   |          |   |
| ANRR                                     | 0         | 0 | ↓      | ↑ | ↑       | ↑ | ↓        | ↓ |
| LibDem                                   | 0         | 0 | ↑      | 0 | ↑       | 0 | ↑        | ↓ |
| Panel (b): Total Capital Inflow Surges   |           |   |        |   |         |   |          |   |
| ANRR                                     | ↓         | ↓ | 0      | ↑ | ↑       | ↓ | ↑        | ↓ |
| LibDem                                   | ↑         | ↑ | ↑      | ↓ | ↑       | 0 | ↑        | ↑ |

Notes: This table provides a schematic presentation of the qualitative conclusions from the analysis presented in Appendix Figure C-5, based on robust mean estimates. For each column (trigger variable) and row (definition of democracy) there are two qualitative conclusions: the first for the trigger effect *under autocracy*, the second for the relative trigger effect *under democracy*. The central tendency is estimated from the respective results using an outlier-robust mean: if this average is positive (negative) and significant at the 10% level, we use ↑ (↓), for an insignificant effect we use 0. A (·,↑) pair suggests democracy increased the propensity of an excessive capital inflow episode, (·,↓) that democracy reduced the propensity. The former is marked in red, the latter in green to highlight the respective detrimental and beneficial effect of democratic regime change. Triggers are: deflation, capital account (CA) deterioration, exchange rate (ER) appreciation and reserve accumulation.

It is fair to say that *no clear pattern* of the relationship between democratic regime change and propensity of excessive capital inflows emerges from this analysis: studying individual definitions for democracy, the patterns between the bonanza and surge results are frequently in direct contradiction. Similarly, different definitions of democracy yield different qualitative results for the same trigger variable within the analysis of bonanzas and surges, respectively.

**Results: Introducing Geography** We present results for our analysis differentiated by ‘good’ and ‘poor’ geography in Appendix Figures C-6 and C-7, adopting simple and more complex specifications (in terms of controls included). Since these results are quite overwhelming and difficult to read, we again create a schematic presentation of our findings in Table 3, drawing qualitative conclusions from the scatter graphs in the two figures. We construct these in the same fashion as described above — the underlying mean estimates and associated  $p$ -values are presented in Appendix Table C-1. Separate panels are for (a) bonanzas and (b) surges, we also consider a ‘simple’ model without additional controls and a ‘full specification’ which includes all controls.<sup>39</sup>

A number of conclusions can be drawn from our analysis: (i) the magnitudes of effects during autocratic or democratic periods for a common one standard deviation in the ‘trigger’ variable are typically larger for ‘good’ than for ‘poor’ geography countries (greater spread of pink than blue markers in the Appendix plots; equivalently, higher mean effects across all specifications

<sup>39</sup>Recall that only one trigger variable is ever interacted, using the same strategy for all triggers in a single regression model would be too demanding in a moderate  $T$  sample.

in Appendix Table C-1); (ii) ‘good’ geography countries are frequently subject to large positive trigger effects on the propensity of excessive capital inflow episodes during autocracy (pink + markers are often positive), but, typically, these vulnerabilities are substantially reduced following democratic regime change (negative pink □ markers); this equates to a large number of marker pairs in the *left* half of Table 3 appearing in green and markers for a small number of specifications appearing in red; (iii) ‘bad’ geography countries, in contrast, frequently experience (additional) positive trigger effects during democracy (positive blue o markers); as the right half of Table 3 indicates, this is particularly the case for CA deterioration and reserve accumulation.

**Table 3:** Schematic Presentation of the Vulnerability Results by Geography

|                                                 | (1)<br>Countries with ‘Good’ Geography |        |         |          | (5)<br>Countries with ‘Poor’ Geography |        |         |          |
|-------------------------------------------------|----------------------------------------|--------|---------|----------|----------------------------------------|--------|---------|----------|
|                                                 | Deflation                              | CA Det | ER Appr | Reserves | Deflation                              | CA Det | ER Appr | Reserves |
| <b>Panel (a): Total Capital Inflow Bonanzas</b> |                                        |        |         |          |                                        |        |         |          |
| <i>Simple Specification</i>                     |                                        |        |         |          |                                        |        |         |          |
| ANRR                                            | 0 ↓                                    | 0 ↑    | ↑ ↓     | ↑ ↓      | ↑ ↓                                    | ↓ ↑    | ↑ ↑     | ↑ ↓      |
| LibDem                                          | ↓ 0                                    | ↑ ↓    | ↑ 0     | ↑ ↓      | ↑ ↑                                    | 0 ↑    | ↑ ↓     | ↓ ↑      |
| <i>Full Specification</i>                       |                                        |        |         |          |                                        |        |         |          |
| ANRR                                            | ↑ 0                                    | ↑ ↑    | ↑ ↓     | 0 0      | 0 0                                    | ↓ ↑    | ↑ ↑     | ↑ ↓      |
| LibDem                                          | ↑ ↑                                    | ↑ ↓    | ↑ 0     | 0 ↓      | ↑ ↑                                    | 0 ↑    | ↑ 0     | ↑ ↑      |
| <b>Panel (b): Total Capital Inflow Surges</b>   |                                        |        |         |          |                                        |        |         |          |
| <i>Simple Specification</i>                     |                                        |        |         |          |                                        |        |         |          |
| ANRR                                            | 0 ↓                                    | ↑ ↑    | ↑ ↓     | ↑ 0      | ↑ ↓                                    | ↑ ↑    | ↑ ↓     | ↑ ↑      |
| LibDem                                          | ↑ ↓                                    | ↑ ↓    | ↑ ↓     | ↑ ↓      | ↑ ↓                                    | ↑ ↓    | ↑ ↓     | ↓ ↑      |
| <i>Full Specification</i>                       |                                        |        |         |          |                                        |        |         |          |
| ANRR                                            | ↓ ↓                                    | ↑ ↑    | ↑ ↓     | ↑ 0      | ↓ ↓                                    | 0 ↑    | ↑ ↓     | ↑ ↓      |
| LibDem                                          | ↑ ↑                                    | ↑ ↓    | ↑ ↑     | ↑ ↓      | ↑ 0                                    | ↑ ↑    | 0 0     | ↑ ↑      |

Notes: This table presents an attempt at a schematic presentation of the qualitative conclusions from the analysis presented in Figures C-6 and C-7. For each column (trigger variable, see below) and row (definition of democracy) there are two qualitative conclusions: the first for the trigger effect under autocracy, the second for the relative trigger effect under democracy. The central tendency is estimated from the eighteen results (3 factor augmentations, 6 geography proxies) using an outlier-robust mean: if this average is positive (negative) and significant at the 10% level, we use ↑ (↓), for an insignificant effect we use 0. A (0,0) pair suggests that there were no statistically significant effect in either period, (·,↑) suggests democracy increased the propensity of an excessive capital inflow episode, (·,↓) that democracy reduced the propensity. The former is marked in red, the latter in green to highlight the respective detrimental and beneficial effect of democratic regime change. If democratic regime change does not alter the trigger effect under autocracy then we do not use any highlight colour. Triggers are: deflation, capital account deterioration, exchange rate appreciation and reserve accumulation.

Studying the relative frequency of different effects for the ANRR and liberal democracy definitions in our schematic representation in Table 3, in ‘good’ geography countries 56% of

all results indicate that trigger variable effects are *significantly lowered* after democratic regime change, and 22% indicate they are *significantly increased* (10% level significance level). In ‘bad’ geography countries these proportions are 38% and 50%, respectively. Hence, while democratic regime change is in the majority associated with reduced vulnerability in the ‘good’ geography sample, it is in the majority associated with increased vulnerability in the ‘bad’ geography sample.<sup>40</sup>

## 4 Geographic Patterns, Historical Reasons?

### 4.1 Empirical Exploration of Geographic Channels

In this section we illustrate that economies with poor geography suffer from disadvantageous ‘structural’ characteristics: their exports are less diversified and represent primary commodities subject to higher volatility in world prices, while domestic production lacks economic complexity and goods trade costs are higher. In all cases we compare these differences based on geography to those when we use proxies for legal origin, culture or colonial history to split the sample. All analysis is limited to countries which experienced democratic regime change (adopting the two definitions for democracy we use throughout as well as the split of liberal democracy into ‘polyarchy’ (electoral democracy) and the ‘liberal component’ — see Section 3.3).

#### 4.1.1 Diversity and Quality of Exports

The dominant paradigm for economic development in the second half of the twentieth century places significant emphasis on trade, predominantly labour-intensive manufactured goods for export (such as apparel) to (i) overcome the limits set by domestic markets, (ii) exploit low labour costs, and (iii) initiate a process of moving up the value chain and/or diversifying into more ‘sophisticated’ (higher value-added) products. Studying the diversity as well as the quality of exports can provide insights into the scope for structural transformation and the potential for countries to reap the benefits from diversification (Hausmann et al., 2007; Henn et al., 2013).

We adopt data from an IMF database on ‘export diversification and quality’ covering 1962–2014 (Henn et al., 2013), which is available for all our sample countries. The Theil index we use combines the concentration in the number of export products by a country and the concentration in its export volumes across products actively exported. A *higher* value for this index marks out a country with a *lower* level of diversification.

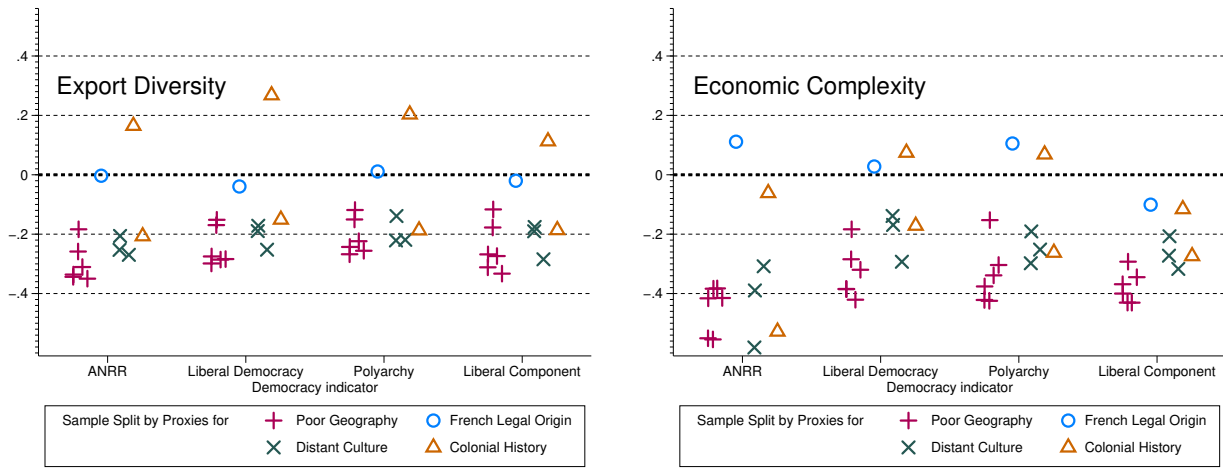
We compute the difference in the level of export diversification for countries with good and poor geography, along with a formal *t*-test for this difference for each of our six proxies for geography and similarly for the other proxies of deep determinants — in the vast majority of mean

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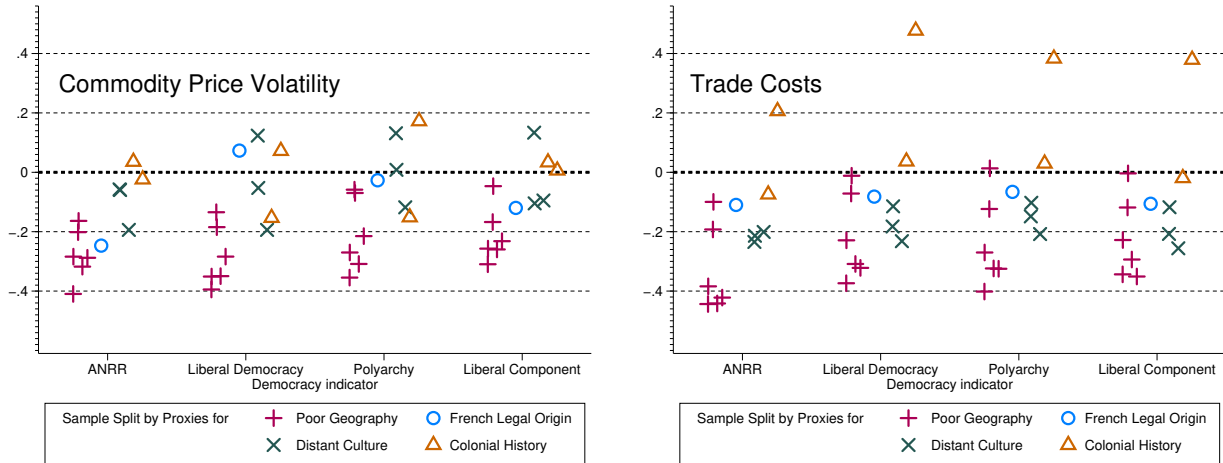
<sup>40</sup>If we carry out the same exercise for all definitions of democracy, in the good geography sample we have again 56% reduced and now 17% increased vulnerability. In contrast these figures are 45% and 36% for ‘bad’ geography countries.

differences studied the  $t$ -tests and associated  $p$ -values indicate that these differences are statistically significant. Results for mean differences are presented in the left plot of panel (a), Figure 4. Across all definitions of democratic regime change, we see that democratising countries with ‘poor’ geography have 10-35% worse (lower) export diversity than their ‘good’ geography peers. For legal origin the differences are negligible, whereas for our cultural proxies they are of similar magnitude (albeit somewhat smaller) than those for geography. For the comparison based on colonial history we see mixed results: ‘extractive colonialisation’ is associated with worse diversity, whereas colonialisation per se correlates with higher diversity.

**Figure 4:** How Deep Determinants Relate to Economic Structure



(a) Relative Export Diversity (left) and Economic Complexity — negative = lower/worse



(b) Relative Aggregate Commodity Price Volatility (left) and Trade Cost — negative = higher/worse

**Notes:** We present mean differences for export diversity, economic complexity, aggregate commodity price (ACP) volatility and trade costs between the samples deemed to have ‘good’ versus ‘poor’ deep determinants of comparative development. Interpretation: in the left plot of panel (a), a result of -0.2 indicates that export diversity is 20% worse in countries with ‘poor’ relative to those with ‘good’ deep determinants. The definition of democracy determines the sample size — we only consider countries which experienced democratic regime change (treated sample).

### 4.1.2 Economic Complexity

Continuing with the notion of product ‘sophistication’ we hypothesise that the narrow(er) range of products produced for export in ‘poor geography’ countries is furthermore of lower complexity. Again results are for democratising countries only and compared with those adopting alternative deep determinants as sample splits.

We adopt data on economic complexity from [Hidalgo and Hausmann \(2009\)](#), who provide rankings across 133 economies in the level of complexity on the basis of the HS (Harmonized System, 1992) product classification. There is a caveat for these data since the series only start in 1995, and some of the sample sizes for ‘treated countries’ in ‘good’ geography locations only feature around 200 observations — these results should be taken with a grain of salt. A higher number implies a lower rank, and hence worse (lower) complexity.

Results for mean differences by deep determinant are presented in the right plot of panel (a), Figure 4. The patterns closely match those of the export diversity analysis above: sample splits by geography and culture show very large differences (lower complexity for ‘poor’ geography or ‘distance’ culture), those for legal origin are comparatively small, while only those for ‘extractive colonisation’ suggest worse complexity.

### 4.1.3 Aggregate Commodity Price Volatility

The analysis of export concentration and economic complexity focused on the basket of goods produced and exported by countries with different deep determinants. But what if, through luck or foresight, countries managed to ‘pick winners’ for their export baskets, goods with advantageous terms of trade and low price volatility? We investigate the economic uncertainty of the basket of primary commodities produced and exported by countries: we ask whether the primary goods exported by countries with ‘poor’ geography are subject to greater exogenous price movements than those with ‘good’ geography (and similarly for the other deep determinants).

For primary commodity price (PCP) volatility we use monthly data from [Gruss and Kebhaj \(2019\)](#) which employs 1962-2018 average net export/GDP weights to aggregate 44 global primary commodity prices from the *IMF Primary Commodity Price Database*: the variations captured hence relate to windfall gains and losses due to changes in exogenous world prices — see [Ciccone \(2018\)](#) and [Eberhardt and Presbitero \(2021\)](#). Primary commodity price shocks are defined as the first difference of the monthly PCP index,  $\Delta PCP_{it\tau} = PCP_{it\tau} - PCP_{it,\tau-1}$  for month  $\tau$  of year  $t$  in country  $i$ . We construct a time-varying measure of PCP volatility following [Bleaney and Greenaway \(2001\)](#): the conditional volatility  $\sigma_{ACP,it\tau}^2$  is predicted from a GARCH(1,1) model of the monthly data for 1975-2018 using a regression of the PCP shocks,  $\Delta PCP_{it\tau}$ , on an intercept term. We convert the monthly estimates to annual frequency by taking the average of monthly volatility within each year.

Results for mean volatility differences by deep determinant are presented in the left plot of Figure 4, panel (b). The patterns match those discussed earlier, although in this instance volatility is substantially higher for the geography split than when we use culture.

#### 4.1.4 Trade costs

Geography in the context of trade is traditionally interpreted as *distance* to or *remoteness* from large markets. We laid out above that the trade gravity literature has a close relation in the capital flow literature (Portes et al., 2001; Bergstrand and Egger, 2007; Head and Ries, 2008; Lane and Milesi-Ferretti, 2008; Pellegrino et al., 2021) and now investigate whether our climatic and disease environment proxies for geography (as well as the proxies for alternative deep determinants) correlate with goods trade costs.

We compute trade costs following the methodology introduced in Novy (2013) using annual bilateral goods trade data from IMF DOTS and GDP data from the World Bank WDI (we adopt  $\sigma = 8$ ). We follow Milner and McGowan (2013) in creating country-specific time-varying trade cost averages from these bilateral estimates.

Results for trade cost differences by deep determinant are presented in the right plot of Figure 4, panel (b). The patterns match those discussed earlier, with sample splits by geography and, to a lesser extent, culture associated with very substantial mean differences. This aside, (i) colonisation is now commonly associated with lower trade costs, even in the 'exploitative colonisation' case, and (ii) French legal origin now has a uniformly negative (albeit small) association with trade costs.

#### 4.1.5 Geographic vs Alternative Channels

In summary, investigating four channels through which deep determinants can correlate with economic structure we find that across the board the geographic explanation is associated with much more substantial differences between country groups ('good', 'poor') than any of the alternative deep determinants offered in the literature.

## 4.2 Deconstructing Geography

In our treatment analysis we have emphasised that the causal effect we identify is for democratic regime change on capital flows using different samples, but that the striking patterns we reveal are mere correlations between countries with 'good'/'poor' geography and higher/lower inflows. In terms of policy advice, this correlation may suffice, but why do we see these patterns?

We know that geography can affect present-day economic prosperity in direct and indirect ways, the latter by establishing specific cultures or leading to specific historical experiences such as colonialism or slave exports. One of the most prominent examples of 'geography via history' is the

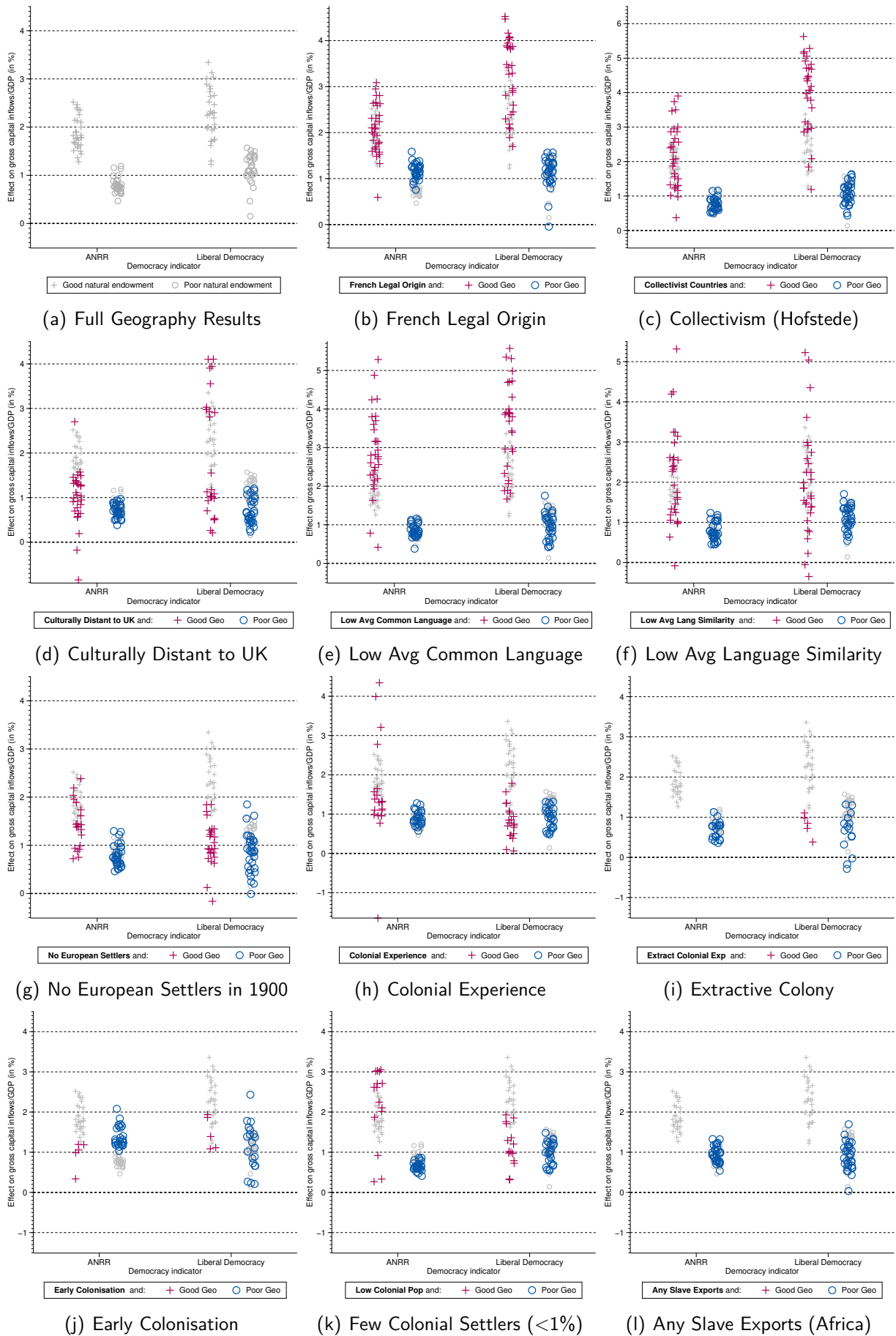
study by [Nunn and Puga \(2012\)](#), who demonstrate the negative direct effect of one geographic feature, ruggedness, and its positive indirect effect with historically fewer slave exports from more rugged African countries. On balance, they conclude, history (indirect effect) accounts for about two thirds and their specific geographic feature (direct effect) for only one third of the income differences across countries. We cannot make such a detailed quantitative assessment, but in the following we study the deep determinants for legal origin, culture and history to get closer to an answer to the question why these patterns persist.

We take the country-specific treatment estimates ( $ITET, \hat{\delta}_i$ ) underlying our main ATET results presented in Figure 3 (the specification including export/trade as additional control) and, using an M-estimator ([Rousseeuw and Leroy, 1987](#)), compute robust mean effects for *specific sub-samples of countries* within these geographical groups. These sub-samples always represent countries with ‘unfavourable’ deep determinants. Panel (a) of Figure 5 (this time in gray) is a reminder of our above results for total capital flows by definition of democracy (ANRR, liberal democracy) and geography (good, poor). In each of the other plots presented in the Figure, we superimpose (in colour) the robust mean effects of each specification (six proxies for geography times five different factor augmentations) for all countries *within these treated samples* which have the unfavourable deep determinant indicated. For instance, in panel (b) we compute the robust mean effects for all countries by geography which have French legal origin: of the 20 to 26 countries in the samples that make up the thirty ANRR ‘good’ geography estimates, 11 to 16 are for countries with French legal origin and we compute the robust mean estimates for the latter subgroups (in dark pink +s) within each specification. We proceed analogously for the ‘poor’ geography sample. The intuition for this exercise is that if French legal origin is a strong drag on economic prosperity (here: capital flows), even following democratic regime change, then we would expect that the effect for treated countries that have French legal origin would be *lower* in magnitude compared with the full sample case including countries with French as well as other, more favourable, legal origins. If results are qualitatively unchanged or better than in the full sample benchmark (in gray), then this is evidence against the hypothesis that the widely-suggested negative effect of French legal origin works through geography in the present context.

Since we are no longer bound by having sufficiently large country samples in the treatment and control groups for a ‘good’ and ‘poor’ deep determinant, we can expand our set of proxies. Focusing on culture, we now study (c) Hofstede’s index for individualism (from [Gorodnichenko and Roland, 2017](#)) and label countries with below median index value as ‘collectivist’; we use our previous proxies for distant culture based on (d) blood type distance from the UK, and our measures of (e) language similarly and (f) common language use; in (g) we study countries with no European settlers in 1900, although this could also proxy for ‘history’ rather than ‘culture’.

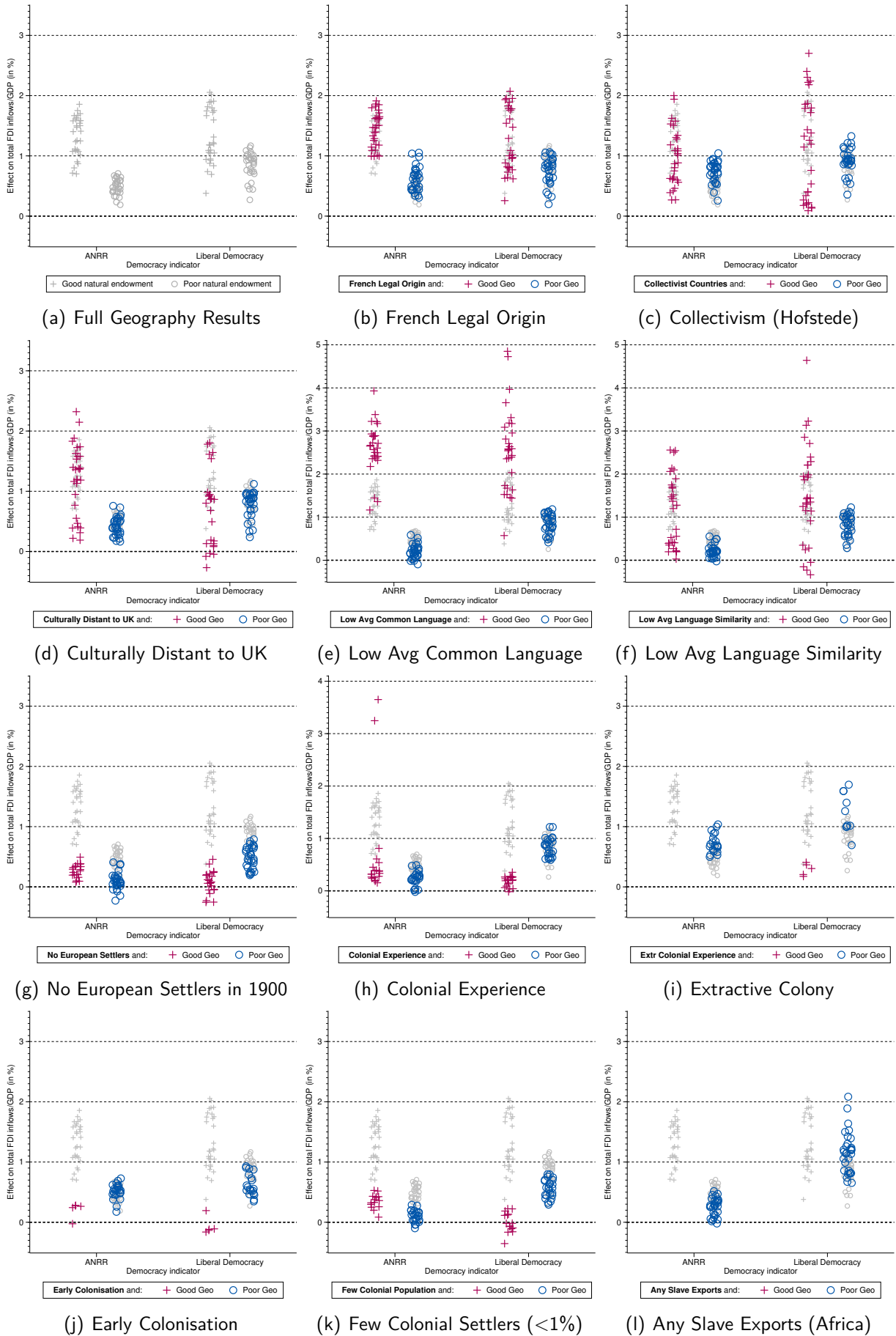
Panels (h) to (l) focus on history, namely colonial experience and (in the African context) experience of slave exports. In addition to the two proxies for colonial experience in (h) and (i) we focus on ‘early colonisation’ (before 1860) in (j), and countries where ‘foreigners’ from the

**Figure 5:** Alternative Deep Determinants 'within' Geography – gross capital inflows



**Notes:** The plots present robust ATET estimates for the causal effect of democracy on capital inflows by geography (in grey, as in Figure 3) and by alternative deep determinants (in colour) *within* these geographical groupings.

**Figure 6:** Alternative Deep Determinants ‘within’ Geography – FDI inflows



**Notes:** The plots present robust ATET estimates for the causal effect of democracy on FDI inflows by geography (in grey, as in Figure 3) and by alternative deep determinants (in colour) *within* these geographical groupings.

colonial power made up less than 1% of the population during the colonial period in (k) — both measures are constructed from data in [Ziltener and Künzler \(2013\)](#). Finally, we single out the countries in Africa with any slave exports (41 out of 52 countries in [Nunn, 2008](#)) in (l).

Outlier-robust means for these sub-samples are only presented if there are treatment estimates for at least four countries. Details of the number of treated countries in the reduced samples are provided in Appendix Table C-2. We present results for gross capital flows and FDI flows in Tables 5 and 6, respectively. Estimates for French Legal Origin in panel (b) are larger or (in the FDI case) qualitatively identical to the full sample results and hence a detrimental effect of geography via French legal origin is not confirmed in the data. Looking at proxies for culture, we clearly see some deviations in our subsample estimates from the full sample results, but in general terms we again see either higher treatment effects or broadly similar effects. The exception here is panel (f) in both figures: having no European settlers in 1900 can be seen to yield lower treatment effects, particularly in countries with good geography. This result is more pronounced in the FDI flows where treatment effects in good and poor geography countries are now on par with each other. All results for colonial experience and slave exports are much less widely available since few if any countries with these characteristics are located in ‘good’ geography regions. Results for various proxies of colonial experience in ‘good’ geography samples are thus based on few observations, but the robust means across many proxies show a very clear tendency for treatment effects to be substantially lower than for the full sample case. In some instances, treatment effects are on par with or, in case of the FDI analysis, even lower in ‘good’ compared with ‘poor’ geography countries. In the analysis of slave exports in panel (l) we only have estimates for countries in the ‘poor’ geography grouping — this is the category that across all other analysis of the current type shows relatively limited deviation from the full sample results, whereas those in ‘good’ geography samples showed at times substantial deviations. Be that as it may, and with a pinch of salt, we might nevertheless point out that the treatment effects of regime change are frequently somewhat higher in this subsample than in the full sample results.

Hence the narrative of a clear distinction in the economic consequences of democratic regime change along geographic lines we have developed in our earlier analysis disappears when we allow for geography to affect outcomes through colonial experience. Geography shapes the patterns of economic consequences, but colonial experience (history) may offer the cause — at least this conclusion is in line with what the data suggest, compared with alternative explanations related to legal origins, culture, or (with the aforementioned caveat in mind) experience of slave exports.

## 5 Concluding remarks

Capital inflows have more recently been viewed with great suspicion, emphasising the perils of excessive inflows in their detrimental effect on economic stability in recipient countries. But the existing literature is largely agreed that ‘normal’ levels of inflows, and in particular inflows of foreign

direct investment, fulfil an important role in furthering economic prosperity. Yet, why capital flows to some countries but not others has long puzzled economists, until improvements in the quality of institutions were motivated and empirically confirmed as an important factor. In this paper, we have connected this literature with the recent work on democracy and growth, asking whether the democratic dividend observed in the latter literature can be isolated in the patterns of capital inflows as well, one of a range of plausible transmission channels for improved economic prosperity following a shift from autocracy to democracy. Our point of departure from this combination of democracy and capital flows is that we argue for strong heterogeneity in the relationship *across countries*. Studying and identifying the underlying causes that shape this heterogeneity is important because policymakers and the populace alike may otherwise have unrealistic expectations of the economic effects of regime change.

We motivate and empirically demonstrate across a range of specifications and robustness checks that geography (proxied by measures of climate and disease environment) appears to capture the differential patterns across countries well, much better than alternative structural ('deep') determinants related to culture, history or legal origins. In countries with favourable geography (temperate climate, low disease environment) democratic regime change gives a substantial boost to total capital inflows as well as inflows of foreign direct investment, whereas when geographic endowment is comparatively worse (tropical climate, higher disease environment) these inflows are substantially lower. 'Good' geography countries are commonly susceptible to excessive capital inflows, but our analysis suggests that these detrimental effects of international capital flows are moderated when countries become democratic. In contrast, countries with comparatively worse geographic dispositions tend to miss out on this effect, and at times even see increased propensity for episodes of excessive inflows after regime change.

Why does democracy aid prosperity in some but not other countries? Our motivation suggests that geography represents a structural determinant of the magnitude and 'volatility' of capital inflows which dominates and hence partially eradicates the benefits of institutional change: *our definition of geography* (climate, disease environment) correlates with the concentration of the export basket, the complexity of production, goods trade costs, and the volatility of aggregate commodity terms of trade of a country — all these represent risk factors which make 'poor' geography countries disproportionately less attractive for foreign capital inflows. While democracy causes economic prosperity, including higher capital inflows, it does so unequally, and our research suggests that these patterns strongly correlate with geographic characteristics.

Our final consideration is for the causal link of unfavourable geography to (in relative terms) worse economic performance after democratic regime change (lower capital inflows, failure to reduce the propensity for excessive capital inflows). Is it 'nature' itself which leads to these outcomes (a direct effect), or does nature affect other deep determinants which in turn lead to the observed patterns (indirect effect)? Although we cannot apportion relative contributions, and are also not able to make any claims for causality, we provide additional analysis which studies

whether alternative unfavourable deep determinants can provide some additional insights related to this question. Adopting a host of unfavourable characteristics, we find that 'good' geography countries often fare even better in terms of the magnitude of effects for capital inflows (e.g. French legal origin, collectivism, other proxies for culture), but that experience of colonialism appears to make the previously substantial differences across geography disappear.

In summary: a significant step change in the quality of institutions unequivocally causes higher rates of capital inflows (total inflows, FDI inflows). However, these benefits are not equally spread across democratising countries. The patterns observed suggest that 'unfavourable' geography can substantially moderate this positive effect of democratic regime change. Yet while geography can provide patterns, a correlation, it appears from our analysis that countries' history, in particular whether they were subject to colonisation, can provide a better candidate for the causes of such unequal outcomes of democratic regime change.

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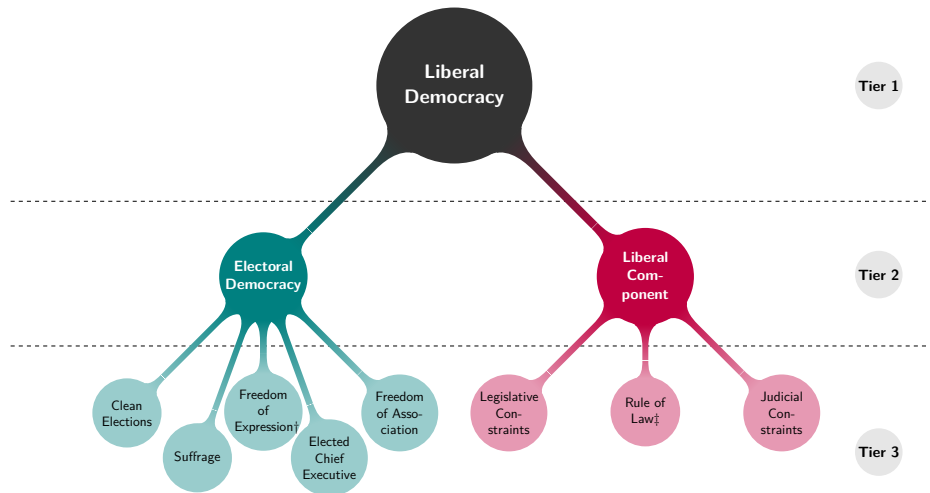
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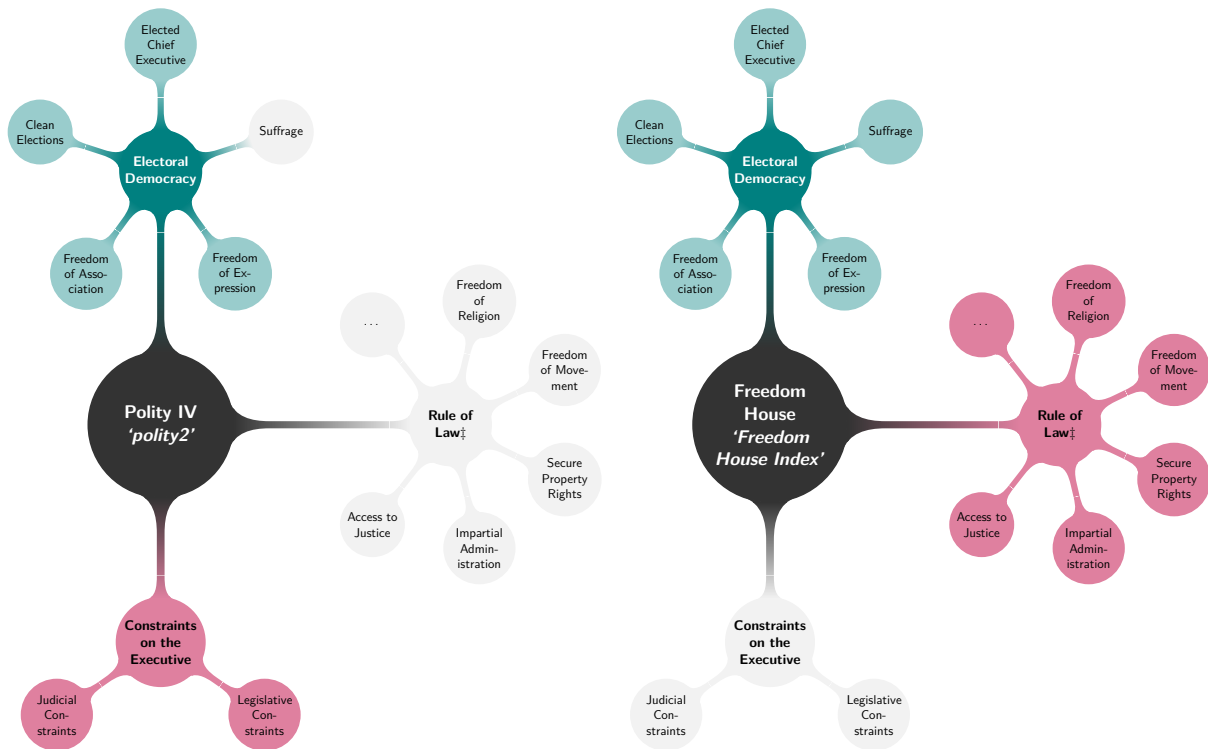
# **Online Appendix – Not Intended for Publication**

## **A Data, Sample Makeup and Descriptives**

**Figure A-1: Definitions of Democratic Institutions**



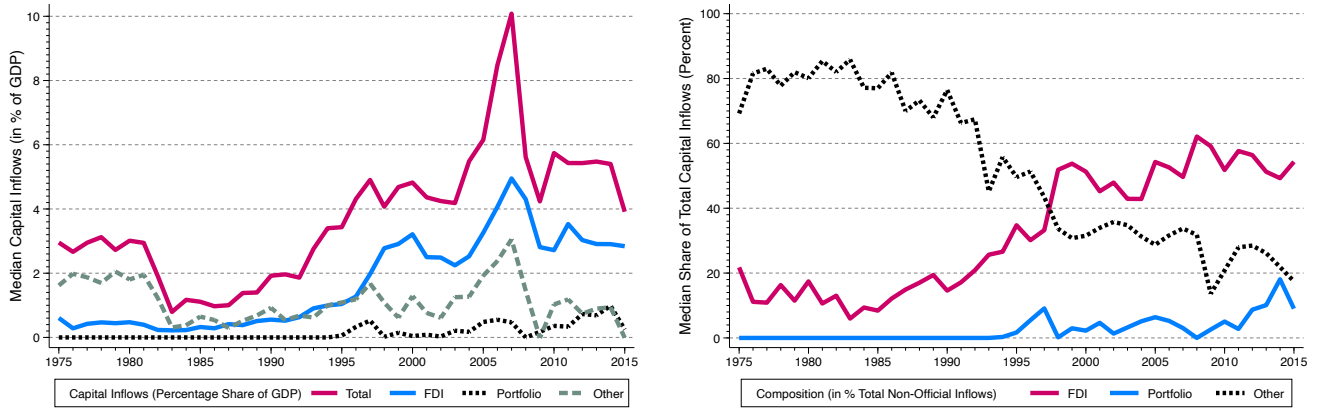
(a) V-Dem Definition of Liberal Democracy and its Components



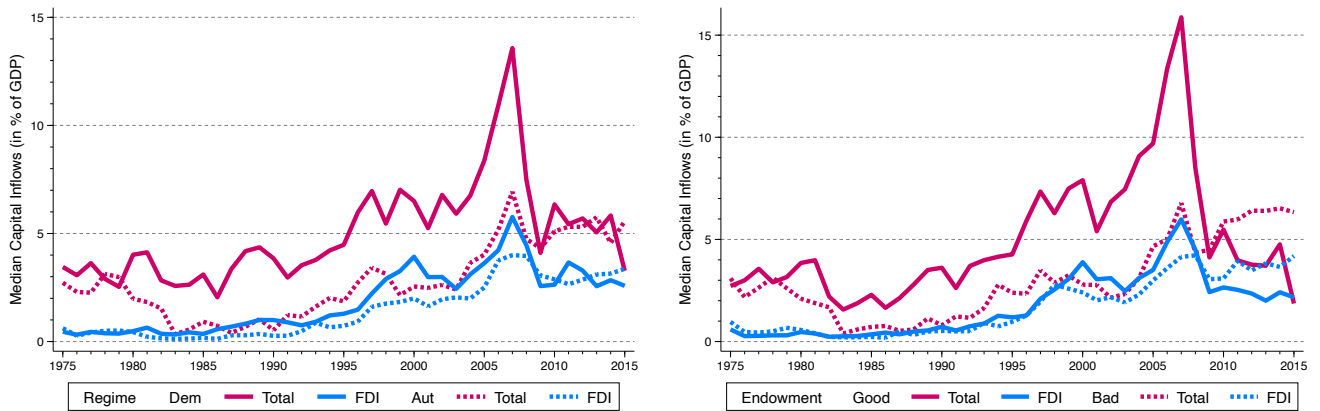
(b) The indices (Polity, FHI) combined in the ANRR definition of Democracy

**Notes:** The images present (a) the V-Dem conceptualisation of liberal democracy, and (b) an attempt at integrating the Polity IV ‘polity2’ and Freedom House FHI into the V-Dem framework. The lower panel provides greater distinction within the ‘Rule of Law’ set of institutions for reference. Institutions, concepts and practices shaded in light grey are not covered by the index in question. † This includes ‘alternative sources of information’. ‡ In its entirety this component covers ‘Individual Liberties and Equality before the Law.’

**Figure A-2: Composition and Evolution of Capital Inflows**



(a) Evolution (left) and Composition of Median Capital Inflows across all countries



(b) Evolution of Median Capital Inflows by Political Regime (left) and Geography

**Notes:** We present median capital flows (in percent of total inflows or in percent of GDP) for all countries in Panel (a) and by political regime and geography in Panel (b). Regime is defined by the V-Dem ERT variable (not countries experiencing democratic regime change), 'poor' geography by being located below the full sample average absolute latitude.

**Table A-1: Sample Makeup – Capital Flow Analysis**

|    | ISO | Country         | Start | End  | Obs | Miss | Income | CapFlow/GDP |       | ANRR Democracy Definition |       |              |    |      | Liberal Democracy Definition |    |        |       |              |    |      |             |    |    |
|----|-----|-----------------|-------|------|-----|------|--------|-------------|-------|---------------------------|-------|--------------|----|------|------------------------------|----|--------|-------|--------------|----|------|-------------|----|----|
|    |     |                 |       |      |     |      |        | Start       | End   | Always                    | Treat | AbsLat<br>Hi | Lo | Ctrl | AbsLat<br>Hi                 | Lo | Always | Treat | AbsLat<br>Hi | Lo | Ctrl | AbsLat<br>H | Lo |    |
| 1  | AGO | Angola          | 1985  | 2015 | 31  |      | LMC    | 3.7         | -1.5  |                           |       |              |    |      | 26                           |    | 26     |       |              |    |      | 31          |    | 31 |
| 2  | ALB | Albania         | 1981  | 2015 | 35  |      | UMC    | 0.0         | 10.5  |                           | 30    | 30           |    |      |                              |    |        | 35    |              | 35 |      |             |    |    |
| 3  | ARG | Argentina       | 1976  | 2015 | 40  |      | UMC    | -1.0        | 1.7   |                           | 35    | 35           |    |      |                              |    |        | 40    |              | 40 |      |             |    |    |
| 4  | ARM | Armenia         | 1993  | 2015 | 23  |      | UMC    | -2.1        | 9.4   |                           | 18    | 18           |    |      |                              |    |        |       |              |    |      | 23          |    | 23 |
| 5  | AUS | Australia       | 1970  | 2015 | 46  |      | HIC    | 2.8         | 5.9   | 41                        |       |              |    |      |                              |    |        | 46    |              |    |      |             |    |    |
| 6  | AUT | Austria         | 1970  | 2015 | 46  |      | HIC    | 3.5         | -2.8  | 41                        |       |              |    |      |                              |    |        | 46    |              |    |      |             |    |    |
| 7  | AZE | Azerbaijan      | 1995  | 2015 | 21  |      | UMC    | 12.9        | 8.5   |                           |       |              |    | 16   |                              | 16 |        |       |              |    |      | 21          |    | 21 |
| 8  | BDI | Burundi         | 1985  | 2015 | 31  |      | LIC    | 0.8         | 8.0   |                           | 26    |              | 26 |      |                              |    |        |       |              |    |      | 31          |    | 31 |
| 9  | BEL | Belgium         | 2002  | 2015 | 14  |      | HIC    | 19.3        | 0.7   | 9                         |       |              |    |      |                              |    |        | 14    |              |    |      |             |    |    |
| 10 | BEN | Benin           | 1974  | 2014 | 41  |      | LMC    | 2.4         | 5.3   |                           | 37    |              | 37 |      |                              |    |        |       | 41           |    | 41   |             |    |    |
| 11 | BFA | Burkina Faso    | 1974  | 2014 | 36  | 5    | LIC    | 1.6         | 18.6  |                           | 32    |              | 32 |      |                              |    |        | 36    |              | 36 |      |             |    |    |
| 12 | BGD | Bangladesh      | 1976  | 2015 | 40  |      | LMC    | 0.0         | 2.0   |                           | 35    | 35           |    |      |                              |    |        |       |              |    |      | 40          |    | 40 |
| 13 | BGR | Bulgaria        | 1981  | 2015 | 35  |      | UMC    | -1.2        | 4.9   |                           | 30    | 30           |    |      |                              |    |        | 35    |              | 35 |      |             |    |    |
| 14 | BIH | Bosnia & Herzeg | 1998  | 2015 | 18  |      | UMC    | 4.0         | 3.5   |                           |       |              |    | 13   |                              | 13 |        | 18    |              | 18 |      |             |    |    |
| 15 | BLR | Belarus         | 1993  | 2015 | 23  |      | UMC    | 1.2         | 4.7   |                           | 18    | 18           |    |      |                              |    |        | 23    |              | 23 |      |             |    |    |
| 16 | BOL | Bolivia         | 1976  | 2015 | 40  |      | LMC    | 2.1         | 6.1   |                           | 35    |              | 35 |      |                              |    |        | 40    |              | 40 |      |             |    |    |
| 17 | BRA | Brazil          | 1975  | 2015 | 41  |      | UMC    | 5.5         | 5.9   |                           | 36    |              | 36 |      |                              |    |        | 41    |              | 41 |      |             |    |    |
| 18 | BRB | Barbados        | 1970  | 2013 | 44  |      | HIC    | 15.2        | 10.6  | 41                        |       |              |    |      |                              |    |        | 44    |              |    |      |             |    |    |
| 19 | BWA | Botswana        | 2000  | 2015 | 16  |      | UMC    | 1.9         | 3.9   | 11                        |       |              |    |      |                              |    |        | 16    |              |    |      |             |    |    |
| 20 | CAF | Central Afr Rep | 1977  | 1994 | 18  |      | LIC    | 1.4         | -0.2  |                           | 18    |              | 18 |      |                              |    |        |       |              |    |      | 18          |    | 18 |
| 21 | CAN | Canada          | 1970  | 2015 | 46  |      | HIC    | 4.4         | 9.3   | 41                        |       |              |    |      |                              |    |        | 46    |              |    |      |             |    |    |
| 22 | CHE | Switzerland     | 1977  | 2015 | 39  |      | HIC    | -0.5        | 3.8   | 34                        |       |              |    |      |                              |    |        | 39    |              |    |      |             |    |    |
| 23 | CHL | Chile           | 1975  | 2015 | 41  |      | HIC    | -1.9        | 10.0  |                           | 36    | 36           |    |      |                              |    |        | 41    |              | 41 |      |             |    |    |
| 24 | CHN | China           | 1982  | 2015 | 34  |      | UMC    | 0.2         | 4.3   |                           |       |              |    | 29   |                              | 29 |        |       |              |    |      | 34          |    | 34 |
| 25 | CIV | Cote d'Ivoire   | 1975  | 2013 | 39  |      | LMC    | 4.7         | 3.7   |                           | 36    |              | 36 |      |                              |    |        |       |              |    |      | 39          |    | 39 |
| 26 | CMR | Cameroon        | 1977  | 2015 | 39  |      | LMC    | 1.7         | 1.3   |                           |       |              |    | 34   |                              | 34 |        |       |              |    |      | 39          |    | 39 |
| 27 | COG | Congo, Rep      | 1978  | 2007 | 30  |      | LMC    | 8.0         | 10.0  |                           | 30    |              | 30 |      |                              |    |        |       |              |    |      | 30          |    | 30 |
| 28 | COL | Colombia        | 1970  | 2015 | 46  |      | UMC    | 2.4         | 7.7   | 41                        |       |              |    |      |                              |    |        | 46    |              | 46 |      |             |    |    |
| 29 | COM | Comoros         | 1981  | 2012 | 25  | 7    | LMC    | 1.0         | 4.7   |                           | 23    |              | 23 |      |                              |    |        |       |              |    |      | 25          |    | 25 |
| 30 | CPV | Cabo Verde      | 1977  | 2015 | 39  |      | LMC    | 0.1         | 6.8   |                           | 34    |              | 34 |      |                              |    |        | 39    |              | 39 |      |             |    |    |
| 31 | CRI | Costa Rica      | 1977  | 2015 | 39  |      | UMC    | 9.6         | 11.2  | 34                        |       |              |    |      |                              |    |        | 39    |              |    |      |             |    |    |
| 32 | CYP | Cyprus          | 1976  | 2015 | 40  |      | HIC    | 4.4         | 27.8  | 35                        |       |              |    |      |                              |    |        | 40    |              |    |      |             |    |    |
| 33 | CZE | Czech Rep       | 1993  | 2015 | 23  |      | HIC    | 7.6         | 5.7   | 18                        |       |              |    |      |                              |    |        | 23    |              |    |      |             |    |    |
| 34 | DEU | Germany         | 1971  | 2015 | 45  |      | HIC    | 2.0         | 0.9   | 40                        |       |              |    |      |                              |    |        | 45    |              |    |      |             |    |    |
| 35 | DJI | Djibouti        | 1991  | 2015 | 25  |      | LMC    | 5.0         | 18.2  |                           | 20    |              | 20 |      |                              |    |        |       |              |    |      | 25          |    | 25 |
| 36 | DNK | Denmark         | 1975  | 2015 | 41  |      | HIC    | 1.7         | 0.7   | 36                        |       |              |    |      |                              |    |        | 41    |              |    |      |             |    |    |
| 37 | DOM | Dominican Rep   | 1977  | 2015 | 39  |      | UMC    | 1.6         | 6.0   |                           | 34    |              | 34 |      |                              |    |        |       |              |    |      | 39          |    | 39 |
| 38 | DZA | Algeria         | 1977  | 2015 | 26  | 13   | LMC    | 9.8         | 1.1   |                           |       |              |    | 21   |                              | 21 |        |       |              |    |      | 26          |    | 26 |
| 39 | ECU | Ecuador         | 1976  | 2015 | 40  |      | UMC    | 0.7         | 1.1   |                           | 35    |              | 35 |      |                              |    |        | 40    |              | 40 |      |             |    |    |
| 40 | EGY | Egypt           | 1977  | 2015 | 39  |      | LMC    | -4.4        | 1.7   |                           |       |              |    | 34   |                              | 34 |        |       |              |    |      | 39          |    | 39 |
| 41 | ESP | Spain           | 1975  | 2015 | 41  |      | HIC    | 2.7         | -1.4  |                           | 36    | 36           |    |      |                              |    |        | 41    |              | 41 |      |             |    |    |
| 42 | EST | Estonia         | 1993  | 2015 | 23  |      | HIC    | 14.6        | 7.1   | 18                        |       |              |    |      |                              |    |        | 23    |              |    |      |             |    |    |
| 43 | ETH | Ethiopia        | 1977  | 2012 | 36  |      | LIC    | 0.1         | 1.7   |                           | 34    |              | 34 |      |                              |    |        |       |              |    |      | 36          |    | 36 |
| 44 | FIN | Finland         | 1975  | 2015 | 41  |      | HIC    | 10.1        | -16.3 | 36                        |       |              |    |      |                              |    |        | 41    |              |    |      |             |    |    |
| 45 | FRA | France          | 1975  | 2015 | 41  |      | HIC    | 3.0         | 0.0   | 36                        |       |              |    |      |                              |    |        | 41    |              |    |      |             |    |    |
| 46 | GAB | Gabon           | 1978  | 2005 | 28  |      | UMC    | -2.6        | 1.0   |                           |       |              |    | 28   |                              | 28 |        |       |              |    |      | 28          |    | 28 |
| 47 | GBR | United Kingdom  | 1970  | 2015 | 46  |      | HIC    | 3.5         | -0.6  | 41                        |       |              |    |      |                              |    |        | 46    |              |    |      |             |    |    |
| 48 | GEO | Georgia         | 1997  | 2015 | 19  |      | UMC    | 7.3         | 11.5  | 14                        |       |              |    |      |                              |    |        |       | 19           | 19 |      |             |    |    |
| 49 | GHA | Ghana           | 1975  | 2015 | 41  |      | LMC    | 0.5         | 7.8   |                           | 36    |              | 36 |      |                              |    |        | 41    |              | 41 |      |             |    |    |
| 50 | GIN | Guinea          | 1986  | 2013 | 28  |      | LIC    | -0.3        | 18.6  |                           | 25    |              | 25 |      |                              |    |        |       |              |    |      | 28          |    | 28 |
| 51 | GMB | Gambia          | 1978  | 2012 | 30  | 5    | LIC    | 1.0         | -3.0  |                           | 28    |              | 28 |      |                              |    |        |       |              |    |      | 30          |    | 30 |
| 52 | GNB | Guinea-Bissau   | 1982  | 2013 | 29  | 3    | LIC    | 0.0         | 2.8   |                           | 26    |              | 26 |      |                              |    |        |       |              |    |      | 29          |    | 29 |
| 53 | GRC | Greece          | 1976  | 2015 | 39  | 1    | HIC    | 3.4         | -9.8  | 34                        |       |              |    |      |                              |    |        | 39    |              |    |      |             |    |    |
| 54 | GTM | Guatemala       | 1977  | 2015 | 39  |      | UMC    | 3.1         | 4.5   |                           | 34    |              | 34 |      |                              |    |        |       | 39           |    | 39   |             |    |    |
| 55 | HKG | Hong Kong       | 1998  | 2015 | 18  |      | HIC    | -85.6       | 64.6  | 13                        |       |              |    |      |                              |    |        |       |              |    |      | 18          |    | 18 |
| 56 | HND | Honduras        | 1974  | 2015 | 42  |      | LMC    | 4.6         | 7.9   |                           | 37    |              | 37 |      |                              |    |        |       |              |    |      | 42          |    | 42 |
| 57 | HRV | Croatia         | 1993  | 2015 | 23  |      | HIC    | 1.9         | 2.7   |                           | 18    | 18           |    |      |                              |    |        | 23    |              | 23 |      |             |    |    |
| 58 | HTI | Haiti           | 1971  | 2015 | 45  |      | LIC    | 0.9         | 2.2   |                           | 40    |              | 40 |      |                              |    |        |       |              |    |      | 45          |    | 45 |
| 59 | HUN | Hungary         | 1982  | 2015 | 34  |      | HIC    | 0.0         | -2.6  |                           | 29    | 29           |    |      |                              |    |        | 34    |              | 34 |      |             |    |    |
| 60 | IDN | Indonesia       | 1981  | 2015 | 35  |      | UMC    | 0.2         | 5.1   |                           | 30    |              | 30 |      |                              |    |        | 35    |              | 35 |      |             |    |    |
| 61 | IND | India           | 1975  | 2015 | 41  |      | LMC    | 0.0         | 6.9   | 36                        |       |              |    |      |                              |    |        |       | 41           | 41 |      |             |    |    |
| 62 | IRL | Ireland         | 1974  | 2015 | 42  |      | HIC    | 13.7        | 76.4  | 37                        |       |              |    |      |                              |    |        | 42    |              |    |      |             |    |    |
| 63 | IRN | Iran            | 1981  | 2000 | 20  |      | UMC    | 0.2         | -0.2  |                           |       |              |    | 20   |                              | 20 |        |       |              |    |      | 20          |    | 20 |
| 64 | ISL | Iceland         | 1976  | 2015 | 40  |      | HIC    | 1.3         | -9.0  | 35                        |       |              |    |      |                              |    |        | 40    |              |    |      |             |    |    |
| 65 | ISR | Israel          | 1970  | 2015 | 46  |      | HIC    | 5.9         | 3.1   | 41                        |       |              |    |      |                              |    |        | 46    |              |    |      |             |    |    |
| 66 | ITA | Italy           | 1970  | 2015 | 46  |      | HIC    | 3.8         | 0.4   | 41                        |       |              |    |      |                              |    |        | 46    |              |    |      |             |    |    |
| 67 | JAM | Jamaica         | 1976  | 2015 | 40  |      | UMC    | -1.4        | 7.2   | 35                        |       |              |    |      |                              |    |        |       | 40           | 40 |      |             |    |    |
| 68 | JOR | Jordan          | 1972  | 2015 | 44  |      | UMC    | 0.0         | 9.8   |                           |       |              |    | 39   |                              | 39 |        |       |              |    |      | 44          |    | 44 |
| 69 | JPN | Japan           | 1977  | 2015 | 39  |      | HIC    | 0.0         | 0.5   | 34                        |       |              |    |      |                              |    |        | 39    |              |    |      |             |    |    |
| 70 | KAZ | Kazakhstan      | 1995  | 2015 | 21  |      | UMC    | 7.3         | 7.7   |                           |       |              |    | 16   |                              | 16 |        |       |              |    |      | 21          |    | 21 |

**Table A-1: Sample Makeup – Capital Flow Analysis (continued)**

|     | ISO | Country         | Start | End  | Obs | Miss | Income | CapFlow/GDP |        | ANRR Democracy Definition |       |                 |      |                 | Liberal Democracy Definition |       |                 |      |                |    |
|-----|-----|-----------------|-------|------|-----|------|--------|-------------|--------|---------------------------|-------|-----------------|------|-----------------|------------------------------|-------|-----------------|------|----------------|----|
|     |     |                 |       |      |     |      |        | Start       | End    | Always                    | Treat | AbsLat<br>Hi Lo | Ctrl | AbsLat<br>Hi Lo | Always                       | Treat | AbsLat<br>Hi Lo | Ctrl | AbsLat<br>H Lo |    |
| 71  | KEN | Kenya           | 1975  | 2014 | 40  |      | LMC    | 2.4         | 6.9    |                           | 36    | 36              |      |                 |                              |       |                 |      | 40             | 40 |
| 72  | KGZ | Kyrgyz Republic | 1993  | 2015 | 23  |      | LMC    | 1.8         | 8.5    |                           | 18    | 18              |      |                 |                              |       |                 |      | 23             | 23 |
| 73  | KHM | Cambodia        | 1992  | 2014 | 23  |      | LMC    | 1.6         | 18.5   |                           | 19    | 19              |      |                 |                              |       |                 |      | 23             | 23 |
| 74  | KOR | South Korea     | 1976  | 2015 | 40  |      | HIC    | 5.2         | -0.2   |                           | 35    | 35              |      |                 |                              | 40    | 40              |      |                |    |
| 75  | KWT | Kuwait          | 1975  | 2015 | 41  |      | HIC    | -0.8        | 2.2    |                           |       |                 | 36   | 36              |                              |       |                 |      | 41             | 41 |
| 76  | LAO | Lao             | 1984  | 2015 | 32  |      | LMC    | 0.4         | 9.3    |                           |       |                 |      |                 | 27                           |       |                 |      | 32             | 32 |
| 77  | LBN | Lebanon         | 2002  | 2015 | 14  |      | UMC    | 10.5        | 15.5   |                           | 9     | 9               |      |                 |                              |       |                 |      | 14             | 14 |
| 78  | LBR | Liberia         | 1979  | 2015 | 21  | 16   | LIC    | 0.1         | 36.5   |                           | 16    | 16              |      |                 |                              | 21    | 21              |      |                |    |
| 79  | LBY | Libya           | 1977  | 2013 | 37  |      | UMC    | -0.1        | 1.1    |                           |       |                 | 34   | 34              |                              |       |                 |      | 37             | 37 |
| 80  | LKA | Sri Lanka       | 1975  | 2015 | 41  |      | LMC    | -0.8        | 6.1    |                           | 36    |                 |      |                 |                              | 41    | 41              |      |                |    |
| 81  | LSO | Lesotho         | 2000  | 2015 | 16  |      | LMC    | 5.3         | 5.0    |                           | 11    |                 |      |                 |                              | 16    | 16              |      |                |    |
| 82  | LTU | Lithuania       | 1993  | 2015 | 23  |      | HIC    | -0.7        | -0.4   |                           | 18    |                 |      |                 | 23                           |       |                 |      |                |    |
| 83  | LUX | Luxembourg      | 2002  | 2015 | 14  |      | HIC    | 701.0       | 1327.6 |                           | 9     |                 |      |                 | 14                           |       |                 |      |                |    |
| 84  | LVA | Latvia          | 1992  | 2015 | 24  |      | HIC    | 4.8         | 10.1   |                           | 19    | 19              |      |                 | 24                           |       |                 |      |                |    |
| 85  | MAR | Morocco         | 1975  | 2015 | 41  |      | LMC    | 0.8         | 7.2    |                           |       |                 | 36   | 36              |                              |       |                 |      | 41             | 41 |
| 86  | MDA | Moldova         | 1994  | 2015 | 22  |      | LMC    | 8.4         | 7.2    |                           | 17    |                 |      |                 |                              | 22    | 22              |      |                |    |
| 87  | MDG | Madagascar      | 1974  | 2013 | 40  |      | LIC    | 0.3         | 8.1    |                           |       | 37              |      |                 |                              |       |                 |      | 40             | 40 |
| 88  | MEX | Mexico          | 1979  | 2015 | 37  |      | UMC    | 4.2         | 7.7    |                           | 32    | 32              |      |                 |                              | 37    | 37              |      |                |    |
| 89  | MLI | Mali            | 1975  | 2014 | 40  |      | LIC    | 0.0         | 3.3    |                           | 36    | 36              |      |                 |                              | 40    | 40              |      |                |    |
| 90  | MLT | Malta           | 1971  | 2015 | 45  |      | HIC    | 6.7         | 42.9   |                           | 40    |                 |      |                 | 45                           |       |                 |      |                |    |
| 91  | MMR | Myanmar         | 1998  | 2015 | 18  |      | LMC    | 4.2         | 3.2    |                           |       |                 | 13   | 13              |                              |       |                 |      | 18             | 18 |
| 92  | MNG | Mongolia        | 1981  | 2015 | 35  |      | LMC    | 0.0         | 29.9   |                           | 30    | 30              |      |                 |                              | 35    | 35              |      |                |    |
| 93  | MOZ | Mozambique      | 1981  | 2015 | 35  |      | LIC    | 0.0         | 40.2   |                           | 30    | 30              |      |                 |                              |       |                 |      | 35             | 35 |
| 94  | MRT | Mauritania      | 1981  | 2015 | 22  | 13   | LMC    | 8.4         | 17.4   |                           |       |                 | 18   | 18              |                              |       |                 |      | 22             | 22 |
| 95  | MUS | Mauritius       | 1976  | 2015 | 40  |      | HIC    | -2.8        | -7.2   |                           | 35    |                 |      |                 | 40                           |       |                 |      |                |    |
| 96  | MWI | Malawi          | 1977  | 2015 | 39  |      | LIC    | 2.5         | 9.3    |                           | 34    | 34              |      |                 |                              | 39    | 39              |      |                |    |
| 97  | MYS | Malaysia        | 1974  | 2015 | 42  |      | UMC    | 5.5         | 6.7    |                           |       |                 | 37   | 37              |                              |       |                 |      | 42             | 42 |
| 98  | NAM | Namibia         | 2000  | 2015 | 16  |      | UMC    | 7.8         | 10.5   |                           | 11    |                 |      |                 | 16                           |       |                 |      |                |    |
| 99  | NER | Niger           | 1974  | 2013 | 40  |      | LIC    | 2.1         | 18.0   |                           |       | 37              | 37   |                 |                              | 40    | 40              |      |                |    |
| 100 | NGA | Nigeria         | 1977  | 2015 | 38  | 1    | LMC    | 1.0         | 4.0    |                           | 33    | 33              |      |                 |                              | 38    | 38              |      |                |    |
| 101 | NIC | Nicaragua       | 1977  | 2015 | 39  |      | LMC    | 8.7         | 12.3   |                           | 34    | 34              |      |                 |                              | 39    | 39              |      |                |    |
| 102 | NLD | Netherlands     | 1970  | 2015 | 46  |      | HIC    | 9.4         | -18.1  |                           | 41    |                 |      |                 | 46                           |       |                 |      |                |    |
| 103 | NOR | Norway          | 1975  | 2015 | 41  |      | HIC    | 8.1         | 10.0   |                           | 36    |                 |      |                 | 41                           |       |                 |      |                |    |
| 104 | NPL | Nepal           | 1981  | 2015 | 35  |      | LMC    | 0.3         | 2.1    |                           | 30    | 30              |      |                 |                              | 35    | 35              |      |                |    |
| 105 | NZL | New Zealand     | 1972  | 2015 | 44  |      | HIC    | 2.3         | 3.1    |                           | 39    |                 |      |                 | 44                           |       |                 |      |                |    |
| 106 | OMN | Oman            | 1974  | 2015 | 42  |      | HIC    | -5.0        | 3.5    |                           |       |                 | 37   | 37              |                              |       |                 |      | 42             | 42 |
| 107 | PAK | Pakistan        | 1976  | 2015 | 40  |      | LMC    | 1.5         | 0.8    |                           | 35    | 35              |      |                 |                              |       |                 |      | 40             | 40 |
| 108 | PAN | Panama          | 1977  | 2015 | 39  |      | HIC    | 120.0       | 18.0   |                           | 34    | 34              |      |                 |                              | 39    | 39              |      |                |    |
| 109 | PER | Peru            | 1977  | 2015 | 39  |      | UMC    | 0.7         | 8.6    |                           | 34    | 34              |      |                 |                              | 39    | 39              |      |                |    |
| 110 | PHL | Philippines     | 1977  | 2015 | 39  |      | LMC    | 4.3         | 2.7    |                           | 34    | 34              |      |                 |                              | 39    | 39              |      |                |    |
| 111 | POL | Poland          | 1976  | 2015 | 40  |      | HIC    | 8.1         | 3.2    |                           | 35    | 35              |      |                 |                              | 40    | 40              |      |                |    |
| 112 | PRT | Portugal        | 1975  | 2015 | 41  |      | HIC    | -0.2        | -5.9   |                           | 36    | 36              |      |                 |                              | 41    | 41              |      |                |    |
| 113 | PRY | Paraguay        | 1975  | 2015 | 41  |      | UMC    | 5.1         | 2.9    |                           | 36    | 36              |      |                 |                              | 41    | 41              |      |                |    |
| 114 | RUS | Russian Fed     | 1994  | 2015 | 22  |      | UMC    | 0.6         | 2.5    |                           | 17    | 17              |      |                 |                              |       |                 |      | 22             | 22 |
| 115 | RWA | Rwanda          | 1976  | 2015 | 40  |      | LIC    | -0.2        | 3.8    |                           |       |                 | 35   | 35              |                              |       |                 |      | 40             | 40 |
| 116 | SAU | Saudi Arabia    | 1971  | 2015 | 45  |      | HIC    | -1.0        | 1.6    |                           |       |                 | 40   | 40              |                              |       |                 |      | 45             | 45 |
| 117 | SDN | Sudan           | 1977  | 2015 | 39  |      | LIC    | 1.1         | 3.3    |                           | 34    | 34              |      |                 |                              |       |                 |      | 39             | 39 |
| 118 | SEN | Senegal         | 1974  | 2014 | 41  |      | LMC    | 3.4         | 7.9    |                           | 37    | 37              |      |                 |                              | 41    | 41              |      |                |    |
| 119 | SGP | Singapore       | 1972  | 2015 | 44  |      | HIC    | 19.1        | 38.2   |                           | 39    |                 |      |                 |                              |       |                 |      | 44             | 44 |
| 120 | SLE | Sierra Leone    | 1977  | 2014 | 35  | 3    | LIC    | 1.9         | 26.6   |                           | 31    | 31              |      |                 |                              |       |                 |      | 35             | 35 |
| 121 | SLV | El Salvador     | 1976  | 2015 | 40  |      | LMC    | 3.6         | 5.4    |                           | 35    | 35              |      |                 |                              | 40    | 40              |      |                |    |
| 122 | STP | Sao Tome & Pr   | 1974  | 2015 | 33  | 9    | LMC    | 4.1         | 4.9    |                           | 27    | 27              |      |                 |                              | 33    | 33              |      |                |    |
| 123 | SVK | Slovak Republic | 1993  | 2015 | 23  |      | HIC    | 6.9         | 7.7    |                           | 18    |                 |      |                 | 23                           |       |                 |      |                |    |
| 124 | SVN | Slovenia        | 1993  | 2015 | 23  |      | HIC    | 0.9         | 2.6    |                           | 18    |                 |      |                 | 23                           |       |                 |      |                |    |
| 125 | SWE | Sweden          | 1970  | 2015 | 46  |      | HIC    | 1.5         | -6.0   |                           | 41    |                 |      |                 | 46                           |       |                 |      |                |    |
| 126 | SWZ | Eswatini        | 2000  | 2015 | 16  |      | LMC    | 7.4         | 0.0    |                           |       |                 | 11   | 11              |                              |       |                 |      | 16             | 16 |
| 127 | SYC | Seychelles      | 1976  | 2015 | 40  |      | HIC    | 17.2        | 28.9   |                           |       |                 | 35   | 35              |                              |       |                 |      | 40             | 40 |
| 128 | SYR | Syria           | 1977  | 2010 | 34  |      | LIC    | 2.0         | 3.8    |                           |       |                 | 34   | 34              |                              |       |                 |      | 34             | 34 |
| 129 | TCD | Chad            | 1981  | 1994 | 14  |      | LIC    | -0.2        | -2.2   |                           |       |                 | 14   | 14              |                              |       |                 |      | 14             | 14 |
| 130 | TGO | Togo            | 1974  | 2015 | 42  |      | LIC    | -10.6       | 6.9    |                           |       |                 | 37   | 37              |                              |       |                 |      | 42             | 42 |
| 131 | THA | Thailand        | 1975  | 2015 | 41  |      | UMC    | 3.1         | 3.9    |                           | 36    | 36              |      |                 |                              | 41    | 41              |      |                |    |
| 132 | TJK | Tajikistan      | 2002  | 2015 | 14  |      | LIC    | 8.7         | 4.3    |                           |       |                 | 9    | 9               |                              |       |                 |      | 14             | 14 |
| 133 | TTO | Trinidad & Tob  | 1975  | 2011 | 37  |      | HIC    | 3.5         | -13.8  |                           | 36    |                 |      |                 | 37                           |       |                 |      |                |    |
| 134 | TUN | Tunisia         | 1976  | 2015 | 40  |      | LMC    | 8.3         | 5.2    |                           |       |                 | 35   | 35              |                              | 40    | 40              |      |                |    |
| 135 | TUR | Turkey          | 1974  | 2015 | 42  |      | UMC    | 0.7         | 8.4    |                           | 37    | 37              |      |                 |                              | 42    | 42              |      |                |    |
| 136 | TZA | Tanzania        | 1976  | 2015 | 40  |      | LMC    | -0.4        | 6.6    |                           |       |                 | 35   | 35              |                              |       |                 |      | 40             | 40 |
| 137 | UGA | Uganda          | 1980  | 2015 | 36  |      | LIC    | -0.1        | 5.5    |                           | 31    | 31              |      |                 |                              |       |                 |      | 36             | 36 |
| 138 | UKR | Ukraine         | 1994  | 2015 | 22  |      | LMC    | 6.8         | 4.4    |                           | 17    |                 |      |                 |                              | 22    | 22              |      |                |    |
| 139 | URY | Uruguay         | 1978  | 2015 | 38  |      | HIC    | 1.8         | 10.6   |                           | 33    | 33              |      |                 |                              | 38    | 38              |      |                |    |
| 140 | USA | United States   | 1970  | 2015 | 46  |      | HIC    | -0.3        | 5.0    |                           | 41    |                 |      |                 | 46                           |       |                 |      |                |    |

**Table A-1: Sample Makeup – Capital Flow Analysis (continued)**

|     | ISO | Country      | Start | End  | Obs | Miss | Income | CapFlow/GDP |      | ANRR Democracy Definition |       |        |    |      |        | Liberal Democracy Definition |        |       |        |    |      |        |    |
|-----|-----|--------------|-------|------|-----|------|--------|-------------|------|---------------------------|-------|--------|----|------|--------|------------------------------|--------|-------|--------|----|------|--------|----|
|     |     |              |       |      |     |      |        | Start       | End  | Always                    | Treat | AbsLat |    | Ctrl | AbsLat |                              | Always | Treat | AbsLat |    | Ctrl | AbsLat |    |
|     |     |              |       |      |     |      |        |             |      |                           |       | Hi     | Lo |      | Hi     | Lo                           |        |       | Hi     | Lo |      | H      | Lo |
| 141 | VEN | Venezuela    | 1970  | 2015 | 45  | 1    | UMC    | 0.2         | 1.7  |                           | 40    | 40     |    |      |        |                              | 45     | 45    |        |    |      |        |    |
| 142 | VNM | Vietnam      | 1996  | 2015 | 20  |      | LMC    | 11.9        | 6.9  |                           |       |        | 15 | 15   |        |                              |        |       | 20     | 20 |      |        |    |
| 143 | YEM | Yemen        | 1990  | 2015 | 26  |      | LIC    | -0.5        | 0.7  |                           |       |        | 21 | 21   |        |                              |        |       | 26     | 26 |      |        |    |
| 144 | ZAF | South Africa | 1998  | 2015 | 18  |      | UMC    | 8.5         | 1.0  | 13                        |       |        |    |      |        |                              |        |       |        |    |      |        |    |
| 145 | ZMB | Zambia       | 1978  | 2015 | 33  | 5    | LMC    | 5.8         | 10.0 |                           | 28    | 28     |    |      |        |                              | 33     | 33    |        |    |      |        |    |
| 146 | ZWE | Zimbabwe     | 1981  | 2015 | 21  | 14   | LMC    | 1.7         | 8.5  |                           | 16    | 16     |    |      |        |                              |        |       | 21     | 21 |      |        |    |

Notes: We present the sample makeup for the capital flow analysis (1975-2015). Income indicates the World Bank Income Level category (Low - LIC, Lower Middle - LMC, Upper Middle - UMC, and High - HIC). We report the gross capital inflow over GDP for the first and last year of the country series, in percent. The remaining columns indicate treated and controls samples (total number of observations, respectively) for two democracy definitions: that by ANRR and the V-Dem Liberal Democracy (sample mean cutoff). 'Always' refers to countries that were democracies throughout the sample period, 'treat' to the treated sample, where absolute latitude ('Abslat') 'Hi' and 'Lo' provide the split for one of the many deep determinants we apply in our analysis. 'Ctrl' is the control sample, again split into 'Hi' and 'Lo' absolute latitude. Absolute latitude is one of the six geography proxies (plus four more in robustness checks) we adopt in the paper, in addition to proxies for culture, history and legal origin.

**Table A-2: Sample Makeup – Excessive Capital Flow Analysis**

|    | ISO | Country             | Start | End  | Obs | Miss | Income | Bonanzas | Surges | ANRR Democracy Definition |       |              |    |            |              | Liberal Democracy Definition |       |              |    |            |              |
|----|-----|---------------------|-------|------|-----|------|--------|----------|--------|---------------------------|-------|--------------|----|------------|--------------|------------------------------|-------|--------------|----|------------|--------------|
|    |     |                     |       |      |     |      |        |          |        | Always                    | Treat | AbsLat<br>Hi | Lo | Ctrl<br>Hi | AbsLat<br>Lo | Always                       | Treat | AbsLat<br>Hi | Lo | Ctrl<br>Hi | AbsLat<br>Lo |
| 1  | AGO | Angola              | 1996  | 2015 | 20  |      | LMC    | 1        | 10     |                           |       |              |    | 15         | 15           |                              |       |              |    | 20         | 20           |
| 2  | ALB | Albania             | 1994  | 2015 | 22  |      | UMC    | 0        | 10     |                           | 17    | 17           |    |            |              |                              | 22    | 22           |    |            |              |
| 3  | ARM | Armenia             | 1994  | 2015 | 22  |      | UMC    | 2        | 7      |                           | 17    | 17           |    |            |              |                              |       |              | 22 | 22         |              |
| 4  | AUS | Australia           | 1990  | 2015 | 26  |      | HIC    | 1        | 10     | 21                        |       |              |    |            |              | 26                           |       |              |    |            |              |
| 5  | AZE | Azerbaijan          | 1996  | 2015 | 20  |      | UMC    | 1        | 7      |                           |       |              |    | 15         | 15           |                              |       |              |    | 20         | 20           |
| 6  | BDI | Burundi             | 1986  | 2015 | 30  |      | LIC    | 0        | 3      |                           | 25    |              | 25 |            |              |                              |       |              | 30 |            | 30           |
| 7  | BGD | Bangladesh          | 1987  | 2015 | 29  |      | LMC    | 1        | 0      |                           | 24    | 24           |    |            |              |                              |       |              | 29 | 29         |              |
| 8  | BGR | Bulgaria            | 1992  | 2015 | 24  |      | UMC    | 1        | 11     | 19                        |       |              |    |            |              | 24                           |       |              |    |            |              |
| 9  | BHR | Bahrain             | 1981  | 2014 | 34  |      | HIC    | 0        | 11     |                           |       |              |    | 30         | 30           |                              |       |              |    | 34         | 34           |
| 10 | BHS | Bahamas             | 1980  | 2015 | 36  |      | HIC    | 0        | 12     | 31                        |       |              |    |            |              |                              |       |              | 36 | 36         |              |
| 11 | BLR | Belarus             | 1995  | 2015 | 21  |      | UMC    | 2        | 5      |                           |       |              |    | 16         | 16           |                              |       |              |    | 21         | 21           |
| 12 | BLZ | Belize              | 1985  | 2015 | 31  |      | UMC    | 0        | 9      | 26                        |       |              |    |            |              |                              |       |              | 31 |            | 31           |
| 13 | BOL | Bolivia             | 1980  | 2015 | 36  |      | LMC    | 1        | 8      |                           | 31    |              | 31 |            |              |                              | 36    |              | 36 |            |              |
| 14 | BRA | Brazil              | 1981  | 2015 | 35  |      | UMC    | 3        | 6      |                           | 30    |              | 30 |            |              |                              | 35    |              | 35 |            |              |
| 15 | BRB | Barbados            | 1980  | 2013 | 34  |      | HIC    | 1        | 10     | 31                        |       |              |    |            |              | 34                           |       |              |    |            |              |
| 16 | BWA | Botswana            | 1980  | 2015 | 36  |      | UMC    | 0        | 9      | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 17 | CAF | Central African Rep | 1981  | 1994 | 14  |      | LIC    | 0        | 0      |                           | 14    |              | 14 |            |              |                              |       |              | 14 |            | 14           |
| 18 | CAN | Canada              | 1980  | 2015 | 36  |      | HIC    | 2        | 13     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 19 | CHE | Switzerland         | 1981  | 2015 | 35  |      | HIC    | 2        | 12     | 30                        |       |              |    |            |              | 35                           |       |              |    |            |              |
| 20 | CHL | Chile               | 1980  | 2015 | 36  |      | HIC    | 2        | 11     |                           | 31    | 31           |    |            |              |                              | 36    | 36           |    |            |              |
| 21 | CHN | China               | 1987  | 2015 | 29  |      | UMC    | 0        | 5      |                           |       |              |    | 24         | 24           |                              |       |              |    | 29         | 29           |
| 22 | CMR | Cameroon            | 1980  | 2015 | 36  |      | LMC    | 1        | 2      |                           |       |              |    | 31         |              | 31                           |       |              |    | 36         |              |
| 23 | COG | Congo, Rep          | 1986  | 2007 | 20  | 2    | LMC    | 1        | 6      |                           | 20    |              | 20 |            |              |                              |       |              | 20 |            | 20           |
| 24 | COL | Colombia            | 1980  | 2015 | 36  |      | UMC    | 2        | 8      | 31                        |       |              |    |            |              |                              | 36    |              | 36 |            |              |
| 25 | CPV | Cabo Verde          | 1984  | 2015 | 32  |      | LMC    | 1        | 12     |                           | 27    |              | 27 |            |              |                              | 32    |              | 32 |            |              |
| 26 | CRI | Costa Rica          | 1980  | 2015 | 36  |      | UMC    | 2        | 10     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 27 | CYP | Cyprus              | 1980  | 2015 | 36  |      | HIC    | 1        | 12     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 28 | CZE | Czech Republic      | 1994  | 2015 | 22  |      | HIC    | 2        | 7      | 17                        |       |              |    |            |              | 22                           |       |              |    |            |              |
| 29 | DEU | Germany             | 1980  | 2015 | 36  |      | HIC    | 1        | 12     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 30 | DJI | Djibouti            | 2001  | 2015 | 15  |      | LMC    | 1        | 8      |                           | 10    |              | 10 |            |              |                              |       |              | 15 |            | 15           |
| 31 | DMA | Dominica            | 1981  | 2013 | 33  |      | UMC    | 0        | 12     | 30                        |       |              |    |            |              |                              |       |              | 33 |            | 33           |
| 32 | DNK | Denmark             | 1980  | 2015 | 36  |      | HIC    | 2        | 13     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 33 | DOM | Dominican Rep       | 1980  | 2015 | 36  |      | UMC    | 1        | 3      | 31                        |       |              |    |            |              |                              |       |              | 36 |            | 36           |
| 34 | DZA | Algeria             | 1980  | 2015 | 22  | 14   | LMC    | 0        | 0      |                           |       |              |    | 17         | 17           |                              |       |              | 22 | 22         |              |
| 35 | ECU | Ecuador             | 1980  | 2015 | 36  |      | UMC    | 1        | 6      | 31                        |       |              |    |            |              |                              | 36    |              | 36 |            |              |
| 36 | EGY | Egypt               | 1980  | 2015 | 36  |      | LMC    | 2        | 4      |                           |       |              |    | 31         | 31           |                              |       |              | 36 | 36         |              |
| 37 | ESP | Spain               | 1980  | 2015 | 36  |      | HIC    | 1        | 13     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 38 | EST | Estonia             | 1996  | 2015 | 20  |      | HIC    | 2        | 7      | 15                        |       |              |    |            |              | 20                           |       |              |    |            |              |
| 39 | ETH | Ethiopia            | 1982  | 2012 | 31  |      | LIC    | 2        | 0      |                           | 29    |              | 29 |            |              |                              |       |              | 31 |            | 31           |
| 40 | FIN | Finland             | 1980  | 2015 | 36  |      | HIC    | 2        | 13     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 41 | FJI | Fiji                | 1980  | 2015 | 36  |      | UMC    | 0        | 12     |                           | 31    |              | 31 |            |              |                              | 36    |              | 36 |            |              |
| 42 | FRA | France              | 1980  | 2015 | 36  |      | HIC    | 0        | 13     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 43 | GAB | Gabon               | 1980  | 2005 | 26  |      | UMC    | 2        | 6      |                           |       |              |    | 26         |              | 26                           |       |              |    | 26         | 26           |
| 44 | GBR | UK                  | 1980  | 2015 | 36  |      | HIC    | 3        | 13     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 45 | GEO | Georgia             | 1998  | 2015 | 18  |      | UMC    | 1        | 6      | 13                        |       |              |    |            |              |                              | 18    | 18           |    |            |              |
| 46 | GHA | Ghana               | 1980  | 2015 | 36  |      | LMC    | 1        | 8      |                           | 31    |              | 31 |            |              |                              | 36    |              | 36 |            |              |
| 47 | GMB | Gambia              | 1980  | 2012 | 27  | 6    | LIC    | 2        | 5      |                           | 25    |              | 25 |            |              |                              |       |              | 27 |            | 27           |
| 48 | GRC | Greece              | 1980  | 2015 | 34  | 2    | HIC    | 2        | 8      | 29                        |       |              |    |            |              | 34                           |       |              |    |            |              |
| 49 | GRD | Grenada             | 1980  | 2013 | 34  |      | UMC    | 0        | 12     |                           | 31    |              | 31 |            |              |                              |       |              | 34 |            | 34           |
| 50 | GTM | Guatemala           | 1980  | 2015 | 36  |      | UMC    | 3        | 5      |                           | 31    |              | 31 |            |              |                              | 36    |              | 36 |            |              |
| 51 | GUY | Guyana              | 1995  | 2015 | 21  |      | UMC    | 0        | 3      | 16                        |       |              |    |            |              |                              | 21    |              | 21 |            |              |
| 52 | HKG | Hong Kong           | 1999  | 2015 | 15  | 2    | HIC    | 1        | 4      |                           |       |              |    |            |              |                              |       |              |    | 15         | 15           |
| 53 | HND | Honduras            | 1980  | 2015 | 25  | 11   | LMC    | 2        | 8      |                           | 20    |              | 20 |            |              |                              |       |              | 25 |            | 25           |
| 54 | HRV | Croatia             | 1996  | 2015 | 20  |      | HIC    | 0        | 7      |                           | 15    | 15           |    |            |              |                              | 20    | 20           |    |            |              |
| 55 | HTI | Haiti               | 1980  | 2015 | 36  |      | LIC    | 5        | 0      |                           | 31    |              | 31 |            |              |                              |       |              | 36 |            | 36           |
| 56 | HUN | Hungary             | 1992  | 2015 | 24  |      | HIC    | 2        | 11     | 19                        |       |              |    |            |              | 24                           |       |              |    |            |              |
| 57 | IDN | Indonesia           | 1982  | 2015 | 34  |      | UMC    | 1        | 1      |                           | 29    |              | 29 |            |              |                              | 34    |              | 34 |            |              |
| 58 | IND | India               | 1980  | 2015 | 36  |      | LMC    | 1        | 4      | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 59 | ISL | Iceland             | 1980  | 2015 | 36  |      | HIC    | 1        | 12     | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 60 | ISR | Israel              | 1996  | 2015 | 20  |      | HIC    | 2        | 7      | 15                        |       |              |    |            |              | 20                           |       |              |    |            |              |
| 61 | ITA | Italy               | 1980  | 2015 | 36  |      | HIC    | 4        | 9      | 31                        |       |              |    |            |              | 36                           |       |              |    |            |              |
| 62 | JAM | Jamaica             | 1980  | 2015 | 36  |      | UMC    | 1        | 12     | 31                        |       |              |    |            |              |                              | 36    |              | 36 |            |              |
| 63 | JOR | Jordan              | 1980  | 2015 | 36  |      | UMC    | 3        | 14     |                           |       |              |    | 31         | 31           |                              |       |              |    | 36         | 36           |
| 64 | JPN | Japan               | 1997  | 2015 | 19  |      | HIC    | 2        | 0      | 14                        |       |              |    |            |              | 19                           |       |              |    |            |              |
| 65 | KAZ | Kazakhstan          | 1996  | 2015 | 20  |      | UMC    | 1        | 7      |                           |       |              |    | 15         | 15           |                              |       |              |    | 20         | 20           |
| 66 | KEN | Kenya               | 1980  | 2014 | 35  |      | LMC    | 2        | 3      |                           | 31    |              | 31 |            |              |                              |       |              | 35 |            | 35           |
| 67 | KGZ | Kyrgyz Republic     | 1996  | 2015 | 20  |      | LMC    | 2        | 7      |                           | 15    | 15           |    |            |              |                              |       |              | 20 | 20         |              |
| 68 | KHM | Cambodia            | 1995  | 2014 | 20  |      | LMC    | 1        | 7      |                           |       |              |    | 16         |              | 16                           |       |              |    | 20         | 20           |
| 69 | KNA | St. Kitts and Nevis | 1982  | 2013 | 32  |      | HIC    | 0        | 11     | 28                        |       |              |    |            |              |                              |       |              | 32 |            | 32           |
| 70 | KOR | South Korea         | 1980  | 2015 | 36  |      | HIC    | 1        | 4      |                           | 31    | 31           |    |            |              |                              | 36    | 36           |    |            |              |

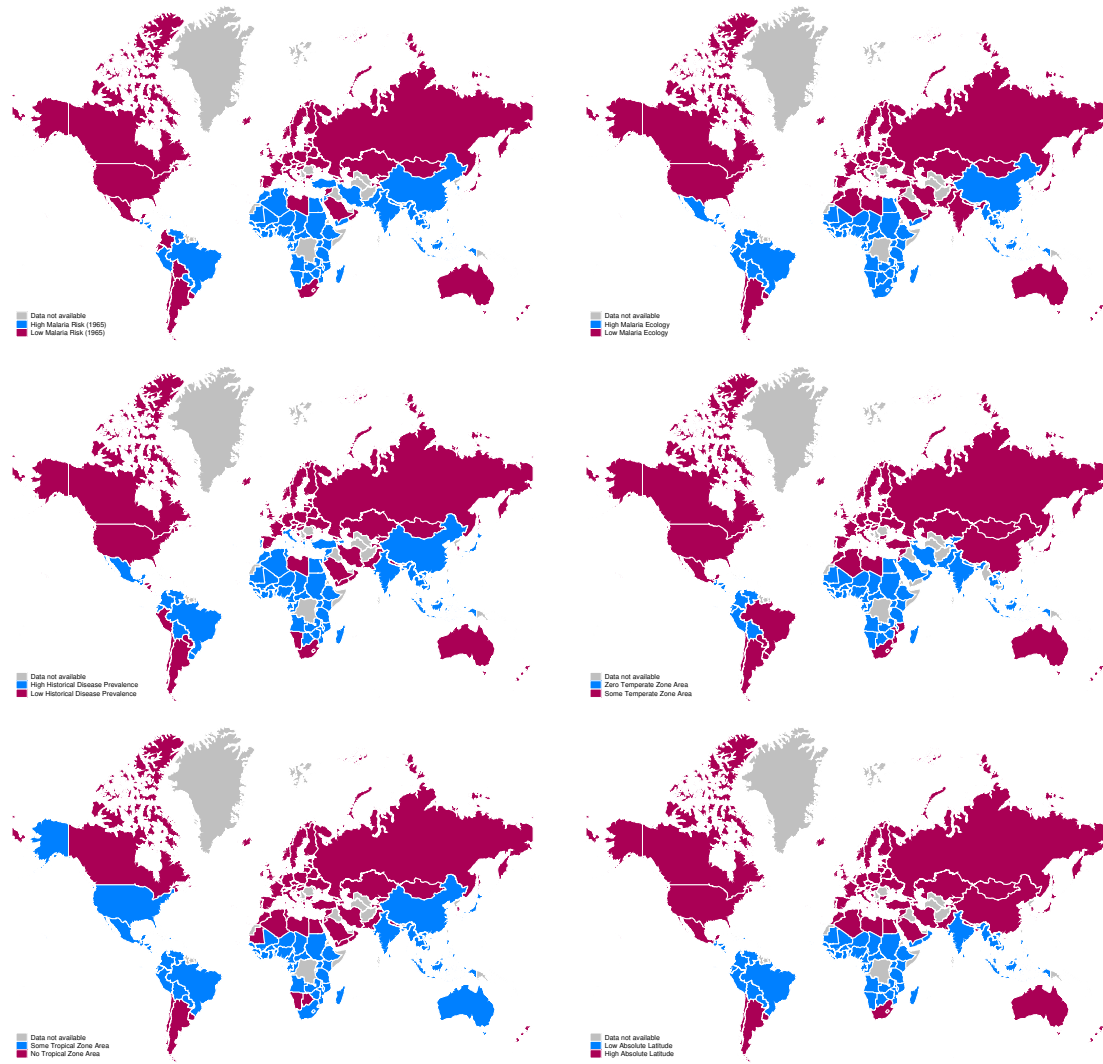
(Continued overleaf)

**Table A-2: Sample Makeup – Excessive Capital Flow Analysis (continued)**

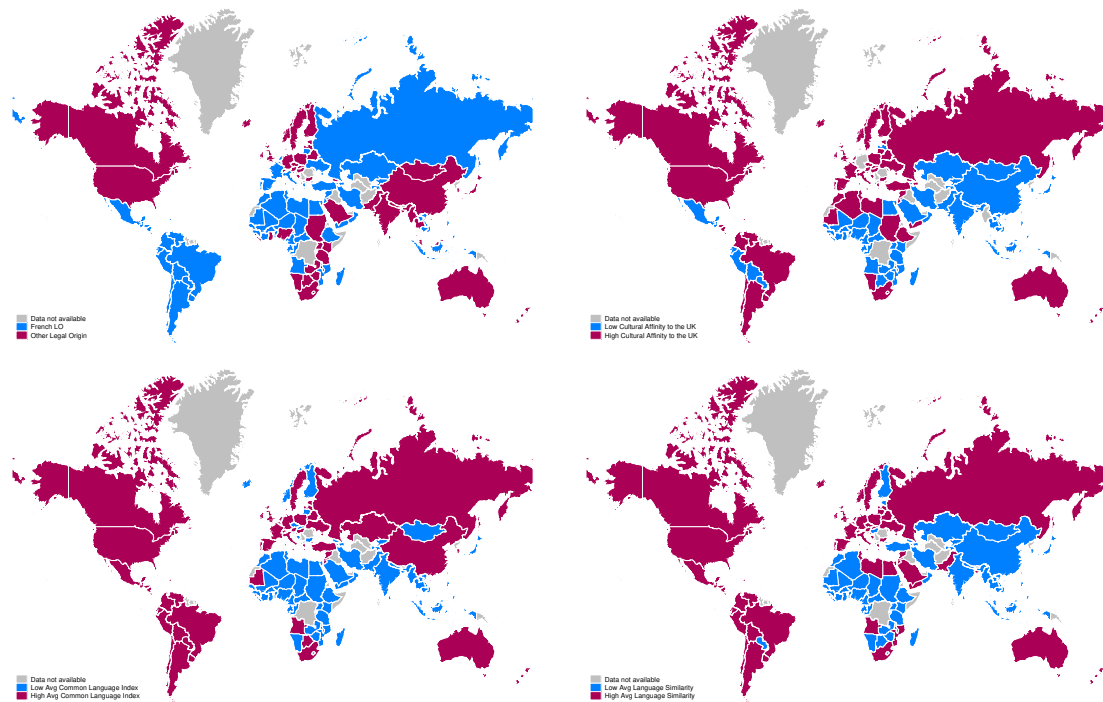
|     | ISO | Country                        | Start | End  | Obs | Miss | Income | Bonanzas | Surges | ANRR Democracy Definition |       |                 |       |                 | Liberal Democracy Definition |       |                 |      |                 |
|-----|-----|--------------------------------|-------|------|-----|------|--------|----------|--------|---------------------------|-------|-----------------|-------|-----------------|------------------------------|-------|-----------------|------|-----------------|
|     |     |                                |       |      |     |      |        |          |        | Always                    | Treat | AbsLat<br>Hi Lo | Ctrl  | AbsLat<br>Hi Lo | Always                       | Treat | AbsLat<br>Hi Lo | Ctrl | AbsLat<br>Hi Lo |
| 71  | KWT | Kuwait                         | 1980  | 2015 | 34  | 2    | HIC    |          | 2      | 4                         |       |                 |       | 29 29           |                              |       |                 | 34   | 34              |
| 72  | LAO | Lao                            | 1989  | 2015 | 27  |      | LMC    |          | 0      | 7                         |       |                 |       | 22 22           |                              |       |                 | 27   | 27              |
| 73  | LBY | Libya                          | 1991  | 2013 | 23  |      | UMC    |          | 2      | 1                         |       |                 |       | 20 20           |                              |       |                 | 23   | 23              |
| 74  | LCA | St. Lucia                      | 1981  | 2013 | 33  |      | UMC    |          | 0      | 9                         | 30    |                 |       |                 |                              |       |                 | 33   | 33              |
| 75  | LKA | Sri Lanka                      | 1980  | 2015 | 36  |      | LMC    |          | 3      | 1                         | 31    |                 |       |                 |                              |       |                 | 36   | 36              |
| 76  | LSO | Lesotho                        | 1981  | 2015 | 32  | 3    | LMC    |          | 1      | 5                         |       | 27 27           |       |                 |                              | 32 32 |                 |      |                 |
| 77  | LTU | Lithuania                      | 1996  | 2015 | 20  |      | HIC    |          | 1      | 7                         | 15    |                 |       |                 | 20                           |       |                 |      |                 |
| 78  | LUX | Luxembourg                     | 2002  | 2015 | 14  |      | HIC    |          | 2      | 5                         | 9     |                 |       |                 | 14                           |       |                 |      |                 |
| 79  | LVA | Latvia                         | 1996  | 2015 | 20  |      | HIC    |          | 2      | 7                         | 15    |                 |       |                 | 20                           |       |                 |      |                 |
| 80  | MAR | Morocco                        | 1980  | 2015 | 36  |      | LMC    |          | 2      | 3                         |       |                 | 31 31 |                 |                              |       |                 | 36   | 36              |
| 81  | MDA | Moldova                        | 1996  | 2015 | 20  |      | LMC    |          | 4      | 7                         | 15    |                 |       |                 |                              | 20 20 |                 |      |                 |
| 82  | MDG | Madagascar                     | 1980  | 2013 | 34  |      | LIC    |          | 1      | 7                         |       | 31 31           |       |                 |                              |       |                 | 34   | 34              |
| 83  | MEX | Mexico                         | 1980  | 2015 | 36  |      | UMC    |          | 1      | 8                         |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 84  | MKD | Macedonia                      | 1997  | 2015 | 19  |      | UMC    |          | 0      | 6                         | 14    |                 |       |                 |                              | 19 19 |                 |      |                 |
| 85  | MLT | Malta                          | 1980  | 2015 | 36  |      | HIC    |          | 1      | 14                        | 31    |                 |       |                 | 36                           |       |                 |      |                 |
| 86  | MMR | Myanmar                        | 1998  | 2015 | 18  |      | LMC    |          | 1      | 1                         |       |                 | 13 13 |                 |                              |       |                 | 18   | 18              |
| 87  | MNG | Mongolia                       | 1993  | 2015 | 23  |      | LMC    |          | 1      | 9                         | 18    |                 |       |                 | 23                           |       |                 |      |                 |
| 88  | MRT | Mauritania                     | 1986  | 2015 | 16  | 14   | LMC    |          | 0      | 2                         |       |                 | 13 13 |                 |                              |       |                 | 16   | 16              |
| 89  | MUS | Mauritius                      | 1980  | 2015 | 36  |      | HIC    |          | 2      | 11                        | 31    |                 |       |                 | 36                           |       |                 |      |                 |
| 90  | MWI | Malawi                         | 1981  | 2015 | 35  |      | LIC    |          | 1      | 3                         |       | 30 30           |       |                 |                              | 35 35 |                 |      |                 |
| 91  | MYS | Malaysia                       | 1980  | 2015 | 36  |      | UMC    |          | 4      | 13                        |       |                 | 31 31 |                 |                              |       |                 | 36   | 36              |
| 92  | NGA | Nigeria                        | 1980  | 2015 | 36  |      | LMC    |          | 1      | 2                         |       | 31 31           |       |                 |                              |       |                 | 36   | 36              |
| 93  | NIC | Nicaragua                      | 2000  | 2015 | 16  |      | LMC    |          | 0      | 9                         | 11    |                 |       |                 |                              |       |                 | 16   | 16              |
| 94  | NLD | Netherlands                    | 1980  | 2015 | 36  |      | HIC    |          | 2      | 14                        | 31    |                 |       |                 | 36                           |       |                 |      |                 |
| 95  | NOR | Norway                         | 1980  | 2015 | 36  |      | HIC    |          | 1      | 11                        | 31    |                 |       |                 | 36                           |       |                 |      |                 |
| 96  | NPL | Nepal                          | 1980  | 2015 | 36  |      | LMC    |          | 4      | 0                         |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 97  | NZL | New Zealand                    | 2001  | 2015 | 15  |      | HIC    |          | 3      | 5                         | 10    |                 |       |                 | 15                           |       |                 |      |                 |
| 98  | OMN | Oman                           | 2001  | 2015 | 15  |      | HIC    |          | 1      | 5                         |       |                 | 10 10 |                 |                              |       |                 | 15   | 15              |
| 99  | PAK | Pakistan                       | 1980  | 2015 | 36  |      | LMC    |          | 2      | 0                         |       | 31 31           |       |                 |                              |       |                 | 36   | 36              |
| 100 | PAN | Panama                         | 1980  | 2015 | 36  |      | HIC    |          | 2      | 9                         |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 101 | PER | Peru                           | 1980  | 2015 | 36  |      | UMC    |          | 2      | 10                        |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 102 | PHL | Philippines                    | 1980  | 2015 | 36  |      | LMC    |          | 1      | 5                         |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 103 | PNG | Papua New Guinea               | 1980  | 2015 | 36  |      | LMC    |          | 0      | 5                         | 31    |                 |       |                 |                              |       |                 | 36   | 36              |
| 104 | POL | Poland                         | 1991  | 2015 | 25  |      | HIC    |          | 0      | 8                         | 20    |                 |       |                 | 25                           |       |                 |      |                 |
| 105 | PRT | Portugal                       | 1980  | 2015 | 36  |      | HIC    |          | 2      | 13                        | 31    |                 |       |                 | 36                           |       |                 |      |                 |
| 106 | PRY | Paraguay                       | 1980  | 2015 | 36  |      | UMC    |          | 2      | 1                         |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 107 | RUS | Russian Federation             | 1997  | 2015 | 19  |      | UMC    |          | 2      | 6                         |       | 14 14           |       |                 |                              |       |                 | 19   | 19              |
| 108 | SAU | Saudi Arabia                   | 1980  | 2015 | 36  |      | HIC    |          | 0      | 3                         |       |                 | 31 31 |                 |                              |       |                 | 36   | 36              |
| 109 | SDN | Sudan                          | 1980  | 2015 | 36  |      | LIC    |          | 0      | 2                         |       | 31 31           |       |                 |                              |       |                 | 36   | 36              |
| 110 | SGP | Singapore                      | 1980  | 2015 | 36  |      | HIC    |          | 2      | 14                        |       |                 |       |                 |                              |       |                 | 36   | 36              |
| 111 | SLB | Solomon Islands                | 1980  | 2015 | 36  |      | LMC    |          | 0      | 12                        |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 112 | SLV | El Salvador                    | 1980  | 2015 | 36  |      | LMC    |          | 1      | 5                         |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 113 | STP | Sao Tome and Principe          | 2002  | 2015 | 14  |      | LMC    |          | 1      | 10                        | 9     |                 |       |                 | 14                           |       |                 |      |                 |
| 114 | SVK | Slovak Republic                | 1994  | 2015 | 22  |      | HIC    |          | 1      | 7                         | 17    |                 |       |                 | 22                           |       |                 |      |                 |
| 115 | SVN | Slovenia                       | 1996  | 2015 | 20  |      | HIC    |          | 2      | 8                         | 15    |                 |       |                 | 20                           |       |                 |      |                 |
| 116 | SWE | Sweden                         | 1980  | 2015 | 36  |      | HIC    |          | 3      | 7                         | 31    |                 |       |                 | 36                           |       |                 |      |                 |
| 117 | SWZ | Eswatini                       | 1980  | 2015 | 36  |      | LMC    |          | 1      | 11                        |       |                 | 31 31 |                 |                              |       |                 | 36   | 36              |
| 118 | SYC | Seychelles                     | 1980  | 2015 | 36  |      | HIC    |          | 3      | 11                        |       |                 | 31 31 |                 |                              |       |                 | 36   | 36              |
| 119 | SYR | Syria                          | 1980  | 2010 | 31  |      | LIC    |          | 1      | 3                         |       |                 | 31 31 |                 |                              |       |                 | 31   | 31              |
| 120 | THA | Thailand                       | 1980  | 2015 | 36  |      | UMC    |          | 1      | 13                        |       | 31 31           |       |                 |                              |       |                 | 36   | 36              |
| 121 | TON | Tonga                          | 1980  | 2013 | 26  | 8    | UMC    |          | 0      | 4                         |       |                 | 23 23 |                 |                              |       |                 | 26   | 26              |
| 122 | TTO | Trinidad and Tobago            | 1980  | 2011 | 32  |      | HIC    |          | 1      | 3                         | 31    |                 |       |                 | 32                           |       |                 |      |                 |
| 123 | TUN | Tunisia                        | 1984  | 2015 | 32  |      | LMC    |          | 1      | 4                         |       |                 | 27 27 |                 |                              | 32 32 |                 |      |                 |
| 124 | TUR | Turkey                         | 1980  | 2015 | 36  |      | UMC    |          | 0      | 10                        |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 125 | TZA | Tanzania                       | 1989  | 2015 | 27  |      | LMC    |          | 1      | 2                         |       |                 | 22 22 |                 |                              |       |                 | 27   | 27              |
| 126 | UGA | Uganda                         | 1994  | 2015 | 22  |      | LIC    |          | 0      | 2                         |       |                 | 17 17 |                 |                              |       |                 | 22   | 22              |
| 127 | UKR | Ukraine                        | 1995  | 2015 | 21  |      | LMC    |          | 1      | 7                         | 16    |                 |       |                 |                              |       |                 | 21   | 21              |
| 128 | URY | Uruguay                        | 1980  | 2015 | 36  |      | HIC    |          | 2      | 12                        |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 129 | USA | United States                  | 1980  | 2015 | 36  |      | HIC    |          | 2      | 11                        | 31    |                 |       |                 | 36                           |       |                 |      |                 |
| 130 | VCT | St. Vincent and the Grenadines | 1980  | 2013 | 34  |      | UMC    |          | 0      | 11                        | 31    |                 |       |                 |                              |       |                 | 34   | 34              |
| 131 | VNM | Vietnam                        | 1997  | 2015 | 19  |      | LMC    |          | 1      | 5                         |       |                 | 14 14 |                 |                              |       |                 | 19   | 19              |
| 132 | VUT | Vanuatu                        | 1983  | 2015 | 33  |      | LMC    |          | 0      | 10                        | 28    |                 |       |                 | 33                           |       |                 |      |                 |
| 133 | WSM | Samoa                          | 1983  | 2015 | 28  | 5    | UMC    |          | 0      | 2                         |       |                 | 23 23 |                 |                              |       |                 | 28   | 28              |
| 134 | ZAF | South Africa                   | 1976  | 2015 | 40  |      | UMC    |          | 2      | 5                         |       | 31 31           |       |                 |                              | 36 36 |                 |      |                 |
| 135 | ZMB | Zambia                         | 1974  | 2015 | 42  |      | LMC    |          | 1      | 11                        |       | 19 19           |       |                 |                              | 24 24 |                 |      |                 |

Notes: We present the sample makeup for the excessive capital flow analysis (bonanzas, surges; 1980-2015). Income indicates the World Bank Income Level category (Low - LIC, Lower Middle - LMC, Upper Middle - UMC, and High - HIC). We report the bonanza and surge count adopting the 1SD and 1-year definitions, respectively. The remaining columns indicate treated and controls samples (total number of observations, respectively) for two democracy definitions: that by ANRR and the V-Dem Liberal Democracy (sample mean cutoff). 'Always' refers to countries that were democracies throughout the sample period, 'treat' to the treated sample, where absolute latitude ('Abslat') 'Hi' and 'Lo' provide the split for one of the many deep determinants we apply in our analysis. 'Ctrl' is the control sample, again split into 'Hi' and 'Lo' absolute latitude. Absolute latitude is one of the six geography proxies (plus four more in robustness checks) we adopt in the paper, in addition to proxies for culture, history and legal origin.

**Figure A-3: Distribution of (Dichotomised) Deep Determinants**



(a) Proxies for Geography



(b) French Legal Origin (top left) and Proxies for Culture

**Notes:** We present the distribution of countries for all deep determinant proxies. Grey indicates data are not available, dark pink (blue) shaded countries are those with supposedly beneficial (detrimental) deep determinants.

**Table A-3: Deep Determinants: Pairwise Correlation**

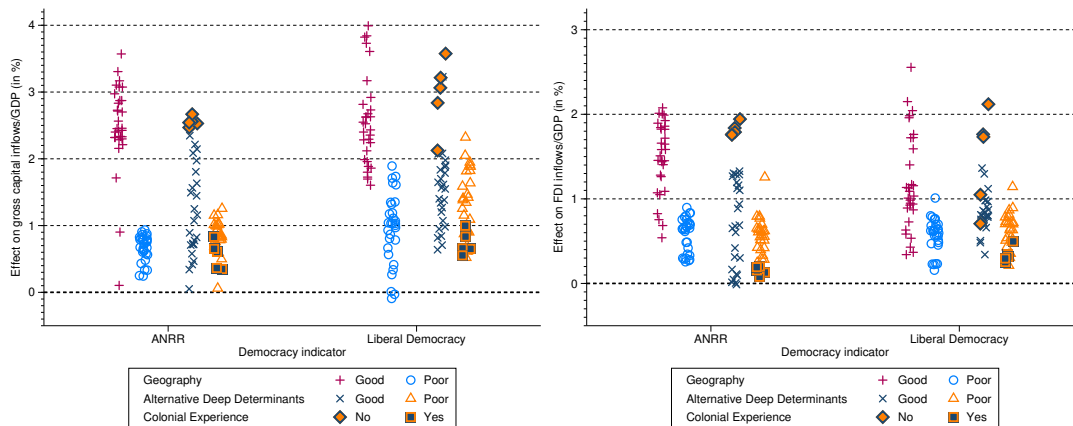
|                        | Geography       |              |                    |                |              |                   | LO                | Culture             |                 |                     |
|------------------------|-----------------|--------------|--------------------|----------------|--------------|-------------------|-------------------|---------------------|-----------------|---------------------|
|                        | Malaria Ecology | Malaria Risk | Disease Prevalence | Zero Temperate | Some Tropics | Absolute Latitude | French Leg.Origin | Cult. Dist. from UK | Common Language | Language Similarity |
| Malaria Ecology        | 1.00<br>108     |              |                    |                |              |                   |                   |                     |                 |                     |
| Malaria Risk           | 0.58<br>108     | 1.00<br>109  |                    |                |              |                   |                   |                     |                 |                     |
| Disease Prevalence     | 0.52<br>108     | 0.50<br>109  | 1.00<br>109        |                |              |                   |                   |                     |                 |                     |
| Zero Temperate         | 0.60<br>99      | 0.48<br>99   | 0.39<br>99         | 1.00<br>99     |              |                   |                   |                     |                 |                     |
| Some Tropics           | 0.79<br>108     | 0.63<br>109  | 0.54<br>109        | 0.60<br>99     | 1.00<br>109  |                   |                   |                     |                 |                     |
| Absolute Latitude      | 0.78<br>108     | 0.55<br>109  | 0.49<br>109        | 0.71<br>99     | 0.81<br>109  | 1.00<br>109       |                   |                     |                 |                     |
| French Legal Origin    | 0.09<br>108     | -0.04<br>109 | 0.03<br>109        | 0.02<br>99     | -0.02<br>109 | 0.09<br>109       | 1.00<br>109       |                     |                 |                     |
| Cult. Distance from UK | 0.47<br>102     | 0.41<br>102  | 0.26<br>102        | 0.56<br>99     | 0.53<br>102  | 0.41<br>102       | -0.16<br>102      | 1.00<br>102         |                 |                     |
| Common Language        | 0.13<br>108     | 0.31<br>109  | 0.21<br>109        | 0.41<br>99     | 0.15<br>109  | 0.18<br>109       | -0.15<br>109      | 0.38<br>102         | 1.00<br>109     |                     |
| Language Similarity    | 0.28<br>108     | 0.35<br>109  | 0.30<br>109        | 0.35<br>99     | 0.31<br>109  | 0.38<br>109       | -0.24<br>109      | 0.47<br>102         | 0.48<br>109     | 1.00<br>109         |

Notes: We present the pairwise correlation coefficients and sample sizes for the time-invariant deep determinants: geography (×6 proxies), French legal origin and culture (×3). This is for the treatment and control samples only, using Liberal Democracy as the regime change definition. Results are virtually identical if we use the full sample (including countries which have been democratic throughout the sample period). The mean (median) for 15 geography correlations is 0.60 (0.58), and for 3 culture correlations 0.44 (0.47). The mean (median) of 18 correlation between geography and culture is 0.32 (0.33), of 6 correlations between geography and legal origin is 0.03 (0.02), and of 3 correlations between legal origin and culture is -0.18 (-0.16).

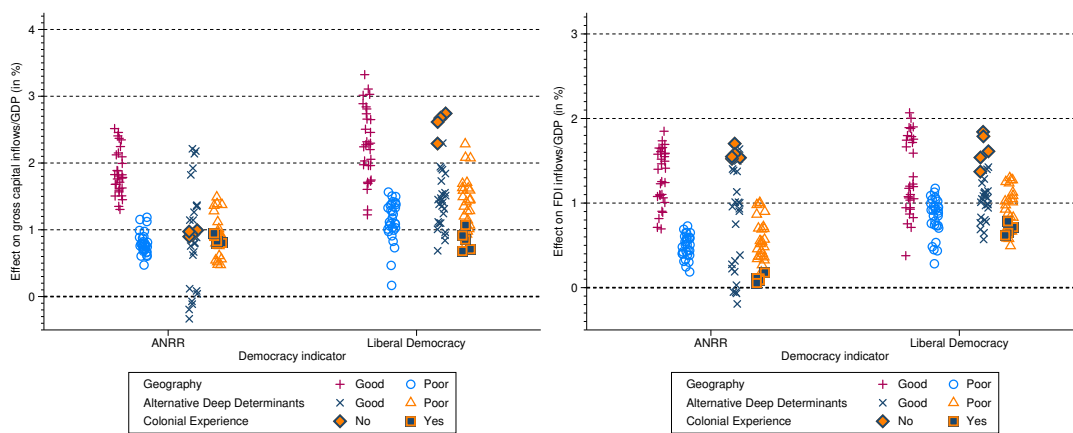
## B Results Tables, Diagnostics and Robustness Checks

### B.1 Main Analysis — Results highlighting Colonial Experience

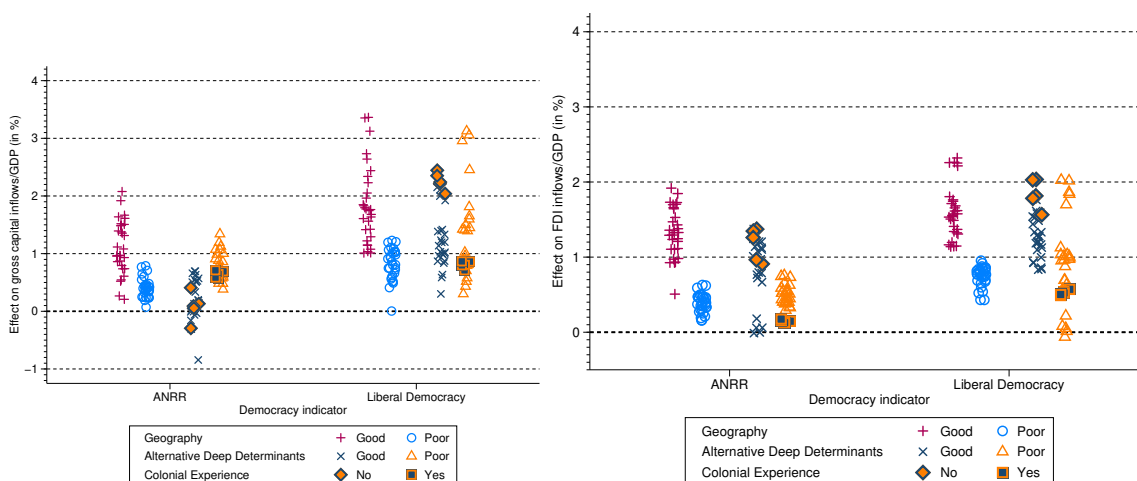
Figure B-1: Democracy, Deep Determinants and Capital Inflows



(a) Total Non-Official Capital (left) and FDI (right) Inflows – no controls



(b) dto – export/trade as control (not a bad control)



(c) dto – full controls (bad controls)

**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by geography (+ and o for good and poor geography, respectively) and alternative deep determinants (x for non-French LO/proximate culture/no colonial experience and  $\triangle$  for French LO/distant culture/colonial experience). This set of results further distinguishes colonial experience. For all other aspects see Figure 3 in the maintext.

## B.2 Main Analysis — Result tables

**Table B-1:** Democracy, Geography and Total Non-Official Capital Inflows (1975-2015)

| Disease Environment             |                     |                     |                     |                     |                     |                    |                     |                    |                     |                     |                     | Climate            |  |  |  |
|---------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--------------------|---------------------|--------------------|---------------------|---------------------|---------------------|--------------------|--|--|--|
| (1)                             | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                | (8)                 | (9)                | (10)                | (11)                | (12)                |                    |  |  |  |
| Malaria Ecology                 |                     | Malaria Risk        |                     | Hist. Disease Index |                     | Tropical Land      |                     | No Temperate Land  |                     | Latitude  > MD      |                     |                    |  |  |  |
| Low                             | High                | Low                 | High                | Low                 | High                | None               | Some                | No                 | Yes                 | Yes                 | No                  |                    |  |  |  |
| ANRR democracy                  |                     |                     |                     |                     |                     |                    |                     |                    |                     |                     |                     |                    |  |  |  |
| 2.694***<br>[0.883]             | 0.768*<br>[0.416]   | 3.551***<br>[1.076] | 0.558<br>[0.381]    | 2.405***<br>[0.876] | 0.734*<br>[0.412]   | 2.440**<br>[1.095] | 0.760**<br>[0.378]  | 1.761**<br>[0.867] | 0.856**<br>[0.371]  | 2.894***<br>[0.956] | 0.866**<br>[0.435]  |                    |  |  |  |
| Treated Countries               | 22                  | 47                  | 23                  | 46                  | 26                  | 43                 | 46                  | 21                 | 44                  | 23                  | 46                  |                    |  |  |  |
| Treated Observations            | 617                 | 1470                | 626                 | 1461                | 763                 | 1324               | 637                 | 1450               | 616                 | 1368                | 650                 |                    |  |  |  |
| Control Countries               | 13                  | 17                  | 10                  | 21                  | 11                  | 20                 | 17                  | 14                 | 9                   | 18                  | 15                  |                    |  |  |  |
| Control Observations            | 342                 | 448                 | 265                 | 560                 | 267                 | 558                | 451                 | 374                | 252                 | 491                 | 416                 |                    |  |  |  |
| Alpha test ( <i>t</i> )         | -1.23               | -1.98               | -2.56               | -0.56               | -1.25               | -0.84              | -0.70               | -1.00              | -1.07               | -5.55               | -0.35               | -4.63              |  |  |  |
| Alternative factor augmentation |                     |                     |                     |                     |                     |                    |                     |                    |                     |                     |                     |                    |  |  |  |
| 2 factors                       | 2.219**             | 0.572               | 2.826***            | 0.291               | 3.084***            | 0.342              | 2.476***            | 0.695*             | 3.090***            | 0.636*              | 2.126***            | 0.599              |  |  |  |
| 3 factors                       | 2.737***            | 0.874**             | 3.054***            | 0.383               | 3.121***            | 0.685*             | 2.684**             | 0.782**            | 2.545**             | 0.637*              | 2.278**             | 0.784*             |  |  |  |
| 4 factors                       | 2.694***            | 0.768*              | 3.551***            | 0.558               | 2.405***            | 0.734*             | 2.440**             | 0.760**            | 1.761**             | 0.856**             | 2.894***            | 0.866***           |  |  |  |
| 5 factors                       | 2.330**             | 0.836*              | 3.315***            | 0.612               | 2.309***            | 0.781**            | 2.392**             | 0.665*             | 0.959               | 0.991**             | 2.411***            | 0.864*             |  |  |  |
| 6 factors                       | 2.970***            | 0.247               | 2.870***            | 0.281               | 2.293***            | 0.871**            | 2.306***            | 0.452              | 0.109               | 0.730**             | 2.306***            | 0.645*             |  |  |  |
| Liberal Democracy Index         |                     |                     |                     |                     |                     |                    |                     |                    |                     |                     |                     |                    |  |  |  |
| > median                        | 2.605***<br>[0.845] | 1.014*<br>[0.592]   | 3.588***<br>[0.785] | -0.049<br>[0.484]   | 2.608***<br>[0.847] | 1.048**<br>[0.477] | 2.224***<br>[0.858] | 0.985<br>[0.599]   | 2.035***<br>[0.779] | 1.222***<br>[0.452] | 2.377***<br>[0.858] | 1.375**<br>[0.595] |  |  |  |
| Treated Countries               | 24                  | 27                  | 23                  | 28                  | 22                  | 29                 | 22                  | 29                 | 22                  | 26                  | 22                  | 29                 |  |  |  |
| Treated Observations            | 801                 | 1029                | 760                 | 1070                | 709                 | 1121               | 718                 | 1112               | 745                 | 996                 | 716                 | 1114               |  |  |  |
| Control Countries               | 17                  | 40                  | 16                  | 42                  | 18                  | 40                 | 22                  | 36                 | 11                  | 40                  | 20                  | 38                 |  |  |  |
| Control Observations            | 498                 | 1262                | 440                 | 1360                | 551                 | 1249               | 630                 | 1170               | 321                 | 1288                | 588                 | 1212               |  |  |  |
| Alpha test ( <i>t</i> )         | -0.13               | -1.74               | -6.72               | 0.10                | 0.63                | -1.90              | -4.53               | 0.02               | -0.91               | 0.10                | 0.48                | -2.48              |  |  |  |
| Alternative factor augmentation |                     |                     |                     |                     |                     |                    |                     |                    |                     |                     |                     |                    |  |  |  |
| 2 factors                       | 1.868***            | 1.618***            | 3.946***            | 0.292               | 2.929***            | 1.745***           | 3.228***            | 0.324              | 1.712***            | 1.204**             | 1.928***            | 1.862***           |  |  |  |
| 3 factors                       | 2.634***            | 1.357**             | 3.773***            | 0.757               | 2.627***            | 1.592***           | 2.715***            | 0.953*             | 2.100***            | 1.299**             | 1.792***            | 1.703**            |  |  |  |
| 4 factors                       | 2.605***            | 1.014*              | 3.588***            | -0.049              | 2.608***            | 1.048**            | 2.224***            | 0.985              | 2.035***            | 1.222***            | 2.377***            | 1.375**            |  |  |  |
| 5 factors                       | 2.559***            | 0.940**             | 3.739***            | 0.019               | 2.391***            | 1.032**            | 2.275**             | 0.557              | 1.942**             | 1.166***            | 2.483***            | 0.827              |  |  |  |
| 6 factors                       | 2.812***            | 0.661               | 3.879***            | -0.054              | 1.599***            | 1.027**            | 2.264**             | 0.385              | 1.731**             | 1.037**             | 2.409***            | 0.760              |  |  |  |

**Notes:** This table is for the analysis by geography. We present robust mean estimates from PCDDID regressions of total non-official capital inflows and a democracy dummy defined as indicated in each result panel — these can be interpreted as Average Treatment Effects on the Treated (ATET). These results are for the model *without any additional controls*. The 12 different models in each panel are for sample splits determined by ‘disease environment’ and ‘climate’, in each case we use three proxies for these factors of geography, separating ‘good’ geography in the odd columns and ‘poor’ geography in the even columns. In a lower part of each panel we report the ATET estimates for specification with two to six factors. See Table 1 for all other details.

Table B-2: Democracy, Alternative Deep Determinants and Total Non-Official Capital Inflows (1975-2015)

| Legal Origin                    |          |          | Culture           |          |                 |         |                     |         | History      |         |                |         |
|---------------------------------|----------|----------|-------------------|----------|-----------------|---------|---------------------|---------|--------------|---------|----------------|---------|
| (1)                             | (2)      |          | (3)               | (4)      | (5)             | (6)     | (7)                 | (8)     | (9)          | (10)    | (11)           | (12)    |
| French LO                       |          |          | Cultural Distance |          | Common Language |         | Language Similarity |         | Colonial Exp |         | Extractive Col |         |
| No                              | Yes      |          | Low               | High     | Low             | High    | High                | Low     | No           | Yes     | No             | Yes     |
| ANRR democracy                  |          |          |                   |          |                 |         |                     |         |              |         |                |         |
| 0.396                           | 0.918*   |          | 1.786*            | 0.825**  | 1.425           | 0.835*  | 1.955***            | 0.776*  | 2.647***     | 0.585   | 0.756          | 0.802   |
| [0.678]                         | [0.512]  |          | [1.019]           | [0.416]  | [0.908]         | [0.497] | [0.756]             | [0.468] | [0.864]      | [0.437] | [0.484]        | [0.901] |
| 23                              | 46       |          | 24                | 42       | 31              | 38      | 30                  | 39      | 33           | 36      | 23             | 14      |
| Treated Countries               |          |          |                   |          |                 |         |                     |         |              |         |                |         |
| 673                             | 1414     |          | 722               | 1281     | 957             | 1130    | 912                 | 1175    | 1000         | 1078    | 694            | 414     |
| Control Countries               |          |          | 10                | 19       | 9               | 22      | 10                  | 21      | 6            | 25      | 11             | 14      |
| Control Observations            |          |          | 264               | 513      | 241             | 584     | 305                 | 520     | 110          | 715     | 334            | 381     |
| Alpha test ( <i>t</i> )         | -1.33    | -2.22    | -2.50             | -3.66    | -4.22           | -2.67   | 0.26                | -5.55   | -8.41        | -2.19   | -2.33          | -5.59   |
| Alternative factor augmentation |          |          |                   |          |                 |         |                     |         |              |         |                |         |
| 2 factors                       | 1.167    | 1.262*** | 1.204             | 0.973**  | 2.155***        | 0.839   | 2.144***            | 1.068*  | 2.537***     | 0.357   | 0.482          | 1.051   |
| 3 factors                       | 0.588    | 0.999*   | 1.444*            | 0.948**  | 1.026           | 0.829*  | 1.616**             | 0.482   | 2.568***     | 0.373   | 0.713          | 1.140   |
| 4 factors                       | 0.396    | 0.918*   | 1.786*            | 0.825**  | 1.425           | 0.835*  | 1.955***            | 0.776*  | 2.647***     | 0.585   | 0.756          | 0.802   |
| 5 factors                       | 0.352    | 1.163**  | 1.574             | 0.953*** | 0.882           | 0.675   | 2.338***            | 0.755   | 2.479***     | 0.662   | 0.833*         | 0.879   |
| 6 factors                       | 0.048    | 1.009*   | 0.755             | 0.097    | 0.818           | 0.319   | 2.185***            | 0.755*  | 2.539***     | 0.828   | 0.724          | 0.599   |
| Liberal Democracy Index         |          |          |                   |          |                 |         |                     |         |              |         |                |         |
| 1.319*                          | 2.300*** |          | 1.364             | 0.881    | 1.766**         | 1.886** | 1.861***            | 1.930** | 3.194***     | 0.814   | 0.997*         | 1.200   |
| [0.773]                         | [0.704]  |          | [0.939]           | [0.624]  | [0.767]         | [0.773] | [0.613]             | [0.760] | [0.753]      | [0.625] | [0.575]        | [0.779] |
| 20                              | 31       |          | 23                | 26       | 27              | 24      | 26                  | 25      | 31           | 20      | 10             | 11      |
| Treated Countries               |          |          |                   |          |                 |         |                     |         |              |         |                |         |
| 653                             | 1177     |          | 770               | 989      | 995             | 835     | 927                 | 903     | 1095         | 735     | 360            | 410     |
| Control Countries               |          |          | 16                | 37       | 18              | 40      | 16                  | 42      | 12           | 46      | 25             | 19      |
| Control Observations            |          |          | 495               | 1163     | 556             | 1244    | 544                 | 1256    | 336          | 1464    | 809            | 596     |
| Alpha test ( <i>t</i> )         | -14.03   | -0.45    | -0.22             | -1.29    | -8.50           | 0.20    | 1.16                | -5.94   | -2.73        | -6.51   | -1.77          | -0.38   |
| Alternative factor augmentation |          |          |                   |          |                 |         |                     |         |              |         |                |         |
| 2 factors                       | 1.911*   | 1.892*** | 3.183***          | 0.567    | 1.988***        | 1.646** | 1.625**             | 1.874** | 3.583***     | 0.659   | 1.284          | 0.923   |
| 3 factors                       | 1.844*   | 2.058*** | 1.352**           | 0.826    | 2.033***        | 1.419** | 1.532**             | 1.805** | 3.072***     | 1.006*  | 0.823          | 1.325   |
| 4 factors                       | 1.319*   | 2.300*** | 1.364             | 0.881    | 1.766**         | 1.886** | 1.861***            | 1.930** | 3.194***     | 0.814   | 0.997*         | 1.200   |
| 5 factors                       | 0.827    | 1.597*** | 1.710*            | 0.645    | 1.638***        | 1.373** | 2.066***            | 1.397** | 2.848***     | 0.669   | 1.129**        | 0.987   |
| 6 factors                       | 0.635    | 1.251**  | 1.240             | 0.557    | 0.976*          | 1.086*  | 1.586**             | 1.391*  | 2.122***     | 0.550   | 0.712          | 0.702   |

**Notes:** This table presents the analysis by alternative deep determinants — French Legal Origin (LO), three measures for cultural clusters, and two for colonial experience. Odd (even) columns are for ‘good’ (‘poor’) deep determinants. See Table B-1 for all other details.

## B.3 Main Analysis — Diagnostic tests

**Table B-3:** Diagnostic Tests — PCDID Capital Flow Analysis

| Democracy Indicator                                             |                   | ANRR |      | LibDem |      | Poly |      | Liberal |      | N  |
|-----------------------------------------------------------------|-------------------|------|------|--------|------|------|------|---------|------|----|
| Deep Determinant Group                                          |                   | 0    | 1    | 0      | 1    | 0    | 1    | 0       | 1    |    |
| <b>Panel A: Total Non-Official Capital Inflows</b>              |                   |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 0.17 | 0.50 | 0.33   | 0.17 | 0.67 | 0.50 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 0.67 | 1.00 | 0.50   | 0.33 | 0.67 | 0.50 | 0.00    | 0.00 | 30 |
| <i>Controls: export/trade</i>                                   |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 0.17 | 0.17 | 0.33   | 0.00 | 0.50 | 0.50 | 0.17    | 0.00 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 0.67 | 0.83 | 0.33   | 0.33 | 0.83 | 0.33 | 0.00    | 0.00 | 30 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.00 | 0.13 | 0.00   | 0.17 | 0.00 | 0.27 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.17 | 0.00 | 0.00   | 0.00 | 0.10 | 0.00 | 0.10    | 0.00 | 30 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 0.17 | 0.67 | 0.33   | 0.00 | 0.67 | 0.17 | 0.33    | 0.00 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 0.67 | 0.67 | 0.50   | 0.33 | 1.00 | 0.67 | 0.33    | 0.00 | 30 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.67 | 0.93 | 0.33   | 0.93 | 0.83 | 1.00 | 1.00    | 1.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.63 | 0.90 | 0.27   | 0.67 | 0.67 | 0.80 | 0.83    | 1.00 | 30 |
| <b>Panel B: FDI Inflows</b>                                     |                   |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 0.33 | 0.67 | 0.00   | 0.00 | 0.67 | 0.17 | 0.33    | 0.17 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 0.50 | 0.33 | 0.33   | 0.00 | 0.50 | 0.00 | 0.17    | 0.00 | 30 |
| <i>Controls: export/trade</i>                                   |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 0.33 | 1.00 | 0.00   | 0.00 | 0.50 | 0.17 | 0.17    | 0.00 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 0.67 | 0.33 | 0.33   | 0.00 | 0.83 | 0.00 | 0.17    | 0.00 | 30 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.07 | 0.13 | 0.00   | 0.17 | 0.00 | 0.17 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.20 | 0.07 | 0.03   | 0.00 | 0.03 | 0.00 | 0.00    | 0.00 | 30 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 0.00 | 1.00 | 0.33   | 0.17 | 0.50 | 0.33 | 0.17    | 0.17 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 0.50 | 0.33 | 0.50   | 0.17 | 0.83 | 0.17 | 0.50    | 0.00 | 30 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.30 | 0.83 | 0.37   | 1.00 | 1.00 | 0.80 | 1.00    | 1.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.43 | 0.67 | 0.23   | 0.77 | 0.73 | 1.00 | 0.83    | 1.00 | 30 |

Notes: The table reports the rejection frequency across the 20 or 30 models analysed for each cell (and presented in Figures 3 and B-2 in the maintext). The ‘Alpha’ test is for weak parallel trends, so if the null hypothesis is rejected the PCDID specification may be misspecified: we want to see very low rejection rates, like for the ‘Liberal Component’. The  $\chi^2$  test is for bad controls, so if the null hypothesis is rejected ( $p < 0.1$ ) we should not include this (set of) control(s): again, we want to see very low rejection rates, like for the models with export/trade as additional control. There are five alternative factor augmentations and four (alternative deep determinants) or six (geography) proxies for deep determinants, hence 20 or 30 models for each of the two deep determinant groups, respectively. These 20 or 30 models are represented in each estimate ‘cloud’ of the aforementioned figures.

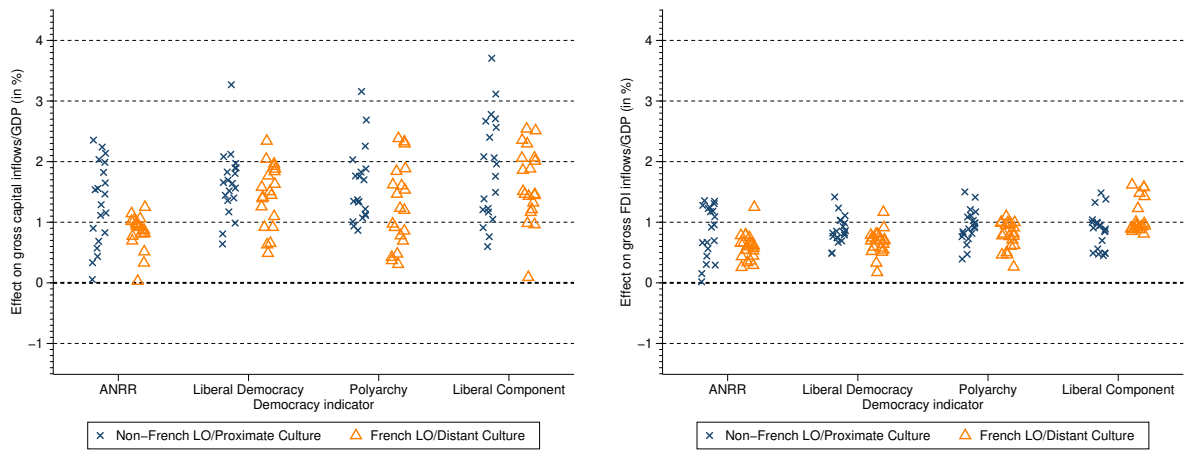
**Table B-4: Statistical Significance — PCDID Capital Flow Analysis**

| Democracy Indicator                                             | ANRR |      | LibDem |      | Poly |      | Liberal |      | N  |
|-----------------------------------------------------------------|------|------|--------|------|------|------|---------|------|----|
| Deep Determinant Group                                          | 0    | 1    | 0      | 1    | 0    | 1    | 0       | 1    |    |
| <b>Panel A: Total Non-Official Capital Inflows</b>              |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.93 | 0.67 | 1.00   | 0.60 | 0.70 | 0.80 | 1.00    | 0.63 | 30 |
| Alternative Deep Det                                            | 0.67 | 0.63 | 0.90   | 0.63 | 0.83 | 0.63 | 0.73    | 1.00 | 30 |
| <i>Controls: export/trade</i>                                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.97 | 0.97 | 1.00   | 0.80 | 0.23 | 1.00 | 1.00    | 0.60 | 30 |
| Alternative Deep Det                                            | 0.67 | 0.57 | 0.97   | 0.83 | 0.90 | 0.97 | 0.83    | 0.97 | 30 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.13 | 0.00 | 0.53   | 0.60 | 0.30 | 0.53 | 0.93    | 0.10 | 30 |
| Alternative Deep Det                                            | 0.67 | 0.50 | 1.00   | 0.70 | 0.90 | 0.83 | 0.80    | 0.63 | 30 |
| <b>Panel B: FDI Inflows</b>                                     |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.93 | 0.97 | 0.90   | 0.73 | 0.83 | 1.00 | 0.83    | 0.77 | 30 |
| Alternative Deep Det                                            | 0.67 | 0.63 | 0.90   | 0.63 | 0.83 | 0.63 | 0.73    | 1.00 | 30 |
| <i>Controls: export/trade</i>                                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 1.00 | 0.97 | 0.97   | 1.00 | 0.87 | 1.00 | 0.93    | 1.00 | 30 |
| Alternative Deep Det                                            | 0.67 | 0.57 | 0.97   | 0.83 | 0.90 | 0.97 | 0.83    | 0.97 | 30 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.90 | 0.53 | 1.00   | 0.87 | 0.93 | 1.00 | 0.97    | 0.83 | 30 |
| Alternative Deep Det                                            | 0.67 | 0.50 | 1.00   | 0.70 | 0.90 | 0.83 | 0.80    | 0.63 | 30 |

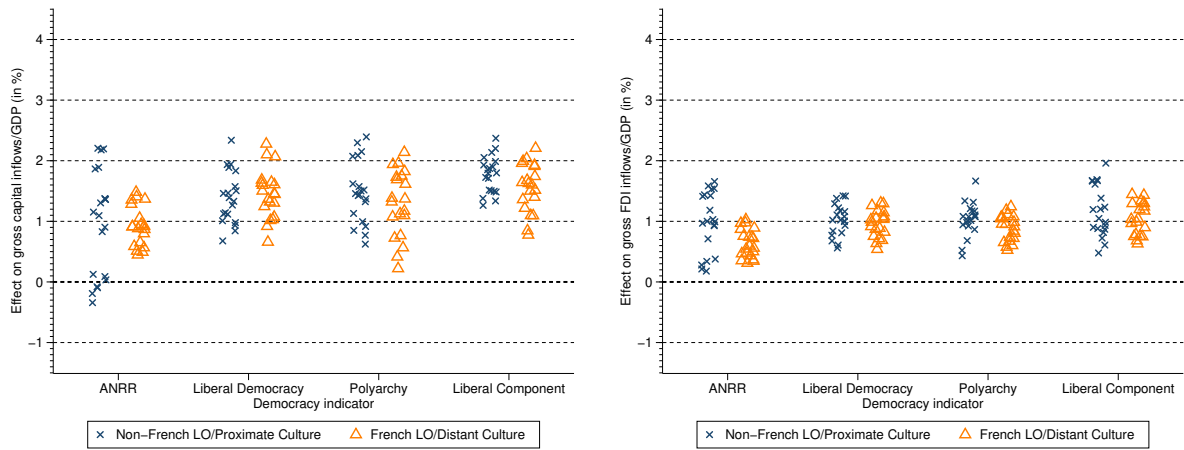
Notes: The table reports the rejection frequency across the 20 or 30 models analysed for each cell (and presented in Figures 3 and B-2 in the maintext). These are for the  $t$ -tests (10% level) of the robust mean PCDID estimates (computed using the non-parametric variance estimator of Pesaran, 2006): if we see very high rejection rates this equates to statistical significance of the ATET presented in aforementioned figures. There are five alternative factor augmentations and four (alternative deep determinants) or six (geography) proxies for deep determinants, hence 20 or 30 models for each of the two deep determinant groups, respectively. These 20 or 30 models are represented in each estimate 'cloud' of the aforementioned figures.

## B.4 Splitting alternative determinants (separate for colonial experience)

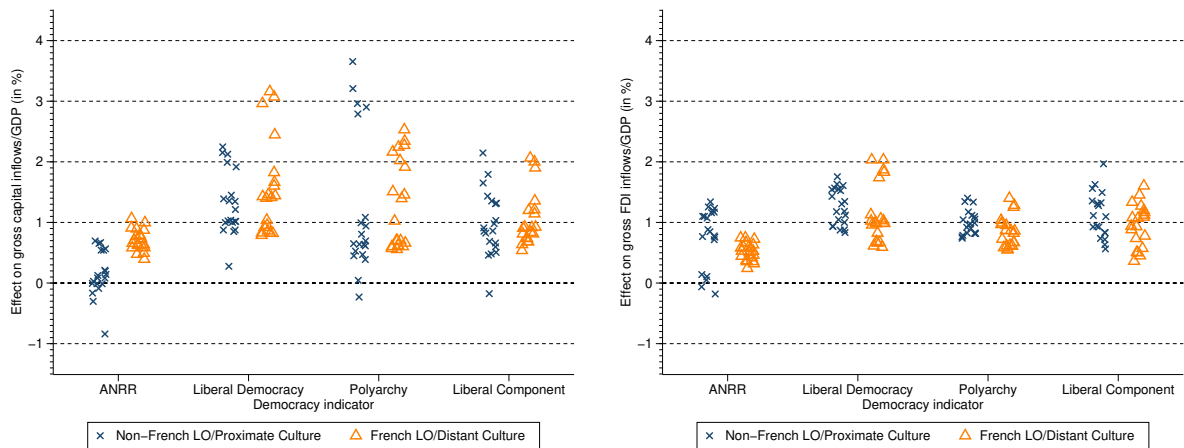
**Figure B-2:** Democracy, Culture/Legal Origin and Capital Inflows



(a) Total Non-Official Capital Inflows (left) and Total FDI Inflows (right) – no controls



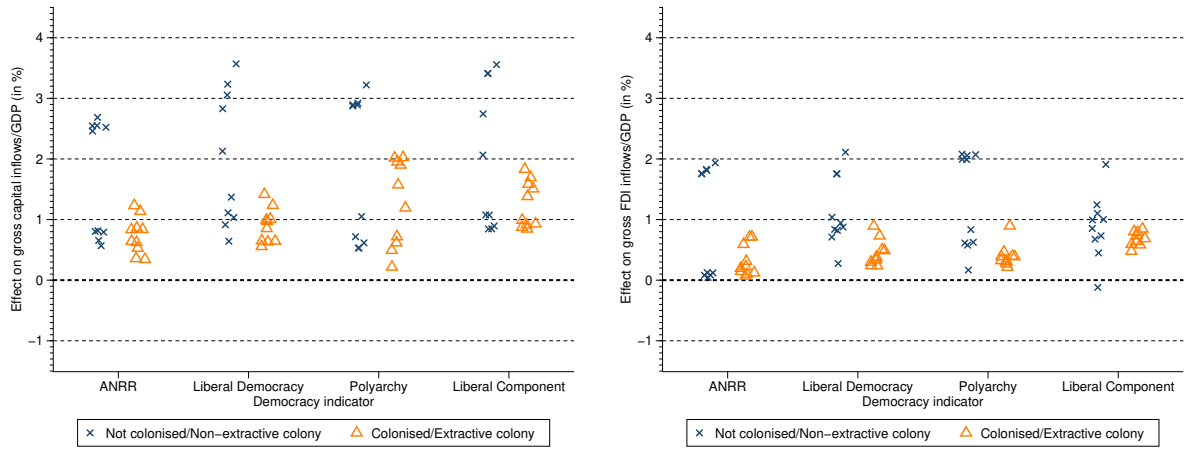
(b) Dto – exports/trade as additional control (not a bad control)



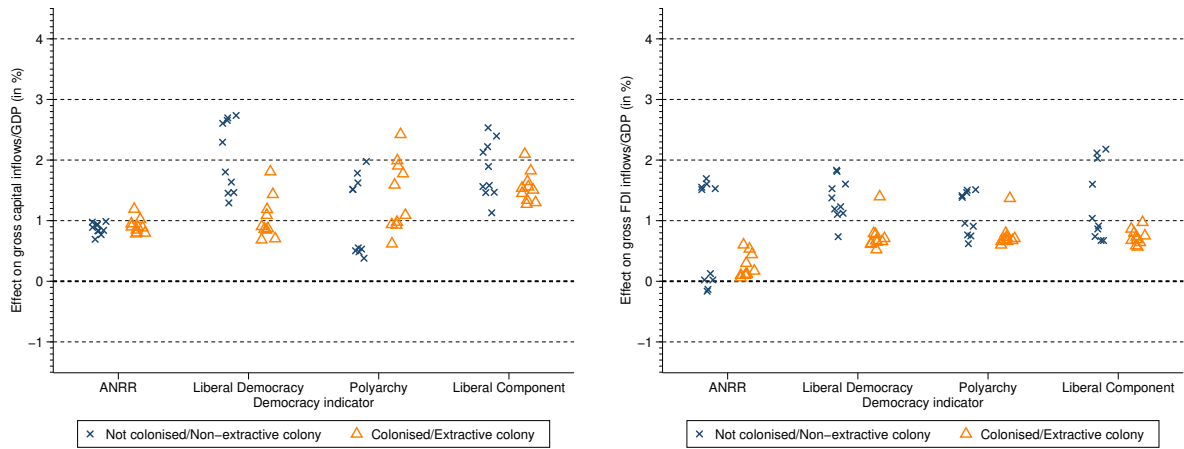
(c) Dto – exports/trade, pop growth and GDPpc growth as controls (bad controls)

**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by Legal Origin and Culture (x for non-French LO/proximate culture and  $\triangle$  for French LO/distant culture), using four different definitions of democratic regime change. See Figure 3 for additional details.

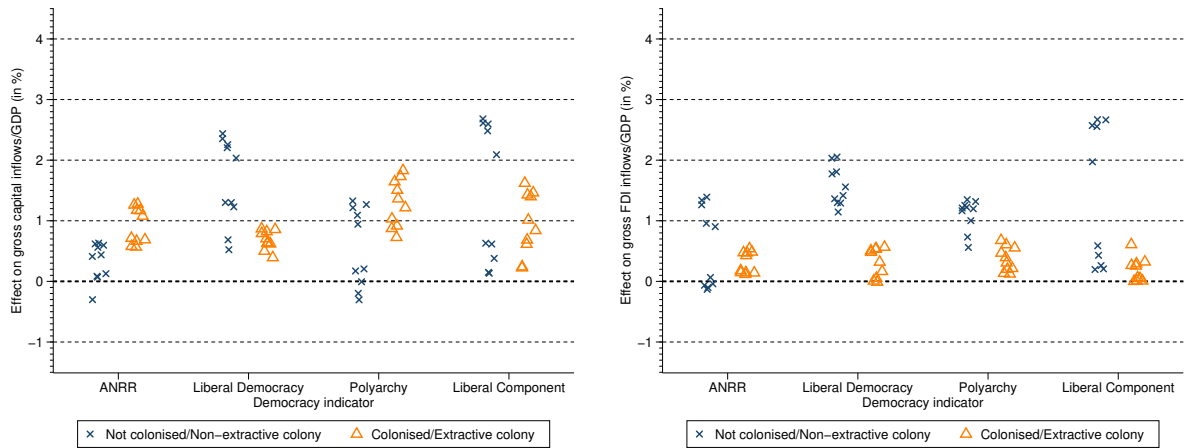
**Figure B-3: Democracy, Colonialism and Capital Inflows**



(a) Total Non-Official Capital Inflows (left) and Total FDI Inflows (right) – no controls



(b) Dto – exports/trade as additional control (not a bad control)



(c) Dto – exports/trade, pop growth and GDPpc growth as controls (bad controls)

**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by Colonial History (x for no colonial history or non-extractive colonialism and  $\triangle$  for colonial history or extractive colonialism), using four different definitions of democratic regime change. See Figure 3 for additional details.

## B.5 Analysis using per capita capital flow definition

**Table B-5:** Diagnostic Tests — PCDID Capital Flow (per capita) Analysis

| Democracy Indicator                                             |                   | ANRR |      | LibDem |      | Poly |      | Liberal |      | N  |
|-----------------------------------------------------------------|-------------------|------|------|--------|------|------|------|---------|------|----|
| Deep Determinant Group                                          |                   | 0    | 1    | 0      | 1    | 0    | 1    | 0       | 1    |    |
| <b>Panel A: Total Non-Official Capital Inflows</b>              |                   |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 0.33 | 0.67 | 0.33   | 0.83 | 0.50 | 1.00 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.25 | 1.00 | 0.75   | 0.50 | 0.75 | 0.75 | 0.50    | 0.00 | 20 |
| <i>Controls: export/trade</i>                                   |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 0.33 | 0.67 | 0.33   | 1.00 | 0.50 | 1.00 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.50 | 1.00 | 0.75   | 0.50 | 0.75 | 0.75 | 0.25    | 0.00 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.07 | 0.13 | 0.00   | 0.07 | 0.00 | 0.13 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 | 0.00 | 0.00    | 0.00 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 0.33 | 1.00 | 0.33   | 1.00 | 0.50 | 1.00 | 0.00    | 0.17 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.50 | 1.00 | 0.50   | 0.75 | 0.75 | 0.75 | 0.00    | 0.00 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.60 | 1.00 | 0.53   | 0.90 | 0.80 | 0.93 | 1.00    | 1.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.70 | 0.95 | 0.90   | 0.40 | 0.90 | 0.55 | 1.00    | 0.75 | 20 |
| <b>Panel B: FDI Inflows</b>                                     |                   |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 0.33 | 0.83 | 0.33   | 1.00 | 0.50 | 1.00 | 0.00    | 0.17 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.50 | 0.75 | 0.50   | 0.75 | 0.50 | 0.75 | 0.25    | 0.25 | 20 |
| <i>Controls: export/trade</i>                                   |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 0.33 | 0.83 | 0.33   | 1.00 | 0.50 | 1.00 | 0.00    | 0.17 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.25 | 0.75 | 0.50   | 0.75 | 0.75 | 0.75 | 0.00    | 0.25 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.00 | 0.00 | 0.00   | 0.07 | 0.00 | 0.00 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.05 | 0.00 | 0.00   | 0.00 | 0.20 | 0.00 | 0.00    | 0.00 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 0.00 | 1.00 | 0.33   | 1.00 | 0.50 | 1.00 | 0.00    | 0.17 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.25 | 0.75 | 0.75   | 0.75 | 0.75 | 0.75 | 0.00    | 0.25 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.17 | 0.83 | 0.33   | 1.00 | 0.77 | 1.00 | 1.00    | 1.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.35 | 0.70 | 0.15   | 0.50 | 1.00 | 0.75 | 1.00    | 0.85 | 20 |

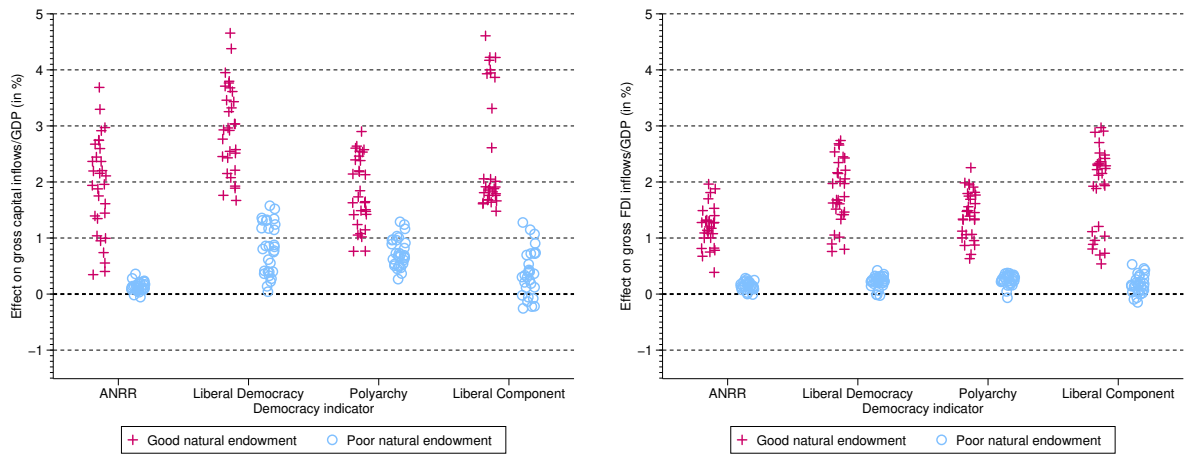
Notes: The table reports the rejection frequency across the 20 or 30 models analysed for each cell (and presented in Figures B-4 and B-5 below). The 'Alpha' test is for weak parallel trends, so if the null hypothesis is rejected the PCDID specification may be misspecified: we want to see very low rejection rates, like for some specifications of the 'Liberal Component'. The  $\chi^2$  test is for bad controls, so if the null hypothesis is rejected ( $p < .1$ ) we should not include this (set of) control(s): again, we want to see very low rejection rates, like for the models with export/trade as additional control. There are five alternative factor augmentations and four (alternative deep determinants) or six (geography) proxies for deep determinants, hence 20 or 30 models for each of the two deep determinant groups, respectively. These 20 or 30 models are represented in each estimate 'cloud' of the aforementioned figures.

**Table B-6:** Statistical Significance — PCDID Capital Flow Analysis (per capita)

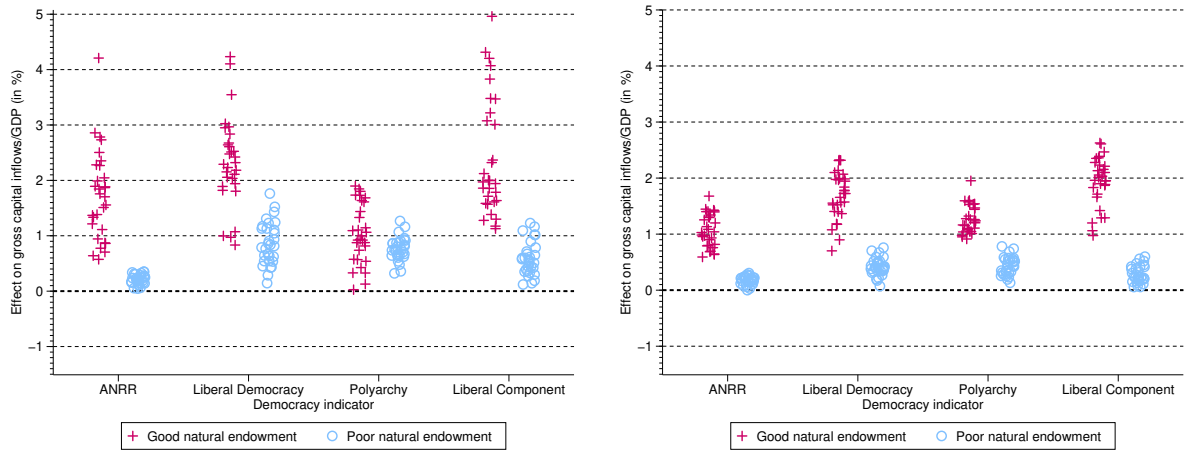
| Democracy Indicator                                             | ANRR |      | LibDem |      | Poly |      | Liberal |      | N  |
|-----------------------------------------------------------------|------|------|--------|------|------|------|---------|------|----|
| Deep Determinant Group                                          | 0    | 1    | 0      | 1    | 0    | 1    | 0       | 1    |    |
| <b>Panel A: Total Non-Official Capital Inflows</b>              |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.83 | 0.27 | 1.00   | 0.83 | 0.90 | 1.00 | 1.00    | 0.70 | 30 |
| Alternative Deep Det                                            | 0.45 | 0.20 | 1.00   | 0.95 | 0.85 | 0.85 | 0.70    | 0.95 | 20 |
| <i>Controls: export/trade</i>                                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.73 | 1.00 | 0.87   | 0.90 | 0.13 | 1.00 | 0.87    | 0.80 | 30 |
| Alternative Deep Det                                            | 0.65 | 0.85 | 0.85   | 0.90 | 0.40 | 0.80 | 0.85    | 1.00 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.00 | 0.07 | 0.70   | 0.80 | 0.47 | 0.50 | 0.70    | 0.87 | 30 |
| Alternative Deep Det                                            | 0.00 | 0.25 | 0.65   | 0.80 | 0.20 | 0.60 | 0.90    | 0.65 | 20 |
| <b>Panel B: FDI Inflows</b>                                     |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 1.00 | 0.80 | 0.97   | 0.60 | 0.97 | 0.80 | 1.00    | 0.33 | 30 |
| Alternative Deep Det                                            | 0.70 | 0.45 | 0.85   | 0.75 | 0.90 | 0.95 | 0.70    | 1.00 | 20 |
| <i>Controls: export/trade</i>                                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.90 | 1.00 | 1.00   | 0.97 | 0.97 | 1.00 | 1.00    | 0.83 | 30 |
| Alternative Deep Det                                            | 0.75 | 0.65 | 0.90   | 1.00 | 0.90 | 1.00 | 0.80    | 1.00 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.70 | 0.07 | 1.00   | 0.93 | 0.97 | 1.00 | 1.00    | 0.90 | 30 |
| Alternative Deep Det                                            | 0.35 | 0.25 | 1.00   | 0.70 | 0.95 | 0.60 | 0.90    | 0.90 | 20 |

Notes: The table reports the rejection frequency across the 20 or 30 models analysed for each cell (and presented in Figures B-4 and B-5 below). These are for the *t*-tests (10% level) of the robust mean PCDID estimates (computed using the non-parametric variance estimator of Pesaran, 2006): if we see very high rejection rates this equates to statistical significance of the ATET presented in the aforementioned figures. There are five alternative factor augmentations and four (alternative deep determinants) or six (geography) proxies for deep determinants, hence 20 or 30 models for each of the two deep determinant groups, respectively. These 20 or 30 models are represented in each estimate ‘cloud’ of the aforementioned figures.

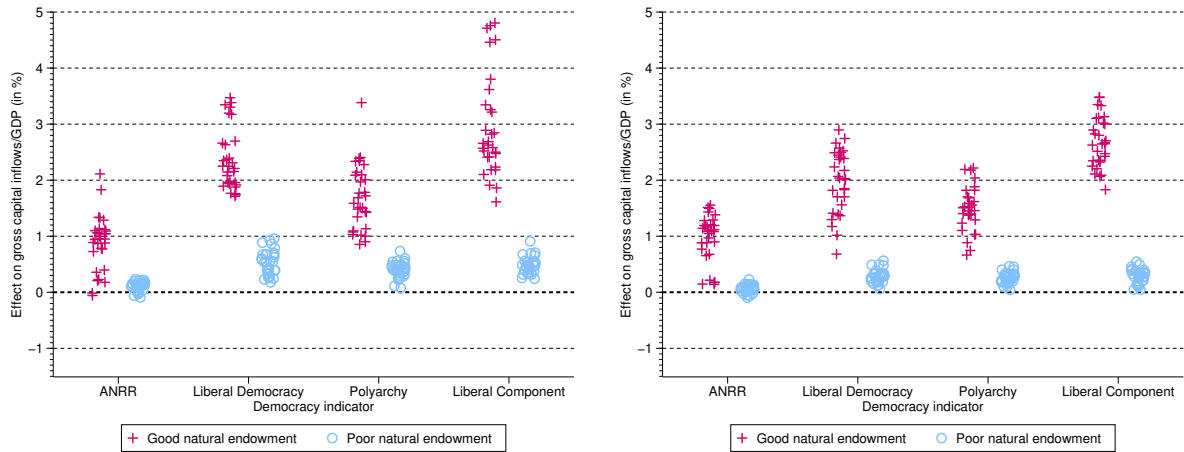
**Figure B-4: Democracy, Geography and Capital Inflows (per capita definition)**



(a) Total Non-Official Capital Inflows (left) and Total FDI Inflows (right) – no controls



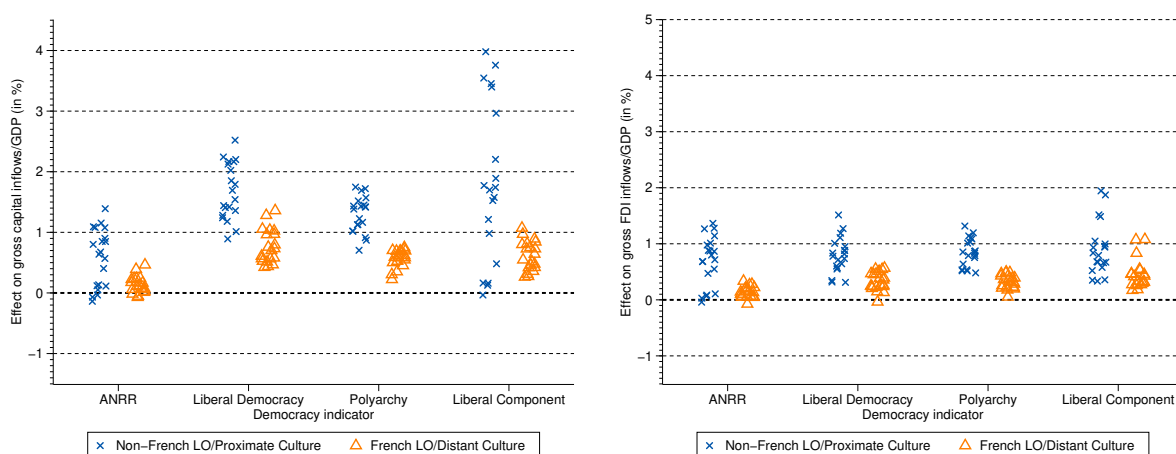
(b) Dto – trade/GDP as additional control



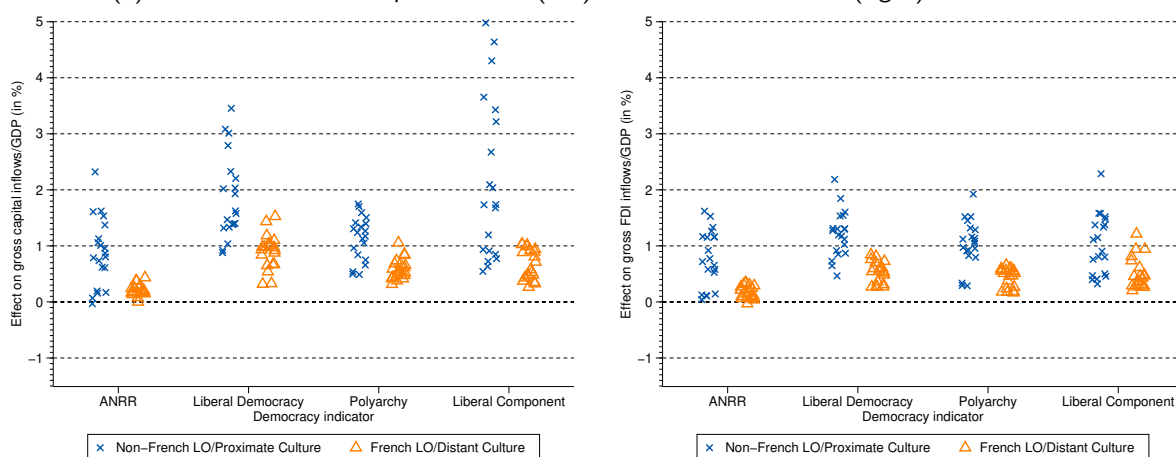
(c) Dto – trade/GDP, pop growth and GDPpc growth as controls

**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by geography (+ and o for good and poor geography, respectively), using four different definitions of democratic regime change. These are the results using per capita capital inflows as dependent variable. See Figure 3 in the maintext for all other details.

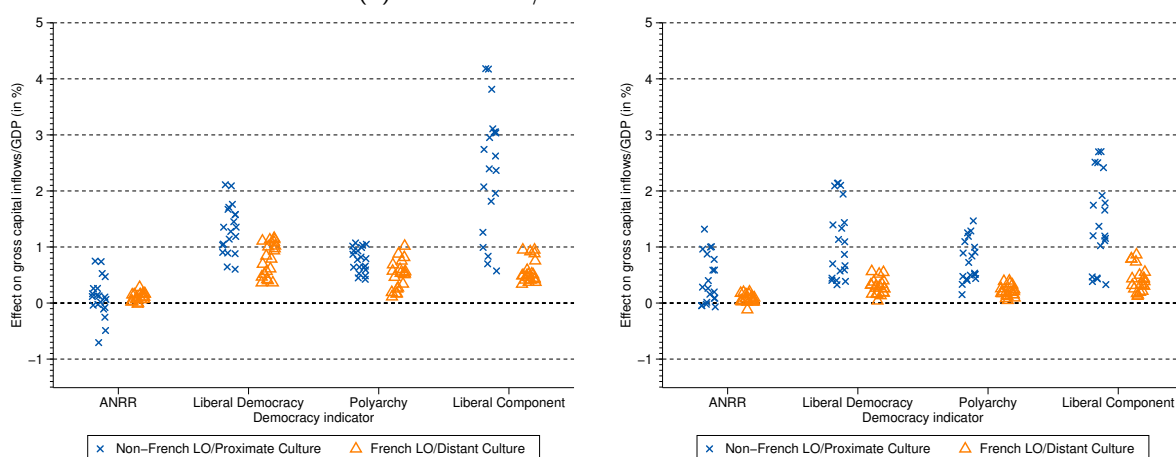
**Figure B-5: Democracy, Alternative Deep Determinants and Capital Inflows (per capita definition)**



(a) Total Non-Official Capital Inflows (left) and Total FDI Inflows (right) – no controls



(b) Dto – trade/GDP as additional control



(c) Dto – trade/GDP, pop growth and GDPpc growth as controls

**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by Legal Origin and Culture (x for non-French LO/proximate culture and  $\Delta$  for French LO/distant culture), using four different definitions of democratic regime change. These are the results using per capita definitions of capital flow measures. See Figure B-2 in the maintext for all other details.

## B.6 Capital flow analysis using WDI data and reduced sample size

**Table B-7:** Diagnostic Tests — PCDID Capital Flow (WDI data) Analysis

| Democracy Indicator                                             |                   | ANRR |      | LibDem |      | Poly |      | Liberal |      | N  |
|-----------------------------------------------------------------|-------------------|------|------|--------|------|------|------|---------|------|----|
| Deep Determinant Group                                          |                   | 0    | 1    | 0      | 1    | 0    | 1    | 0       | 1    |    |
| <b>Panel A: Total Non-Official Capital Inflows</b>              |                   |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 0.67 | 0.33 | 0.50   | 0.50 | 0.83 | 0.00 | 0.50    | 0.50 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.50 | 0.50 | 0.83   | 0.00 | 0.50 | 0.50 | 0.83    | 1.00 | 20 |
| <i>Controls: export/trade</i>                                   |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 1.00 | 1.00 | 0.50   | 0.67 | 0.33 | 0.00 | 0.50    | 0.83 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.50 | 0.67 | 0.33   | 0.00 | 0.50 | 0.83 | 0.67    | 0.33 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.00 | 0.13 | 0.00   | 0.17 | 0.00 | 0.27 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 | 0.00 | 0.00    | 0.00 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 1.00 | 0.83 | 0.67   | 1.00 | 0.33 | 0.83 | 0.67    | 1.00 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.67 | 1.00 | 0.33   | 0.83 | 0.67 | 1.00 | 1.00    | 1.00 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.67 | 0.93 | 0.33   | 0.93 | 1.00 | 1.00 | 0.83    | 1.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.70 | 1.00 | 0.40   | 0.50 | 0.75 | 0.85 | 1.00    | 1.00 | 20 |
| <b>Panel B: FDI Inflows</b>                                     |                   |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 0.83 | 0.33 | 0.33   | 0.17 | 0.67 | 0.50 | 0.67    | 0.33 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.33 | 0.17 | 0.67   | 0.33 | 0.50 | 0.25 | 0.67    | 0.50 | 20 |
| <i>Controls: export/trade</i>                                   |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 1.00 | 0.33 | 0.17   | 0.00 | 0.83 | 0.50 | 0.33    | 0.17 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.17 | 0.00 | 0.33   | 0.17 | 0.83 | 0.33 | 0.83    | 0.50 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.07 | 0.13 | 0.00   | 0.17 | 0.00 | 0.17 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.05 | 0.00 | 0.05   | 0.00 | 0.00 | 0.00 | 0.00    | 0.00 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha t > 1.96    | 1.00 | 0.50 | 0.67   | 0.33 | 0.83 | 0.83 | 0.83    | 0.33 | 30 |
| Alternative Deep Det                                            | Alpha t > 1.96    | 0.67 | 0.33 | 0.83   | 0.33 | 1.00 | 0.33 | 0.83    | 0.83 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.30 | 0.83 | 0.37   | 1.00 | 0.80 | 1.00 | 1.00    | 1.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.45 | 0.75 | 0.35   | 0.65 | 0.85 | 1.00 | 1.00    | 1.00 | 20 |

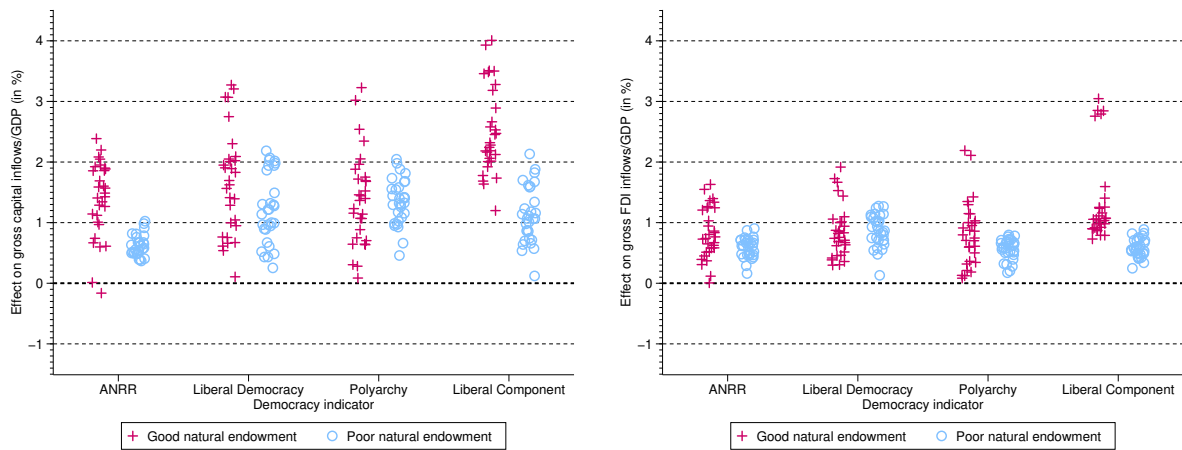
Notes: The table reports the rejection frequency across the 20 or 30 models analysed for each cell (and presented in Figures B-6 and B-7 below). The 'Alpha' test is for weak parallel trends, so if the null hypothesis is rejected the PCDID specification may be misspecified: we want to see very low rejection rates, like for some specifications of the 'Liberal Component'. The  $\chi^2$  test is for bad controls, so if the null hypothesis is rejected ( $p < .1$ ) we should not include this (set of) control(s): again, we want to see very low rejection rates, like for the models with export/trade as additional control. There are five alternative factor augmentations and four (alternative deep determinants) or six (geography) proxies for deep determinants, hence 20 or 30 models for each of the two deep determinant groups, respectively. These 20 or 30 models are represented in each estimate 'cloud' of the aforementioned figures.

**Table B-8:** Statistical Significance — PCDID Capital Flow Analysis (WDI data)

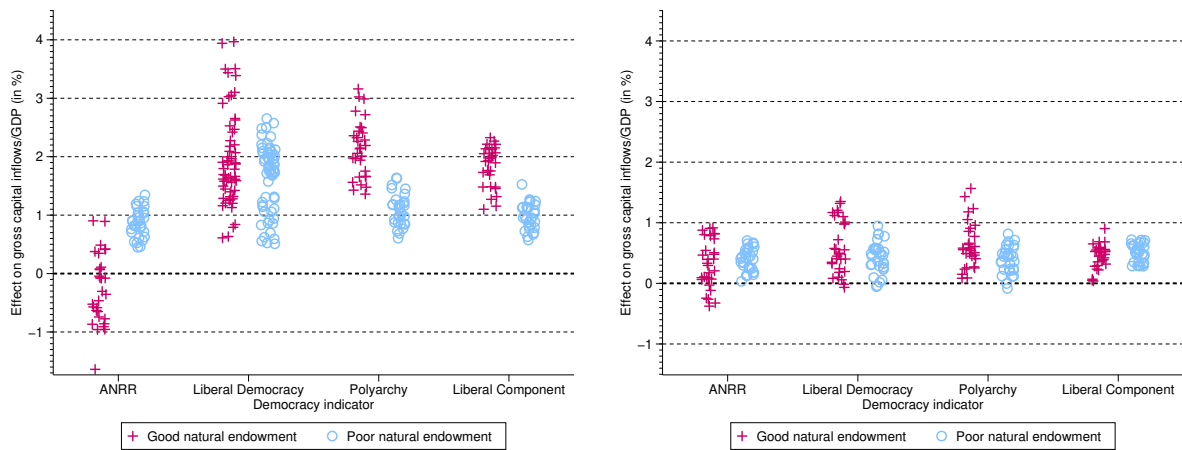
| Democracy Indicator                                             | ANRR |      | LibDem |      | Poly |      | Liberal |      | N  |
|-----------------------------------------------------------------|------|------|--------|------|------|------|---------|------|----|
| Deep Determinant Group                                          | 0    | 1    | 0      | 1    | 0    | 1    | 0       | 1    |    |
| <b>Panel A: Total Non-Official Capital Inflows</b>              |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.33 | 0.33 | 0.57   | 0.43 | 0.40 | 0.93 | 0.93    | 0.73 | 30 |
| Alternative Deep Det                                            | 0.20 | 0.35 | 0.75   | 0.60 | 0.75 | 0.40 | 0.75    | 0.95 | 20 |
| <i>Controls: export/trade</i>                                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.00 | 0.77 | 0.57   | 0.73 | 0.87 | 0.83 | 0.73    | 0.80 | 30 |
| Alternative Deep Det                                            | 0.00 | 0.65 | 0.75   | 0.65 | 0.45 | 0.75 | 0.70    | 0.95 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.00 | 0.07 | 0.30   | 0.10 | 0.80 | 0.00 | 0.17    | 0.00 | 30 |
| Alternative Deep Det                                            | 0.00 | 0.05 | 0.45   | 0.05 | 0.50 | 0.05 | 0.20    | 0.05 | 20 |
| <b>Panel B: FDI Inflows</b>                                     |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.60 | 0.97 | 0.57   | 0.83 | 0.40 | 0.63 | 0.80    | 0.83 | 30 |
| Alternative Deep Det                                            | 0.50 | 0.90 | 0.45   | 0.50 | 0.35 | 0.55 | 0.30    | 0.85 | 20 |
| <i>Controls: export/trade</i>                                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.10 | 0.70 | 0.30   | 0.13 | 0.23 | 0.27 | 0.03    | 0.70 | 30 |
| Alternative Deep Det                                            | 0.15 | 0.90 | 0.15   | 0.45 | 0.00 | 0.20 | 0.25    | 0.60 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |      |      |        |      |      |      |         |      |    |
| Geography                                                       | 0.00 | 0.13 | 0.43   | 0.20 | 0.33 | 0.00 | 0.03    | 0.00 | 30 |
| Alternative Deep Det                                            | 0.05 | 0.15 | 0.60   | 0.25 | 0.00 | 0.00 | 0.15    | 0.10 | 20 |

Notes: The table reports the rejection frequency across the 20 or 30 models analysed for each cell (and presented in Figures B-6 and B-7 below). These are for the  $t$ -tests (10% level) of the robust mean PCDID estimates (computed using the non-parametric variance estimator of Pesaran, 2006): if we see very high rejection rates this equates to statistical significance of the ATET presented in the aforementioned figures. There are five alternative factor augmentations and four (alternative deep determinants) or six (geography) proxies for deep determinants, hence 20 or 30 models for each of the two deep determinant groups, respectively. These 20 or 30 models are represented in each estimate ‘cloud’ of the aforementioned figures.

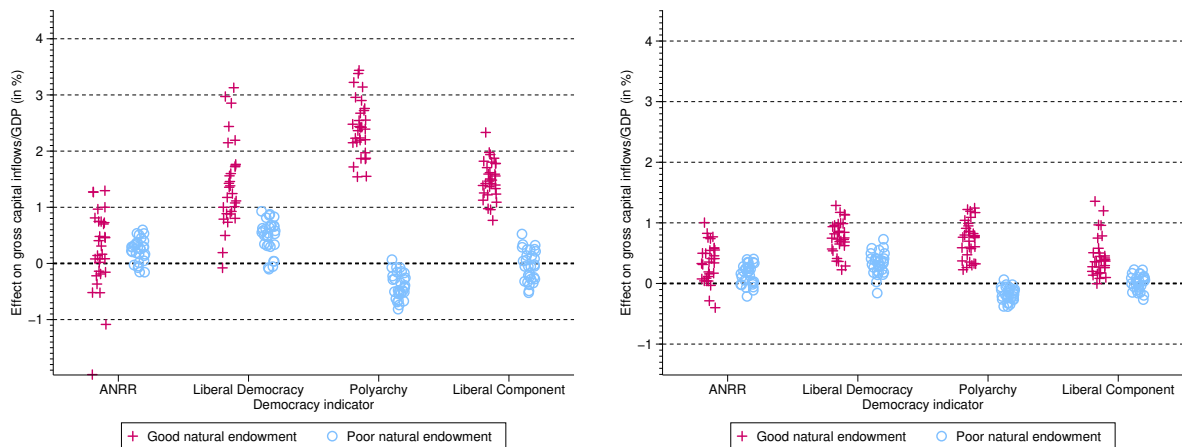
**Figure B-6: Democracy, Geography and Capital Inflows (WDI)**



(a) Total Non-Official Capital Inflows (left) and Total FDI Inflows (right) – no controls



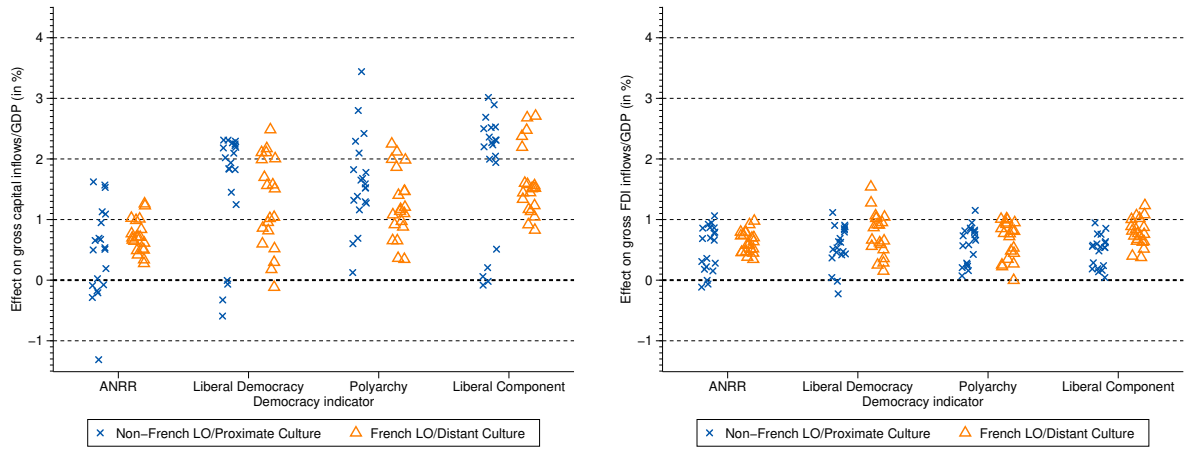
(b) Dto – trade/GDP as additional control



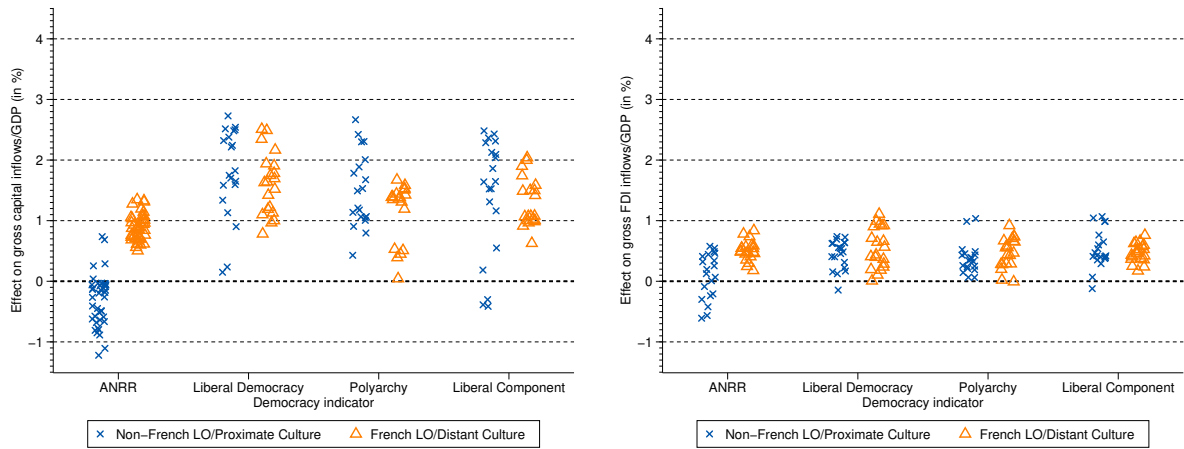
(c) Dto – trade/GDP, pop growth and GDPpc growth as controls

**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by geography (+ and o for good and poor geography, respectively), using four different definitions of democratic regime change. These are the results using WDI data/coverage, which reduces the overall sample (treated, controls, discarded 'always' democracies) by approximately 15 percent, and the treated sample by around 30 percent. See Figure 3 in the maintext for all other details.

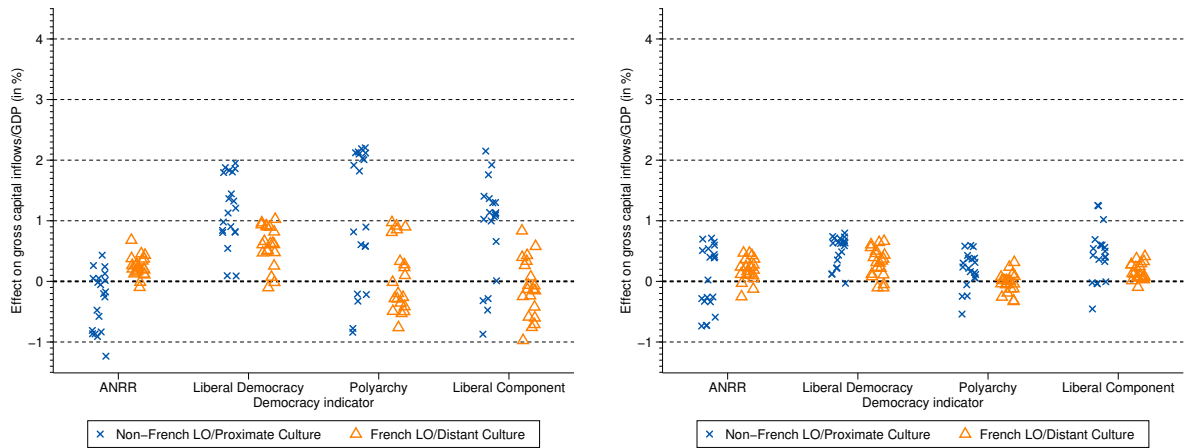
**Figure B-7: Democracy, Alternative Deep Determinants and Capital Inflows (WDI)**



(a) Total Non-Official Capital Inflows (left) and Total FDI Inflows (right) – no controls



(b) Dto – trade/GDP as additional control

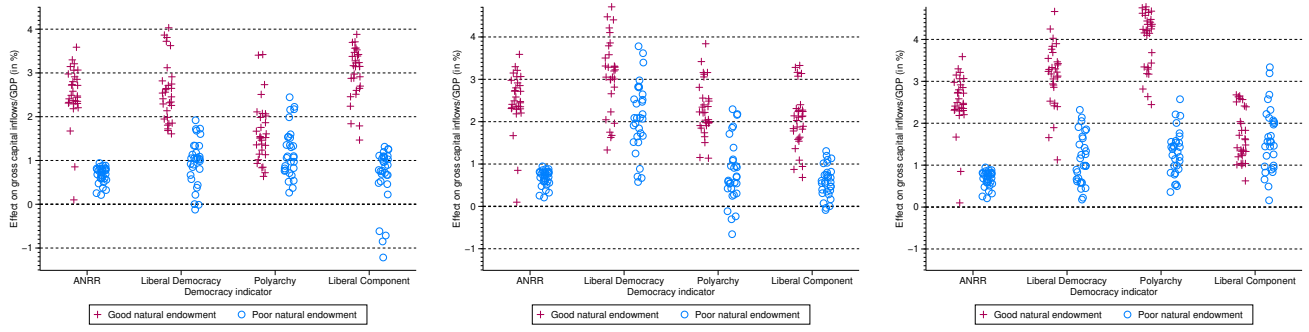


(c) Dto – trade/GDP, pop growth and GDPpc growth as controls

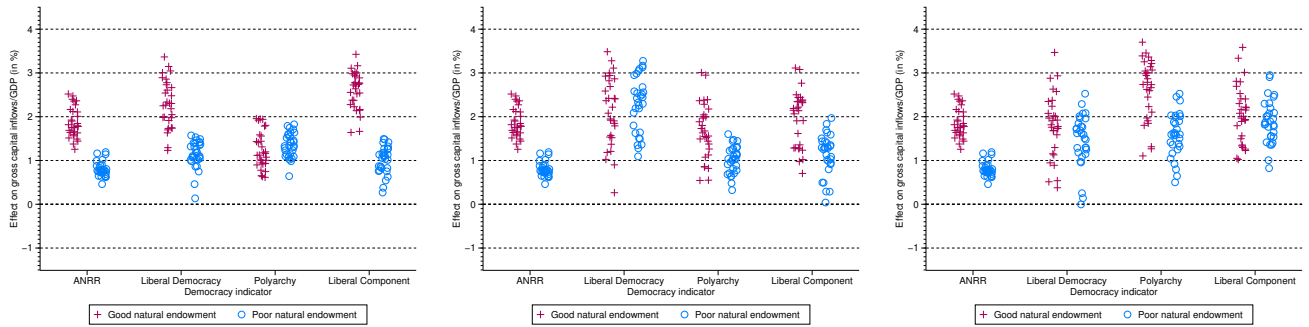
**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by Legal Origin and Culture (x for non-French LO/proximate culture and  $\triangle$  for French LO/distant culture), using four different definitions of democratic regime change. These are the results using WDI data/coverage, which reduces the overall sample (treated, controls, discarded 'always' democracies) by approximately 15 percent, and the treated sample by around 30 percent. See Figure B-2 in the maintext for all other details.

## B.7 Capital flow analysis using tighter democracy thresholds (V-Dem data)

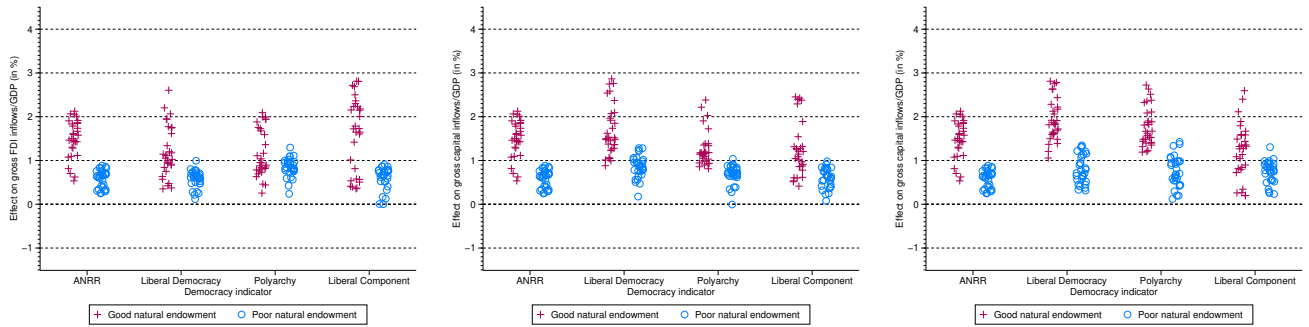
**Figure B-8: Democracy, Geography and Total Capital Inflows (a,b) or FDI (c,d)**



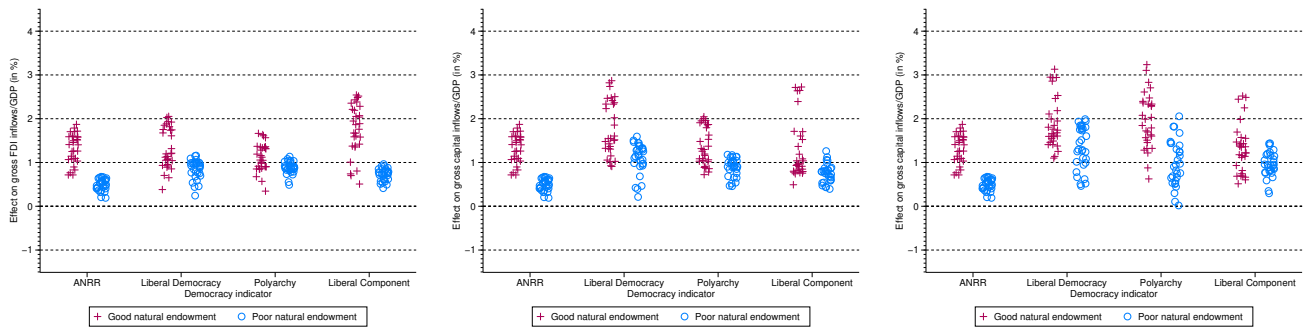
(a) Total Inflows: Benchmark (left), Democracy mean+1/4 SD (middle), and mean+1/2 SD – no controls



(b) Dto – exports/trade as control



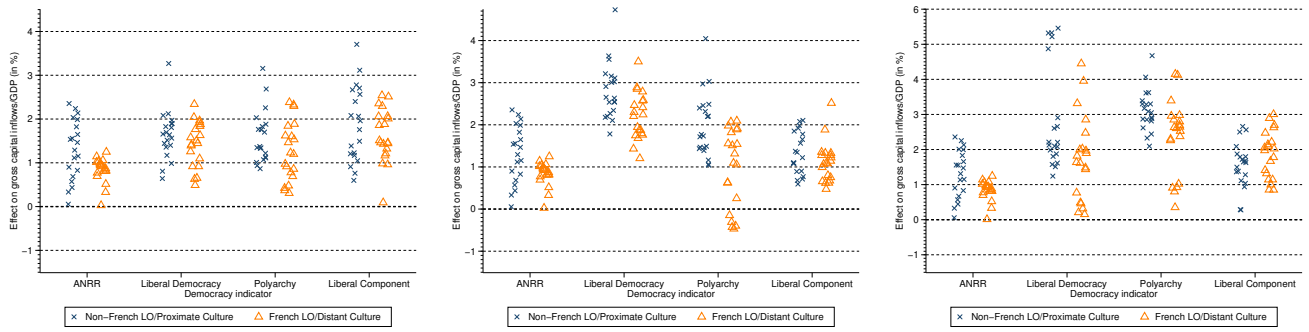
(c) FDI: Benchmark (left), Democracy mean+1/4 SD (middle), and mean+1/2 SD – no controls



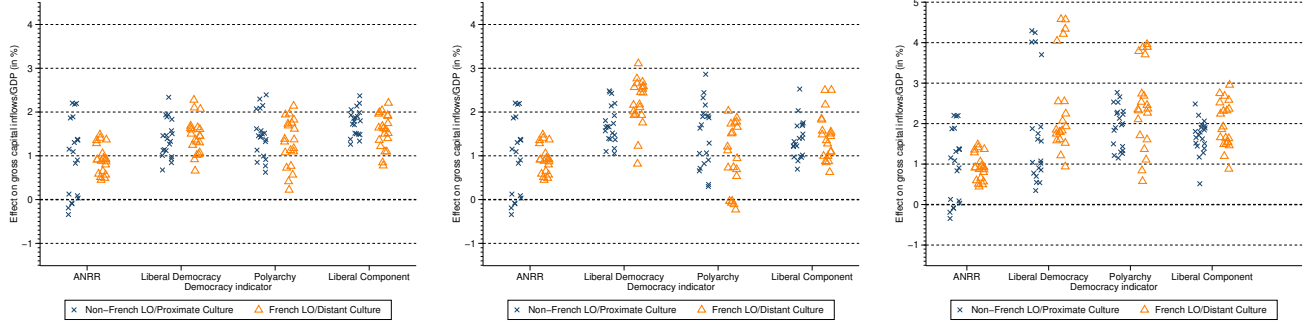
(d) Dto – exports/trade as control

**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by geography (+ and o for good and poor geography, respectively), using four different definitions of democratic regime change. Compared with the benchmark results in Figure 3 in the maintext – replicated in the left plot of each panel – we use tighter definitions for the V-Dem democracy dummies: in the middle (right) plot the threshold is defined as the mean plus 1/4 (1/2) of 1 SD. Panels (a) and (b) are for total inflows/GDP, without controls and with export/trade as controls; similarly for Panels (c) and (d), which are for FDI/GDP. Alpha tests are passed in 90+% of specifications with adjusted democracy definition, in 95+% of specifications in panels (b) and (d) exports/trade is not a ‘bad control’.

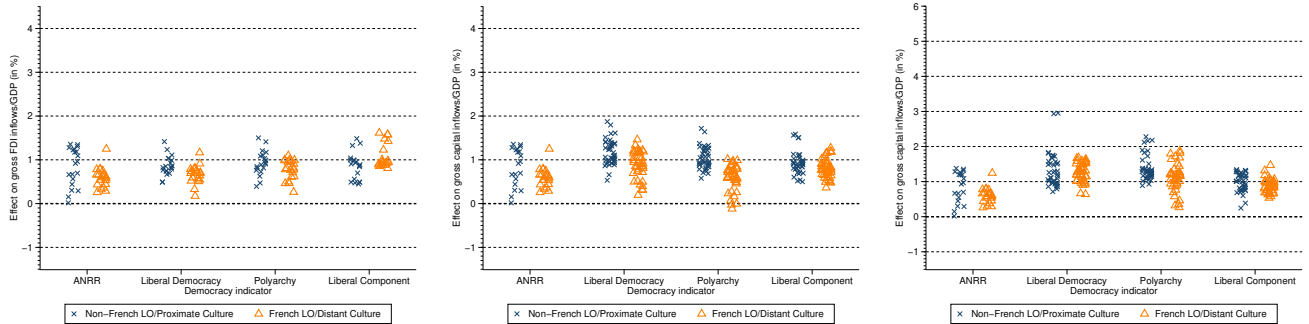
**Figure B-9: Democracy, Alternative Deep Determinants and Total Capital Inflows (a,b) or FDI (c,d)**



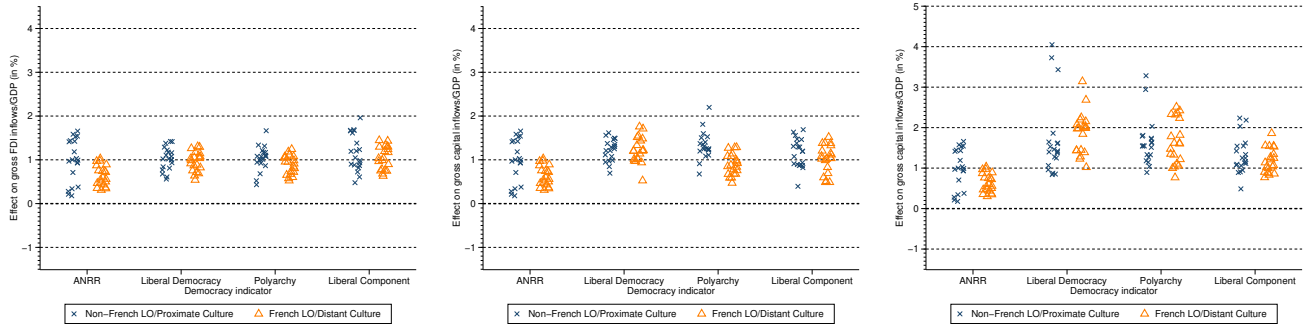
(a) Total Inflows: Benchmark (left), Democracy mean+1/4 SD (middle), and mean+1/2 SD – no controls



(b) Dto – exports/trade as control



(c) FDI: Benchmark (left), Democracy mean+1/4 SD (middle), and mean+1/2 SD – no controls

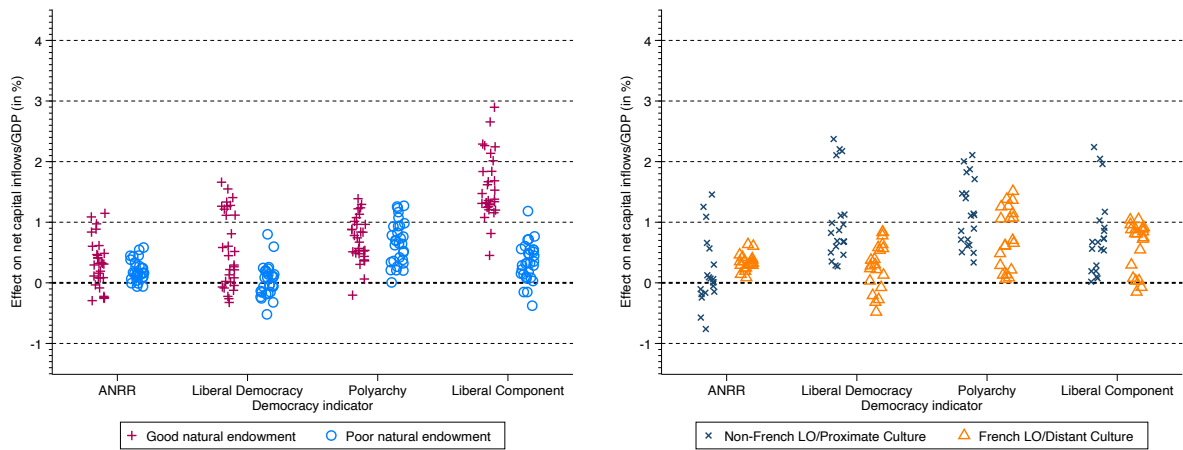


(d) Dto – exports/trade as control

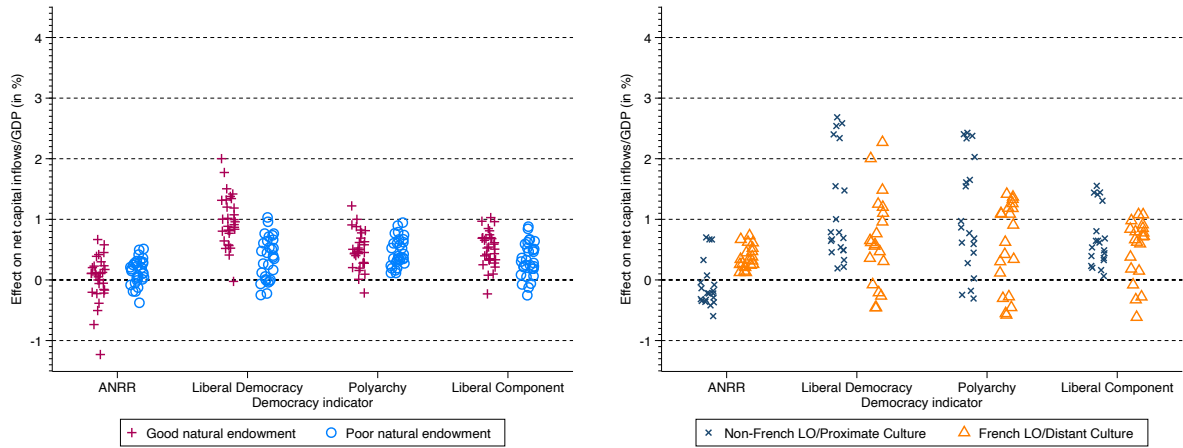
**Notes:** The plots present robust ATET (Mean Group PCDID) estimates for the causal effect of democracy on capital inflows by Legal Origin and Culture (x for non-French LO/proximate culture and  $\triangle$  for French LO/distant culture), using four different definitions of democratic regime change. See Figure B-8 for all other details.

## B.8 Capital flow analysis using net (non-official) capital inflows

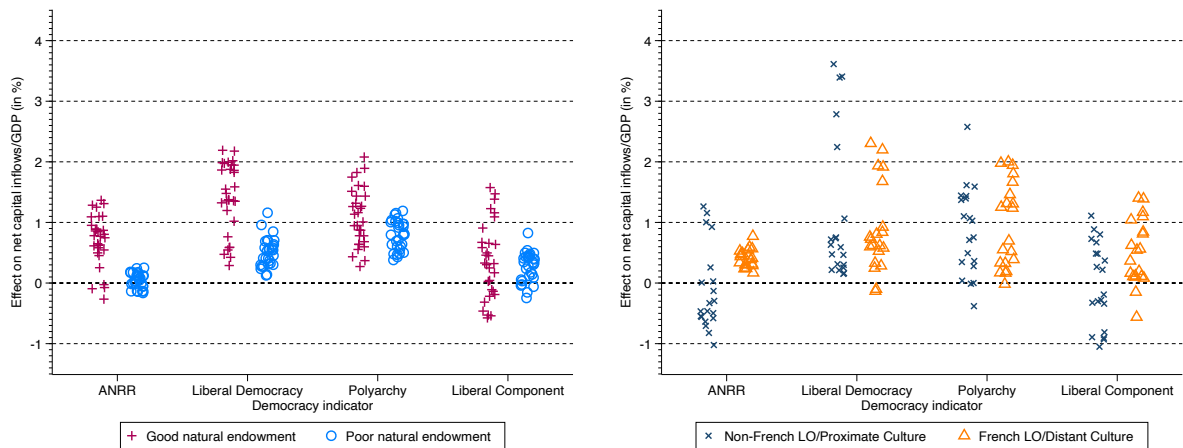
**Figure B-10: Democracy and Net (Non-Official) Capital Inflows**



(a) Geography (left) and alternative deep determinants (right) – no controls



(b) Dto – trade/GDP as additional control



(c) Dto – trade/GDP, pop growth and GDPpc growth as controls

**Notes:** The plots present robust ATET (Mean Group PCIDID) estimates for the causal effect of democracy on net non-official capital inflows by geography (left column) and alternative deep determinants (right column).

**Table B-9:** Diagnostic Tests — Net (Non-official) capital flows

| Democracy Indicator                                             |                   | ANRR |      | LibDem |      | Poly |      | Liberal |      | N  |
|-----------------------------------------------------------------|-------------------|------|------|--------|------|------|------|---------|------|----|
| Deep Determinant Group                                          |                   | 0    | 1    | 0      | 1    | 0    | 1    | 0       | 1    |    |
| <b>Net Non-Official Capital Inflows</b>                         |                   |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 0.83 | 1.00 | 1.00   | 0.33 | 1.00 | 0.00 | 1.00    | 0.67 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 1.00 | 1.00 | 0.50   | 0.50 | 0.75 | 0.25 | 0.75    | 1.00 | 20 |
| <i>Controls: export/trade</i>                                   |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 1.00 | 1.00 | 1.00   | 0.17 | 0.83 | 0.33 | 1.00    | 0.33 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 1.00 | 0.75 | 0.75   | 0.25 | 0.50 | 0.00 | 1.00    | 0.75 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.00 | 0.03 | 0.00   | 0.00 | 0.00 | 0.00 | 0.00    | 0.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.00 | 0.00 | 0.10   | 0.00 | 0.00 | 0.10 | 0.00    | 0.00 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |                   |      |      |        |      |      |      |         |      |    |
| Geography                                                       | Alpha $t > 1.96$  | 1.00 | 1.00 | 1.00   | 0.67 | 1.00 | 0.50 | 1.00    | 1.00 | 30 |
| Alternative Deep Det                                            | Alpha $t > 1.96$  | 1.00 | 0.75 | 1.00   | 0.75 | 1.00 | 0.50 | 0.75    | 1.00 | 20 |
| Geography                                                       | $\chi^2(p) < 0.1$ | 0.47 | 0.73 | 0.17   | 1.00 | 1.00 | 1.00 | 0.80    | 1.00 | 30 |
| Alternative Deep Det                                            | $\chi^2(p) < 0.1$ | 0.45 | 0.85 | 0.50   | 0.60 | 1.00 | 1.00 | 0.95    | 1.00 | 20 |

Notes: The table reports the rejection frequency across the 20 or 30 models analysed for each cell. The ‘Alpha’ test is for weak parallel trends, so if the null hypothesis is rejected the PCDID specification may be misspecified: we want to see *very low* rejection rates. The  $\chi^2$  test is for bad controls, so if the null hypothesis is rejected ( $p < .1$ ) we should not include this (set of) control(s): again, we want to see *very low* rejection rates,. There are five alternative factor augmentations and four (alternative deep determinants) or six (geography) proxies for deep determinants, hence 20 or 30 models for each of the two deep determinant groups, respectively.

**Table B-10:** Statistical Significance — PCDID Net Capital Flow Analysis

| Democracy Indicator                                             |  | ANRR |      | LibDem |      | Poly |      | Liberal |      | N  |
|-----------------------------------------------------------------|--|------|------|--------|------|------|------|---------|------|----|
| Deep Determinant Group                                          |  | 0    | 1    | 0      | 1    | 0    | 1    | 0       | 1    |    |
| <b>Net (Non-Official) Capital Inflows</b>                       |  |      |      |        |      |      |      |         |      |    |
| <i>Controls: none</i>                                           |  |      |      |        |      |      |      |         |      |    |
| Geography                                                       |  | 0.00 | 0.00 | 0.00   | 0.00 | 0.50 | 0.00 | 0.03    | 0.13 | 30 |
| Alternative Deep Det                                            |  | 0.05 | 0.00 | 0.20   | 0.00 | 0.20 | 0.00 | 0.25    | 0.25 | 20 |
| <i>Controls: export/trade</i>                                   |  |      |      |        |      |      |      |         |      |    |
| Geography                                                       |  | 0.00 | 0.00 | 0.03   | 0.00 | 0.00 | 0.00 | 0.00    | 0.03 | 30 |
| Alternative Deep Det                                            |  | 0.00 | 0.00 | 0.25   | 0.10 | 0.25 | 0.10 | 0.25    | 0.15 | 20 |
| <i>Controls: export/trade, GDP pc growth, population growth</i> |  |      |      |        |      |      |      |         |      |    |
| Geography                                                       |  | 0.00 | 0.00 | 0.00   | 0.07 | 0.00 | 0.00 | 0.00    | 0.13 | 30 |
| Alternative Deep Det                                            |  | 0.05 | 0.00 | 0.20   | 0.25 | 0.00 | 0.15 | 0.05    | 0.40 | 20 |

Notes: The table reports the rejection frequency across the 20 or 30 models analysed for each cell. These are for the  $t$ -tests (10% level) of the robust mean PCDID estimates (computed using the non-parametric variance estimator of Pesaran, 2006): if we see *very high* rejection rates this equates to statistical significance of the ATET. There are five alternative factor augmentations and four (alternative deep determinants) or six (geography) proxies for deep determinants, hence 20 or 30 models for each of the two deep determinant groups, respectively.

## B.9 Capital flow analysis using alternative geography proxies

**Table B-11: Total Capital Inflows – Alternative Proxies of Geography (1975-2015)**

|                                                | (1)                 | (2)              | (3)                 | (4)                | (5)                 | (6)                | (7)                       | (8)                 |
|------------------------------------------------|---------------------|------------------|---------------------|--------------------|---------------------|--------------------|---------------------------|---------------------|
|                                                | <b>Landlocked</b>   |                  | <b>High UV</b>      | <b>Radiation</b>   | <b>Few Frost</b>    | <b>Days</b>        | <b>Low Ag-Suitability</b> |                     |
|                                                | No                  | Yes              | No                  | Yes                | No                  | Yes                | No                        | Yes                 |
| <b>ANRR democracy</b>                          | 1.286***<br>[0.390] | 0.129<br>[0.700] | 1.465*<br>[0.799]   | 0.597*<br>[0.339]  | 1.915*<br>[1.039]   | 0.589*<br>[0.333]  | 1.022**<br>[0.465]        | 0.461<br>[0.495]    |
| Treated Countries                              | 50                  | 18               | 23                  | 42                 | 21                  | 44                 | 37                        | 28                  |
| Treated Observations                           | 1564                | 500              | 654                 | 1329               | 582                 | 1401               | 1140                      | 843                 |
| Control Countries                              | 22                  | 7                | 15                  | 14                 | 15                  | 14                 | 8                         | 21                  |
| Control Observations                           | 663                 | 128              | 380                 | 397                | 390                 | 387                | 177                       | 600                 |
| $\chi^2$ test ( $p$ )                          | 0.78                | 0.02             | 0.45                | 0.30               | 0.61                | 0.65               | 0.78                      | 0.73                |
| Alpha test ( $t$ )                             | -0.06               | -7.23            | -0.66               | -2.04              | -0.50               | -3.07              | -1.42                     | -2.20               |
| <i>Alternative factor augmentation</i>         |                     |                  |                     |                    |                     |                    |                           |                     |
| 2 factors                                      | 1.262***            | -0.012           | 1.228*              | 0.573              | 1.715*              | 0.620*             | 1.135**                   | 0.649               |
| 3 factors                                      | 1.324***            | 0.063            | 1.568**             | 0.631*             | 1.803**             | 0.617*             | 1.021**                   | 0.521               |
| 4 factors                                      | 1.286***            | 0.129            | 1.465*              | 0.597*             | 1.915*              | 0.589*             | 1.022**                   | 0.461               |
| 5 factors                                      | 1.112***            | 0.186            | 1.909**             | 0.291              | 1.984**             | 0.599              | 0.683                     | 0.500               |
| 6 factors                                      | 1.137***            | 0.186            | 1.967**             | 0.441              | 1.735*              | 0.704**            | 0.531                     | 0.664               |
| <b>Liberal Democracy<br/>Index &gt; median</b> | 1.617***<br>[0.490] | 0.848<br>[0.913] | 2.321***<br>[0.860] | 0.794<br>[0.569]   | 2.786***<br>[1.078] | 0.827<br>[0.631]   | 1.418***<br>[0.495]       | 1.693**<br>[0.673]  |
| Treated Countries                              | 37                  | 13               | 21                  | 27                 | 21                  | 27                 | 32                        | 16                  |
| Treated Observations                           | 1380                | 428              | 704                 | 1015               | 678                 | 1041               | 1117                      | 602                 |
| Control Countries                              | 42                  | 14               | 19                  | 33                 | 18                  | 34                 | 19                        | 33                  |
| Control Observations                           | 1405                | 346              | 547                 | 1086               | 517                 | 1116               | 574                       | 1059                |
| $\chi^2$ test ( $p$ )                          | 0.62                | 0.06             | 0.97                | 0.15               | 0.81                | 0.40               | 0.63                      | 0.53                |
| Alpha test ( $t$ )                             | -1.17               | -1.32            | 0.73                | 0.51               | 0.34                | 0.54               | 0.52                      | 0.52                |
| <i>Alternative factor augmentation</i>         |                     |                  |                     |                    |                     |                    |                           |                     |
| 2 factors                                      | 1.853***            | 1.368**          | 1.667*              | 1.093**            | 1.633*              | 1.023**            | 1.497***                  | 2.063***            |
| 3 factors                                      | 1.714***            | 0.938            | 1.510*              | 0.603              | 2.372**             | 0.878*             | 1.424***                  | 1.679***            |
| 4 factors                                      | 1.617***            | 0.848            | 2.321***            | 0.794              | 2.786***            | 0.827              | 1.418***                  | 1.693**             |
| 5 factors                                      | 1.712***            | 0.910            | 2.499***            | 0.848              | 3.167***            | 0.734              | 1.270***                  | 1.850***            |
| 6 factors                                      | 1.624***            | 1.130            | 2.394**             | 0.500              | 2.678**             | 0.715              | 1.181***                  | 1.519***            |
| <b>Polyarchy<br/>Index &gt; median</b>         | 1.891***<br>[0.433] | 0.126<br>[0.762] | 0.276<br>[0.732]    | 1.064**<br>[0.414] | 1.684<br>[1.052]    | 1.063**<br>[0.463] | 0.903*<br>[0.516]         | 1.610***<br>[0.605] |
| Treated Countries                              | 45                  | 14               | 24                  | 32                 | 23                  | 33                 | 36                        | 20                  |
| Treated Observations                           | 1665                | 451              | 800                 | 1209               | 734                 | 1275               | 1256                      | 753                 |
| Control Countries                              | 34                  | 13               | 15                  | 28                 | 15                  | 28                 | 14                        | 29                  |
| Control Observations                           | 1120                | 323              | 433                 | 892                | 443                 | 882                | 417                       | 908                 |
| $\chi^2$ test ( $p$ )                          | 0.38                | 0.32             | 1.00                | 0.14               | 0.88                | 0.22               | 0.45                      | 0.79                |
| Alpha test ( $t$ )                             | -3.12               | -1.91            | -0.94               | 0.39               | -1.04               | 0.23               | -0.73                     | -1.62               |
| <i>Alternative factor augmentation</i>         |                     |                  |                     |                    |                     |                    |                           |                     |
| 2 factors                                      | 1.858***            | 0.922            | 0.716               | 1.471***           | 1.063               | 1.361***           | 1.018*                    | 1.685**             |
| 3 factors                                      | 1.541***            | 0.704            | 0.942               | 1.243***           | 1.446               | 1.071**            | 0.881*                    | 1.577**             |
| 4 factors                                      | 1.891***            | 0.126            | 0.276               | 1.064**            | 1.684               | 1.063**            | 0.903*                    | 1.610***            |
| 5 factors                                      | 1.830***            | 0.232            | 0.574               | 1.079**            | 1.513               | 0.700*             | 0.939*                    | 1.846***            |
| 6 factors                                      | 1.743***            | 0.279            | 0.976               | 1.143***           | 1.601               | 0.672*             | 0.954                     | 1.384**             |
| <b>Liberal Component<br/>Index &gt; median</b> | 1.815***<br>[0.669] | 0.574<br>[0.907] | 2.375***<br>[0.872] | 0.816<br>[0.531]   | 2.750***<br>[0.925] | 0.881*<br>[0.534]  | 1.102*<br>[0.668]         | 2.004***<br>[0.567] |
| Treated Countries                              | 43                  | 13               | 24                  | 30                 | 22                  | 32                 | 30                        | 24                  |
| Treated Observations                           | 1606                | 449              | 820                 | 1138               | 745                 | 1221               | 1069                      | 897                 |
| Control Countries                              | 31                  | 12               | 15                  | 27                 | 15                  | 27                 | 18                        | 24                  |
| Control Observations                           | 1000                | 287              | 401                 | 867                | 412                 | 856                | 544                       | 724                 |
| $\chi^2$ test ( $p$ )                          | 0.61                | 0.01             | 0.33                | 0.83               | 0.41                | 0.84               | 0.48                      | 0.92                |
| Alpha test ( $t$ )                             | 0.59                | -2.81            | 0.02                | 0.67               | -0.21               | 0.62               | 0.67                      | -0.86               |
| <i>Alternative factor augmentation</i>         |                     |                  |                     |                    |                     |                    |                           |                     |
| 2 factors                                      | 2.241***            | 1.221            | 2.172**             | 1.118**            | 2.456***            | 1.266**            | 1.551**                   | 2.051***            |
| 3 factors                                      | 1.877***            | 0.676            | 2.550***            | 0.727              | 2.830***            | 0.93               | 1.192*                    | 2.097***            |
| 4 factors                                      | 1.815***            | 0.574            | 2.375***            | 0.816              | 2.750***            | 0.881*             | 1.102*                    | 2.004***            |
| 5 factors                                      | 1.803***            | 0.53             | 2.126**             | 0.803              | 2.640***            | 0.901              | 1.177*                    | 2.273***            |
| 6 factors                                      | 1.726***            | 0.624            | 2.240**             | 0.014              | 2.843***            | 0.218              | 1.032*                    | 1.522***            |

Notes: This table presents the analysis for a number of alternative proxies for geography — whether a country is landlocked, high UV radiation exposure, low number of frost days per year, and low suitability for agriculture. These results are for the model with export/trade as additional control.

## C Analysis of Excessive Capital Inflows

## D Illustrations of Excessive Capital Inflows

Figure C-1 illustrates how the bonanza and surge indicators work and the patterns they create (adopting total capital inflows): the plots for the bonanza indicators in the left column of the Figure ignore global circumstances and focus on *within-country* 'normal' capital inflows (the smoothed line in grey) and significant deviations (the blue and green dashed lines and markers for bonanza years). This means that the entire dataset only features 175 bonanzas when we adopt a definition of 1 standard deviation(s) from the trend, an unconditional bonanza propensity of 5%. Surges, on the other hand, focus on those country-time observations which feature in the country *and* full sample top-30%, hence taking *global* circumstances into account. There are almost 1,000 single-year surges over the 1980-2015 time period, an unconditional propensity of 24%: surges are, by construction, vastly more common than bonanzas. Note that given upward-trending capital inflows roughly half of all bonanzas or surges occur in the last ten years of the sample.

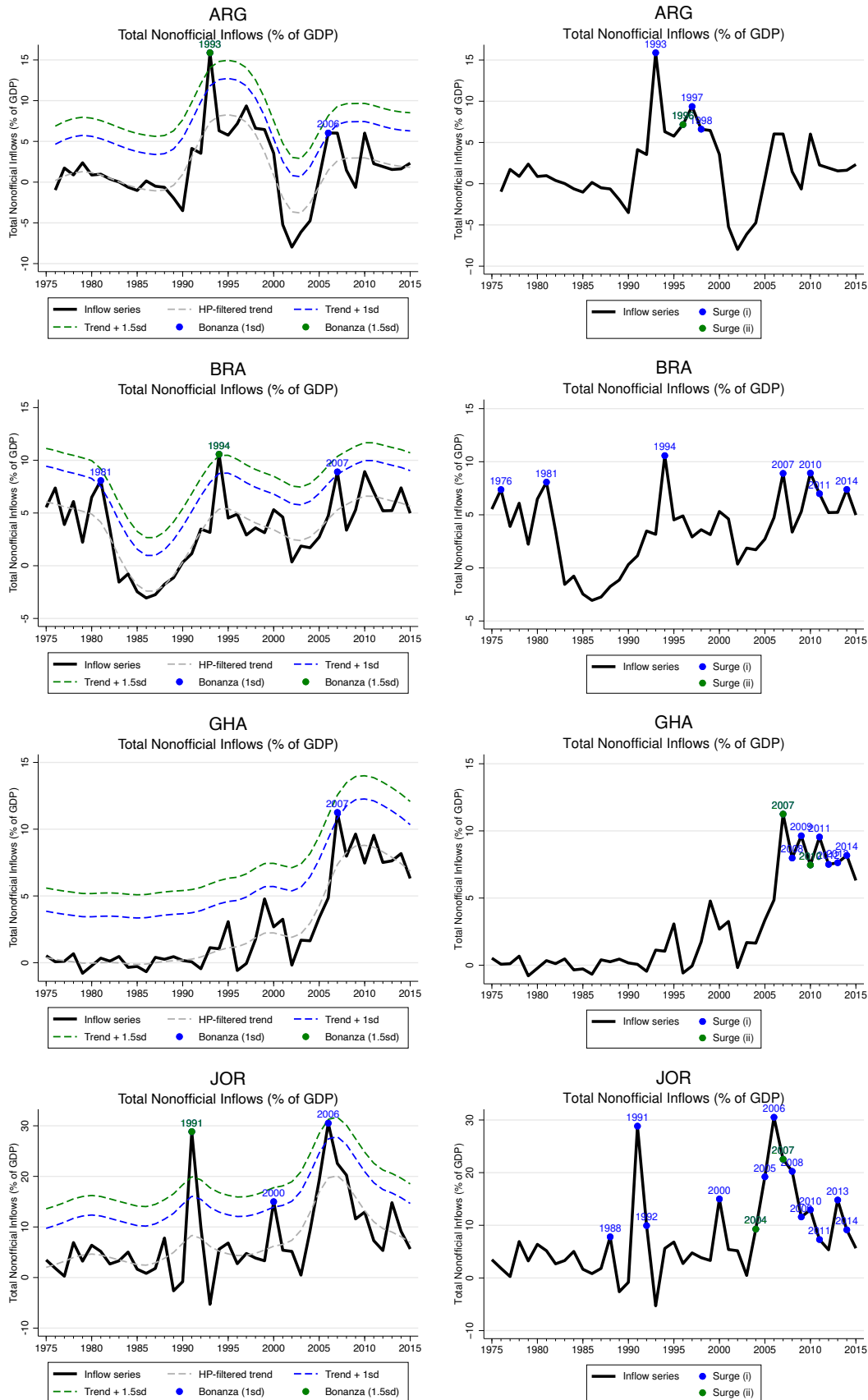
### D.1 Event Analysis

As a descriptive tool, we carry out an event analysis for each of the four domestic determinants of excessive capital inflows proposed in the literature. This is a univariate investigation of variable evolution in proximity to an excessive capital inflow 'event' (see [Gourinchas and Obstfeld, 2012](#); [Eberhardt and Presbitero, 2021](#), for details on the methodology) — we consider bonanza and surge definitions using total gross capital inflows across all countries in our sample, irrespective of geography or political institutions. Figure C-2 presents the event analysis plots for bonanzas and surges in the left and right column, respectively — the blue bars mark 90% confidence intervals. There is only weak evidence for a substantial decline in inflation in the lead-up to excessive capital inflows. In contrast, deterioration in the capital account ratio, exchange rate depreciation, and an increase in international reserves can all be detected in both sets of analysis.

In Figure C-3 we restrict the sample to countries which either experienced democratic regime change or remained autocracies throughout the sample period — patterns are very similar to those presented here, albeit less precisely estimated. Note that event analysis is a crude descriptive device, ignoring for instance that several surges may happen in quick succession (hence with fewer than 5 years prior to and after the event).

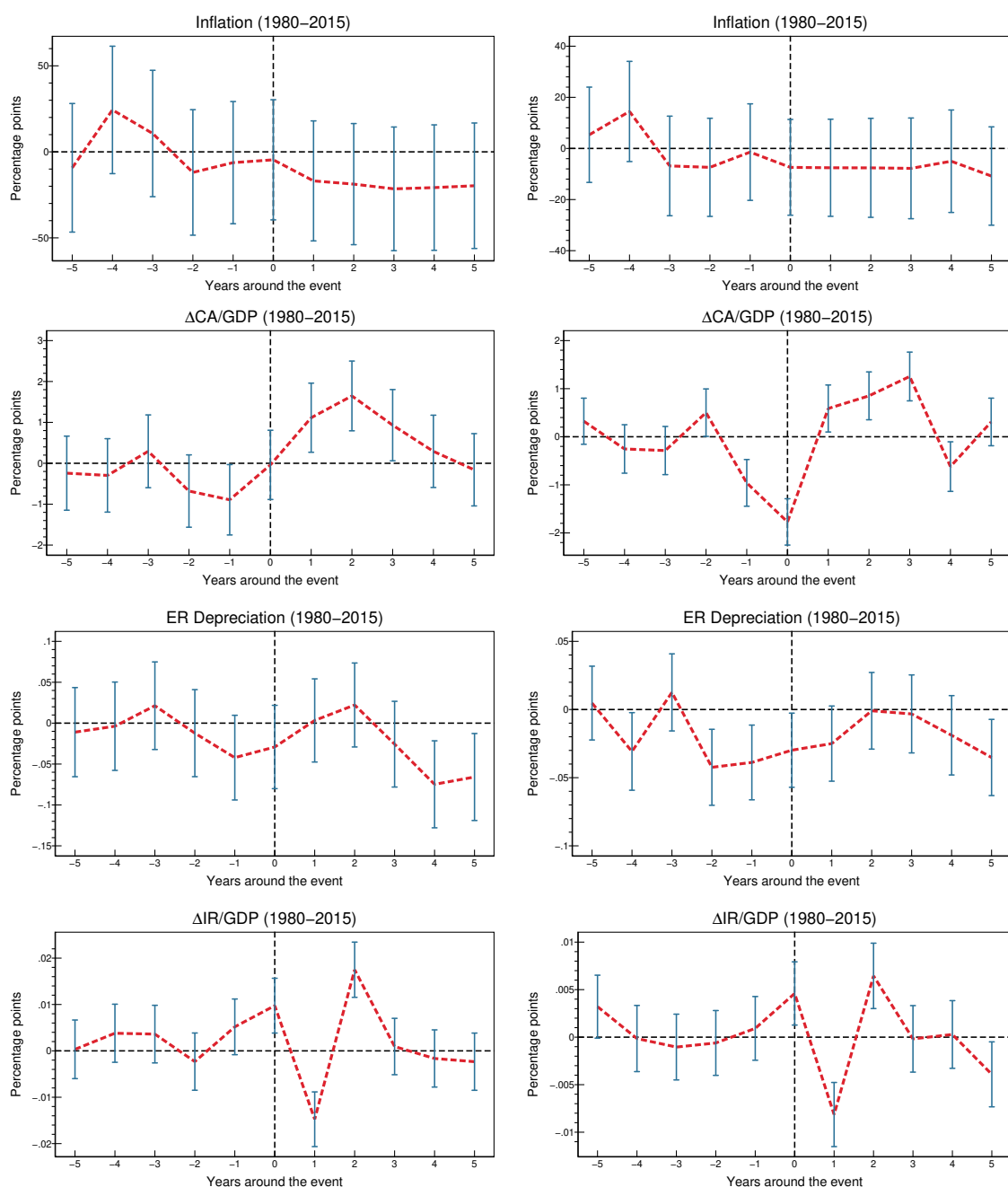
### D.2 Vulnerability analysis — graphical results

**Figure C-1: Capital Inflow Bonanzas (left) and Surges — Examples**



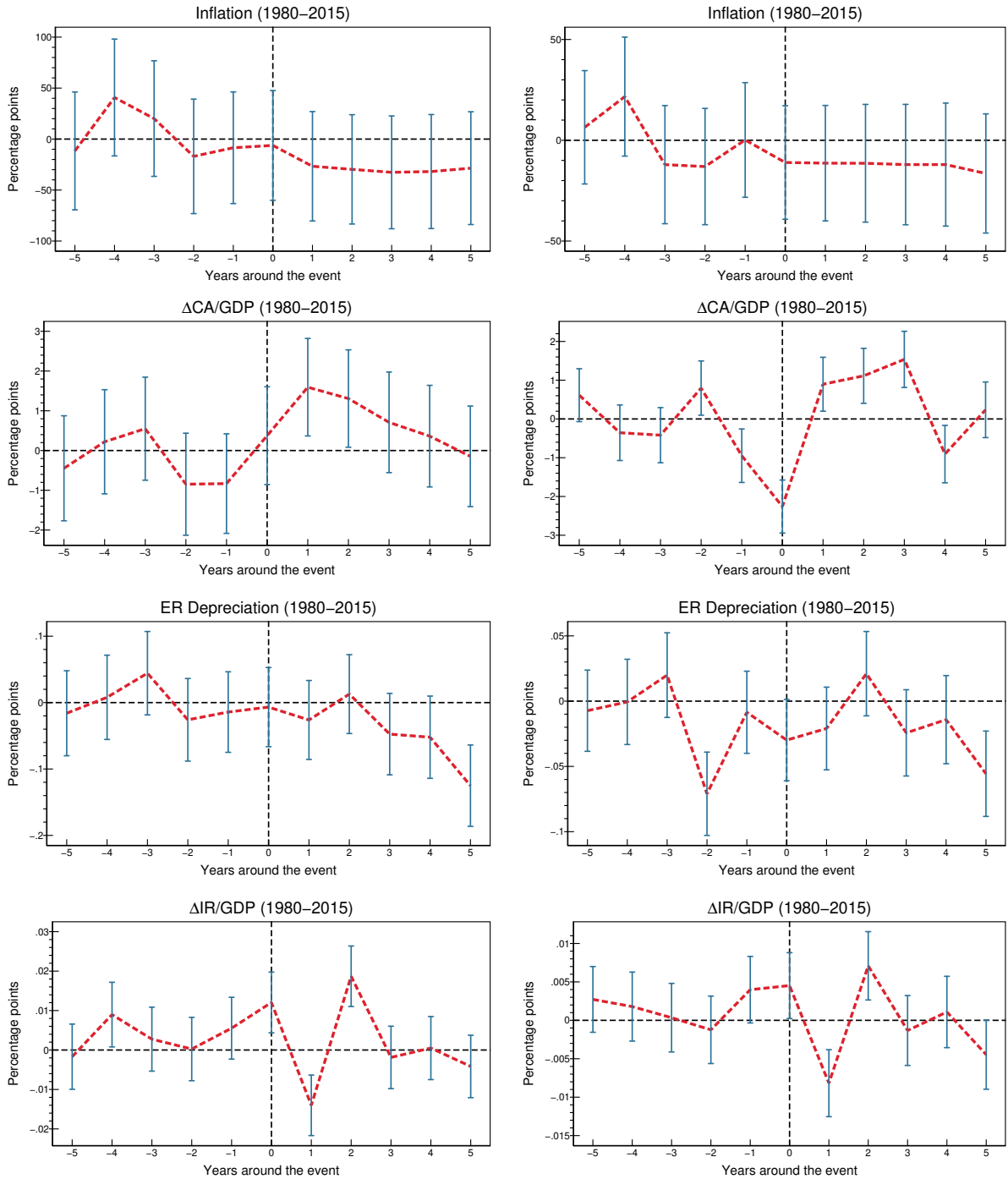
**Notes:** The plots present gross non-official capital inflow series in four countries (Argentina, Brazil, Ghana, and Jordan). The bonanza analysis in the left column adds smoothed (HP-filtered) trend capital inflow and highlights the two definitions we adopt for capital inflow bonanzas (values above 1 or 1.5 standard deviations of the smoothed trend). The surge analysis in the right column adds single-year surges (in blue) and start years of surges that lasted for three years (in green) — any three-year surge is also a one-year surge, by definition. In the empirical analysis we do not exploit the 1.5 SD definition of bonanzas or the three-year surges.

**Figure C-2: Event Analysis, total capital inflows (left: bonanza, right: surge)**



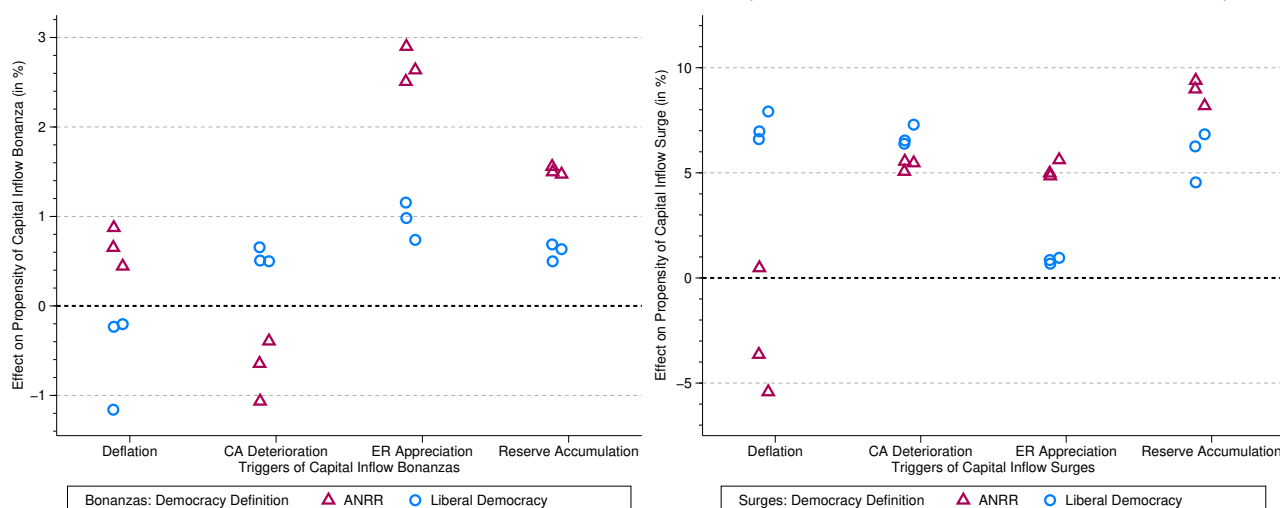
**Notes:** We present selected event analysis plots for the period 1980–2015. The estimates are derived from event dummy lags and leads in a pooled regression with country-fixed effects with 90% confidence intervals (blue bars), where the ‘event’ refers to a capital inflow bonanza (left column) or surge (right column) — we adopt the 1SD bonanza and single-year surge definitions, respectively. Variables investigated are inflation, change in capital account-to-GDP ratio, depreciation, and change in international reserves-to-GDP ratio. In contrast to the data used in our regressions, there is no MA(3) transformation carried out here.

**Figure C-3: Event Analysis, total capital inflows, treated and control only (left: bonanza, right: surge)**



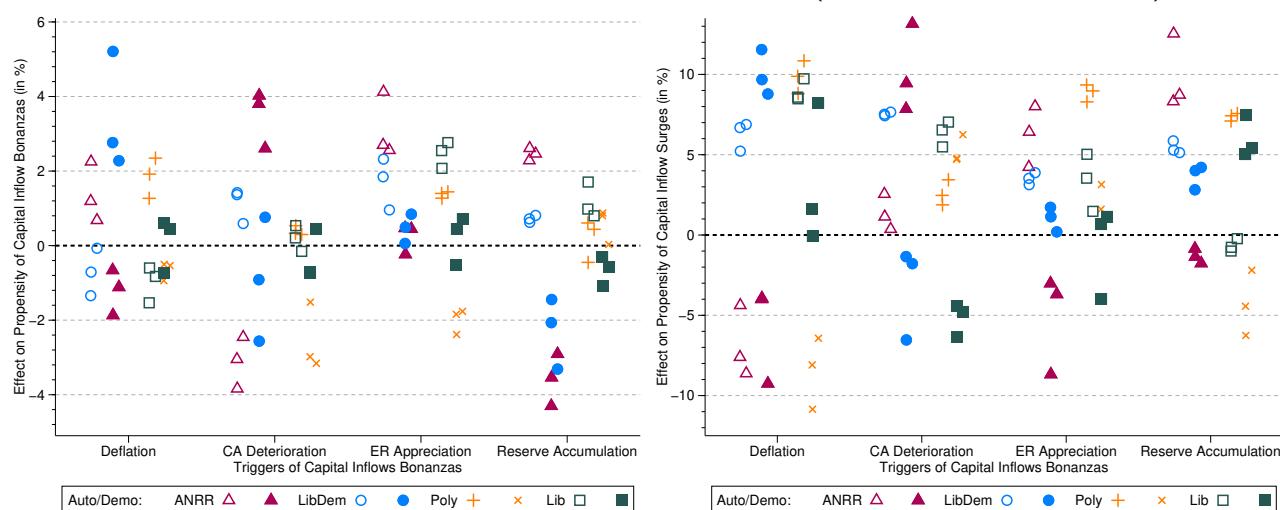
**Notes:** We present selected event analysis plots for the period 1980–2015. The estimates are derived from event dummy lags and leads in a pooled regression with country fixed effects with 90% confidence intervals (blue bars), where the 'event' refers to a capital inflow bonanza (left column) or surge (right column) — we adopt the 1SD bonanza and single-year surge definitions, respectively. The sample is made up of countries which either experienced democratic regime change or remained autocracies throughout the sample period. Variables investigated are inflation, change in capital account-to-GDP ratio, depreciation and change in international reserves-to-GDP ratio. In contrast to the data used in our regressions there is no MA(3) transformation carried out here.

**Figure C-4: Total Capital Inflow Bonanzas and Surges (ignoring geography and regime change)**



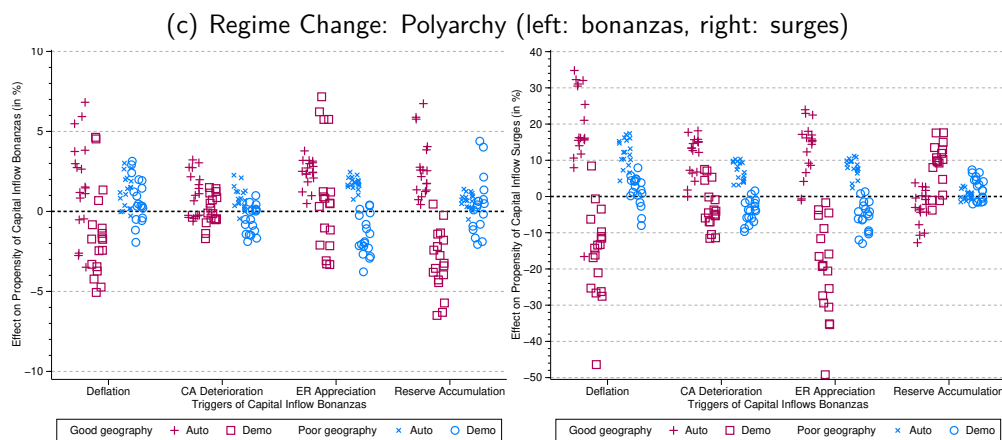
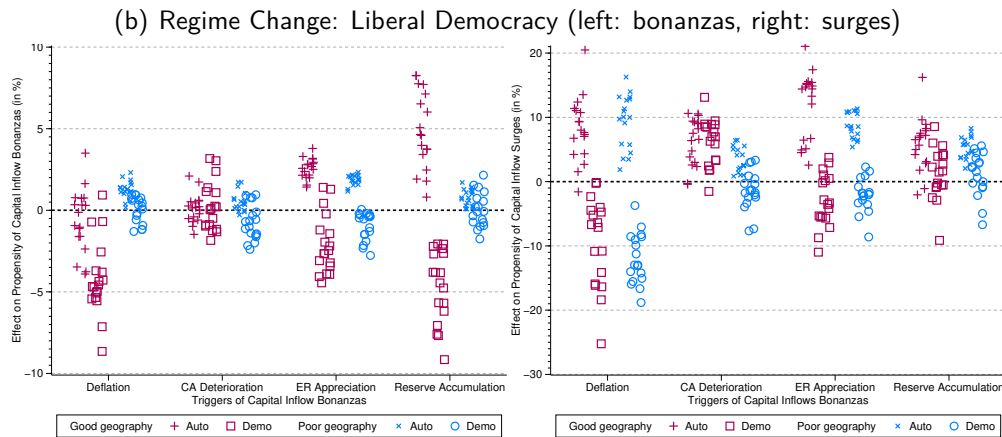
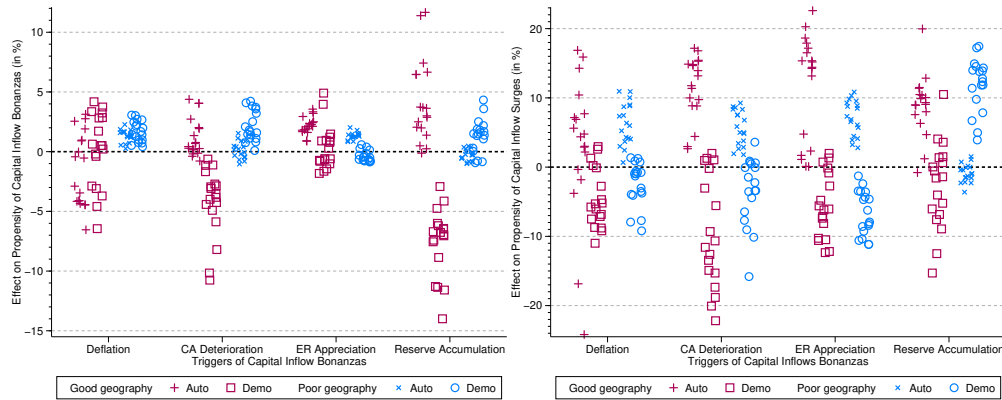
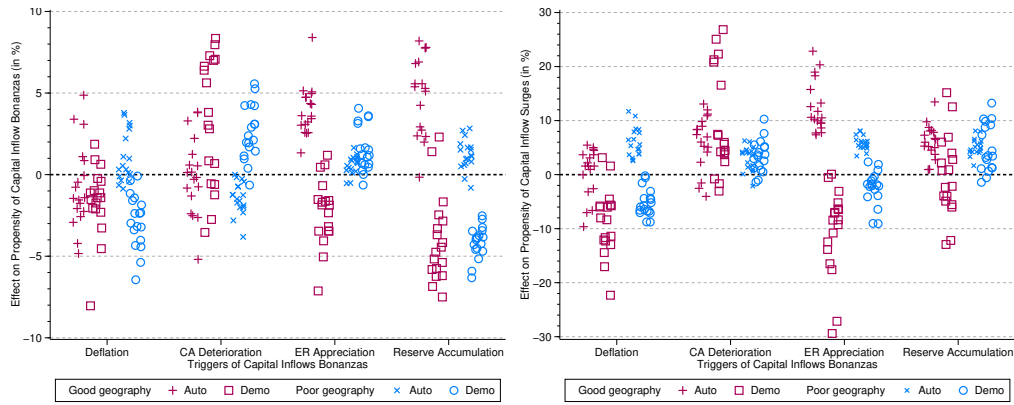
**Notes:** We present the effect of a 1 SD increase in the ‘trigger’ for excessive capital inflows for different samples (countries which at one point became democratic based on the ANRR or Liberal Democracy definition). We ignore whether a country is in democracy or not. Although democracy does not appear in the regression model, it defines the sample of treated countries investigated here. All independent variables are winsorised (top and bottom 1% of observations) and MA(3)-transformed in the regression models. The unconditional propensity of a bonanza is around 5%, that of a surge is almost 25% — hence the substantial difference in scale between the bonanza plot on the left and surge plot on the right.

**Figure C-5: Total Capital Inflow Bonanzas and Surges (ignoring deep determinants)**



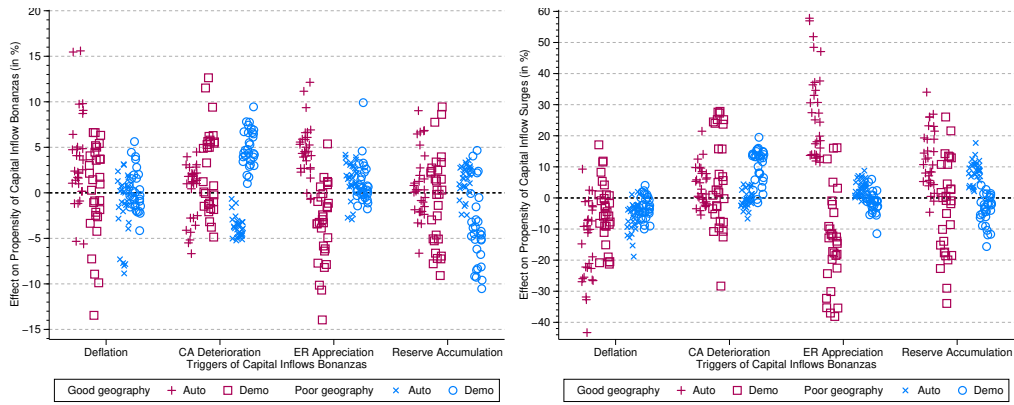
**Notes:** We present the effect of a 1 SD increase in the ‘trigger’ for excessive capital inflows for different samples (countries which at one point became democratic based on the four different definitions). In panel (a) we ignored whether a country is in democracy or not. Although democracy does not appear in the regression model, it defines the sample of treated countries investigated here. In panel (b) we study the differences between autocracy and (in relative terms) democracy for the effect of trigger variables on the propensity of excessive capital inflows (bonanzas or surges). All independent variables are winsorised (top and bottom 1% of observations) and MA(3)-transformed in the regression models. The unconditional propensity of a bonanza is around 5%, that of a surge is almost 25% — hence the substantial difference in scale between the bonanza plot on the left and surge plot on the right.

**Figure C-6: Democracy and Total Capital Inflow Bonanzas and Surges (simple specification)**

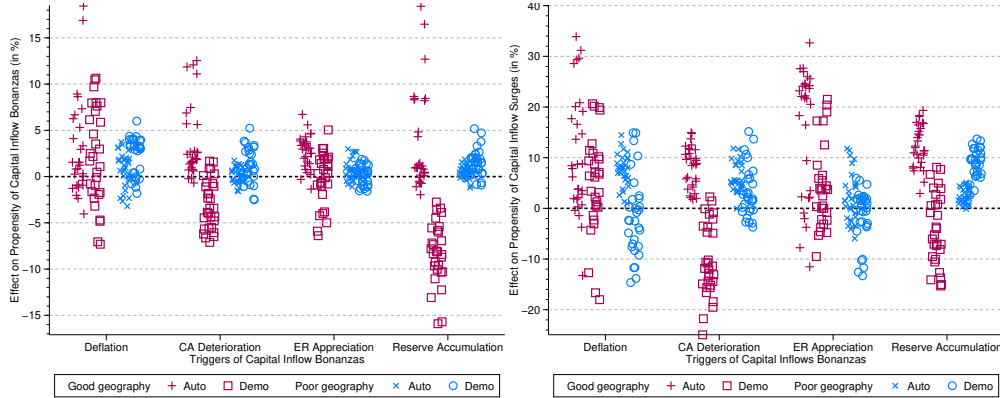


**Notes:** We present the effect of a 1 SD increase in the 'trigger' for excessive capital inflows in Autocracy (+ and x), alongside the *relative* effect in Democracy (□ and o). Panels indicate the regime change definition.

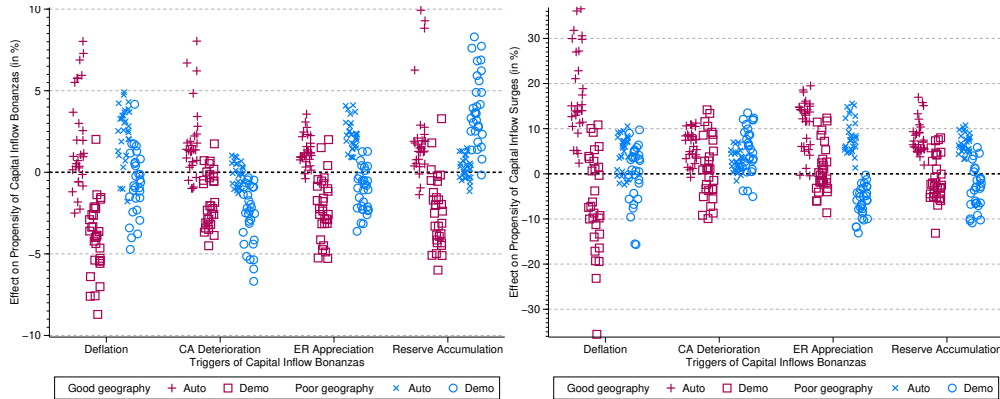
**Figure C-7: Democracy and Total Capital Inflow Bonanzas and Surges (full spec)**



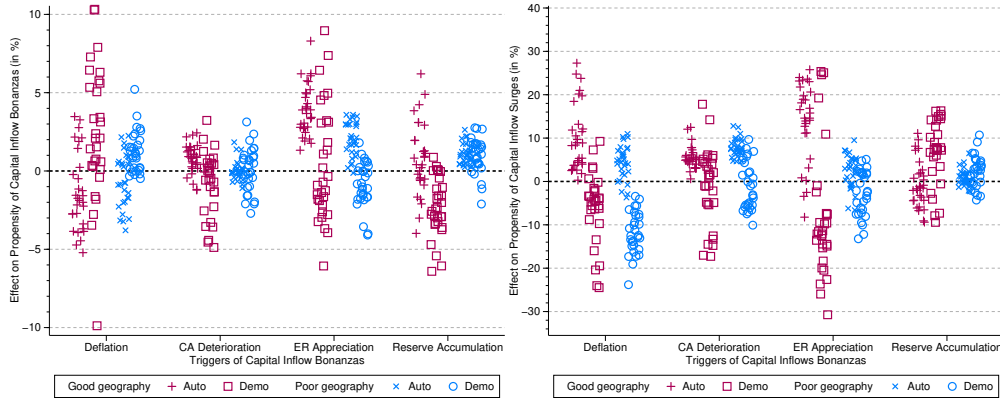
(a) Regime Change: ANRR (left: bonanzas, right: surges)



(b) Regime Change: Liberal Democracy (left: bonanzas, right: surges)



(c) Regime Change: Polyarchy (left: bonanzas, right: surges)



(d) Regime Change: Liberal Component (left: bonanzas, right: surges)

**Notes:** We present the effect of a 1 SD increase in the 'trigger' for excessive capital inflows in Autocracy (+ and x), alongside the *relative* effect in Democracy (□ and o). Panels indicate the regime change definition.

**Table C-1: Central tendency in the Vulnerability Analysis (1980-2015)**

|                      | (1)                             |                | (2)           |                | (3)           |                | (4)           |               | (5)                             |                | (6)           |               | (7)          |               | (8)           |               |
|----------------------|---------------------------------|----------------|---------------|----------------|---------------|----------------|---------------|---------------|---------------------------------|----------------|---------------|---------------|--------------|---------------|---------------|---------------|
|                      | Countries with 'Good' Geography |                |               |                |               |                |               |               | Countries with 'Poor' Geography |                |               |               |              |               |               |               |
|                      | Deflation                       |                | CA Det        |                | ER Appr       |                | Reserves      |               | Deflation                       |                | CA Det        |               | ER Appr      |               | Reserves      |               |
|                      | Auto                            | Demo           | Auto          | Demo           | Auto          | Demo           | Auto          | Demo          | Auto                            | Demo           | Auto          | Demo          | Auto         | Demo          | Auto          | Demo          |
| Panel (A): Bonanzas  |                                 |                |               |                |               |                |               |               |                                 |                |               |               |              |               |               |               |
| Simple Specification |                                 |                |               |                |               |                |               |               |                                 |                |               |               |              |               |               |               |
| ANRR                 | -1.04<br>0.13                   | -1.28<br>0.00  | -0.04<br>0.95 | 3.29<br>0.01   | 3.67<br>0.00  | -2.24<br>0.00  | 4.93<br>0.00  | -4.36<br>0.00 | 0.91<br>0.06                    | -2.83<br>0.00  | -1.44<br>0.00 | 2.46<br>0.00  | 0.61<br>0.00 | 1.39<br>0.00  | 1.24<br>0.00  | -4.08<br>0.00 |
| LibDem               | -1.59<br>0.06                   | 0.17<br>0.82   | 0.78<br>0.06  | -3.60<br>0.00  | 2.14<br>0.00  | -0.13<br>0.70  | 3.79<br>0.00  | -6.85<br>0.00 | 1.26<br>0.00                    | 1.63<br>0.00   | 0.00<br>0.99  | 2.09<br>0.00  | 1.30<br>0.00 | -0.48<br>0.00 | -0.36<br>0.00 | 1.34<br>0.00  |
| Poly                 | -0.62<br>0.23                   | -4.71<br>0.00  | -0.07<br>0.69 | 0.09<br>0.82   | 2.54<br>0.00  | -2.27<br>0.00  | 4.85<br>0.00  | -4.62<br>0.00 | 1.23<br>0.00                    | 0.08<br>0.70   | 0.45<br>0.01  | -0.82<br>0.01 | 1.89<br>0.00 | -0.62<br>0.00 | 0.74<br>0.00  | -0.04<br>0.91 |
| Liberal              | 1.68<br>0.04                    | -1.94<br>0.01  | 1.02<br>0.01  | 0.027<br>0.92  | 2.49<br>0.00  | 0.49<br>0.58   | 2.44<br>0.00  | -3.20<br>0.00 | 1.19<br>0.00                    | 0.68<br>0.08   | 0.75<br>0.00  | -0.56<br>0.02 | 1.73<br>0.00 | -1.58<br>0.00 | 0.74<br>0.00  | 0.20<br>0.64  |
| Full Specification   |                                 |                |               |                |               |                |               |               |                                 |                |               |               |              |               |               |               |
| ANRR                 | 2.92<br>0.00                    | 0.64<br>0.03   | 1.49<br>0.02  | 2.40<br>0.07   | 4.36<br>0.00  | -3.23<br>0.00  | 0.56<br>0.00  | -0.82<br>0.25 | -0.69<br>0.00                   | -0.01<br>0.00  | -3.95<br>0.21 | 4.92<br>0.00  | 1.85<br>0.00 | 0.50<br>0.01  | 1.51<br>0.00  | -4.28<br>0.00 |
| LibDem               | 1.52<br>0.00                    | 2.10<br>0.01   | 1.66<br>0.00  | -3.03<br>0.00  | 2.64<br>0.00  | 0.04<br>0.04   | 0.28<br>0.00  | -8.09<br>0.01 | 0.86<br>0.00                    | 2.87<br>0.17   | 0.06<br>0.00  | 1.10<br>0.01  | 0.64<br>0.10 | 0.03<br>0.67  | 0.44<br>0.00  | 1.43<br>0.00  |
| Poly                 | 1.46<br>0.00                    | -4.07<br>0.01  | 1.09<br>0.00  | -1.68<br>0.43  | 1.15<br>0.00  | -2.24<br>0.84  | 1.26<br>0.00  | -2.55<br>0.21 | 2.66<br>0.00                    | -0.71<br>0.81  | -0.19<br>0.00 | -2.61<br>0.00 | 2.22<br>0.00 | -1.16<br>0.00 | 0.23<br>0.00  | 3.80<br>0.00  |
| Liberal              | -1.52<br>0.00                   | 2.50<br>0.00   | 0.84<br>0.00  | -0.62<br>0.68  | 3.82<br>0.00  | 0.10<br>0.00   | 0.65<br>0.24  | -2.29<br>0.00 | -0.62<br>0.00                   | 1.09<br>0.00   | -0.11<br>0.00 | 0.05<br>0.29  | 2.05<br>0.00 | -0.85<br>0.01 | 0.90<br>0.21  | 1.04<br>0.01  |
| Panel (B): Surges    |                                 |                |               |                |               |                |               |               |                                 |                |               |               |              |               |               |               |
| Simple Specification |                                 |                |               |                |               |                |               |               |                                 |                |               |               |              |               |               |               |
| ANRR                 | 1.24<br>0.25                    | -8.85<br>0.00  | 6.53<br>0.00  | 8.04<br>0.01   | 11.50<br>0.00 | -8.59<br>0.00  | 5.96<br>0.00  | -0.62<br>0.75 | 5.76<br>0.00                    | -5.75<br>0.00  | 3.27<br>0.00  | 3.27<br>0.00  | 5.76<br>0.00 | -2.29<br>0.01 | 4.89<br>0.00  | 4.73<br>0.00  |
| LibDem               | 5.38<br>0.01                    | -4.81<br>0.00  | 11.73<br>0.00 | -9.55<br>0.00  | 12.49<br>0.00 | -5.72<br>0.00  | 9.25<br>0.00  | -2.75<br>0.10 | 6.18<br>0.00                    | -2.24<br>0.02  | 5.61<br>0.00  | -3.04<br>0.02 | 7.00<br>0.00 | -6.75<br>0.00 | -0.95<br>0.01 | 12.15<br>0.00 |
| Poly                 | 7.83<br>0.00                    | -9.03<br>0.00  | 6.29<br>0.00  | 6.26<br>0.00   | 12.96<br>0.00 | -2.99<br>0.01  | 5.91<br>0.00  | 1.51<br>0.11  | 9.39<br>0.00                    | -12.11<br>0.00 | 3.27<br>0.00  | -1.37<br>0.09 | 8.65<br>0.00 | -1.98<br>0.00 | 5.14<br>0.00  | 1.26<br>0.12  |
| Liberal              | 19.82<br>0.00                   | -15.48<br>0.00 | 13.16<br>0.00 | -3.96<br>0.02  | 13.52<br>0.00 | -19.75<br>0.00 | -2.20<br>0.07 | 10.25<br>0.00 | 12.14<br>0.00                   | 1.80<br>0.11   | 6.81<br>0.00  | -4.17<br>0.00 | 7.90<br>0.00 | -5.61<br>0.00 | 0.18<br>0.46  | 2.02<br>0.02  |
| Full Specification   |                                 |                |               |                |               |                |               |               |                                 |                |               |               |              |               |               |               |
| ANRR                 | -14.27<br>0.00                  | -4.88<br>0.03  | 3.09<br>0.02  | 5.45<br>0.07   | 26.01<br>0.00 | -14.29<br>0.00 | 12.37<br>0.00 | -3.39<br>0.25 | -6.11<br>0.00                   | -3.14<br>0.00  | -0.63<br>0.21 | 11.35<br>0.00 | 3.22<br>0.00 | -1.51<br>0.01 | 8.24<br>0.00  | -3.42<br>0.00 |
| LibDem               | 9.93<br>0.00                    | 5.22<br>0.01   | 8.14<br>0.00  | -10.67<br>0.00 | 21.74<br>0.00 | 3.27<br>0.04   | 12.20<br>0.00 | -3.99<br>0.01 | 7.47<br>0.00                    | -2.20<br>0.17  | 5.21<br>0.00  | 2.66<br>0.01  | 1.59<br>0.10 | -0.36<br>0.67 | 2.33<br>0.00  | 9.10<br>0.00  |
| Poly                 | 16.58<br>0.00                   | -5.97<br>0.01  | 6.32<br>0.00  | 1.09<br>0.43   | 12.30<br>0.00 | 0.22<br>0.84   | 7.46<br>0.00  | -1.33<br>0.21 | 5.17<br>0.00                    | 0.28<br>0.81   | 2.61<br>0.00  | 5.34<br>0.00  | 8.00<br>0.00 | -6.18<br>0.00 | 6.56<br>0.00  | -3.59<br>0.00 |
| Liberal              | 9.28<br>0.00                    | -4.71<br>0.00  | 4.71<br>0.00  | -0.70<br>0.68  | 14.79<br>0.00 | -13.43<br>0.00 | -1.27<br>0.24 | 7.46<br>0.00  | 4.49<br>0.00                    | -11.83<br>0.00 | 7.72<br>0.00  | -1.25<br>0.29 | 2.50<br>0.00 | -2.76<br>0.01 | 0.47<br>0.21  | 2.06<br>0.01  |

Notes: This table presents the robust mean estimates for each trigger and interaction variable distinguishing autocratic and democratic periods — these are averages of the estimates visualised in Figures C-6 and C-7. Good (poor) geography countries are presented in the left (right) half of the table. The small numbers in the second row for each definition of democracy are *p*-values for the robust mean estimates. The schematic presentation of the results in Table 3 of the maintext is based on these estimates and inferential statistics.

## D.3 Deep Determinants within Geography Samples

**Table C-2: Alternative Deep Determinants within Geography — Sample size**

| N                             | LO     | Culture |         |     |     |         | History |      |           |           |        |           | Total |
|-------------------------------|--------|---------|---------|-----|-----|---------|---------|------|-----------|-----------|--------|-----------|-------|
|                               | French | Hofst   | Dist UK | CL  | LS  | No Euro | Col     | Extr | Early Col | Few Settl | Slaves | High Slav |       |
| Countries with Good Geography |        |         |         |     |     |         |         |      |           |           |        |           |       |
| 2                             | 0      | 0       | 0       | 0   | 0   | 10      | 0       | 20   | 45        | 15        | 0      | 0         | 90    |
| 3                             | 0      | 0       | 5       | 0   | 0   | 10      | 15      | 25   | 20        | 25        | 0      | 0         | 100   |
| 4                             | 0      | 0       | 20      | 5   | 0   | 25      | 15      | 0    | 15        | 25        | 0      | 0         | 105   |
| 5                             | 0      | 0       | 25      | 0   | 5   | 15      | 15      | 25   | 10        | 15        | 0      | 0         | 110   |
| 6                             | 0      | 0       | 35      | 5   | 30  | 25      | 20      | 5    | 0         | 20        | 0      | 0         | 140   |
| 7                             | 0      | 5       | 25      | 30  | 25  | 25      | 20      | 0    | 0         | 5         | 0      | 0         | 135   |
| 8                             | 0      | 50      | 0       | 30  | 30  | 5       | 10      | 0    | 0         | 0         | 0      | 0         | 125   |
| 9                             | 0      | 35      | 5       | 20  | 15  | 5       | 5       | 0    | 0         | 0         | 0      | 0         | 85    |
| 10                            | 5      | 30      | 5       | 20  | 15  | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 75    |
| 11                            | 17     | 0       | 0       | 10  | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 27    |
| 12                            | 15     | 0       | 0       | 0   | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 15    |
| 13                            | 23     | 0       | 0       | 0   | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 23    |
| 14                            | 30     | 0       | 0       | 0   | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 30    |
| 15                            | 16     | 0       | 0       | 0   | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 16    |
| 16                            | 14     | 0       | 0       | 0   | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 14    |
| Sum                           | 120    | 120     | 120     | 120 | 120 | 120     | 100     | 75   | 90        | 105       | 0      | 0         | 1,090 |
| Countries with Poor Geography |        |         |         |     |     |         |         |      |           |           |        |           |       |
| 6                             | 0      | 0       | 0       | 0   | 0   | 0       | 0       | 5    | 0         | 0         | 0      | 0         | 5     |
| 7                             | 0      | 0       | 0       | 0   | 0   | 0       | 0       | 0    | 5         | 0         | 0      | 0         | 5     |
| 8                             | 0      | 0       | 0       | 0   | 0   | 0       | 0       | 15   | 0         | 0         | 0      | 30        | 45    |
| 9                             | 0      | 0       | 0       | 0   | 0   | 0       | 0       | 5    | 5         | 0         | 0      | 0         | 10    |
| 10                            | 0      | 0       | 0       | 0   | 0   | 5       | 0       | 30   | 20        | 5         | 30     | 0         | 90    |
| 11                            | 0      | 0       | 0       | 0   | 0   | 0       | 0       | 0    | 5         | 0         | 0      | 35        | 40    |
| 12                            | 0      | 0       | 0       | 0   | 0   | 10      | 0       | 15   | 25        | 10        | 0      | 25        | 85    |
| 13                            | 0      | 0       | 0       | 0   | 0   | 10      | 5       | 5    | 20        | 10        | 30     | 0         | 80    |
| 14                            | 0      | 0       | 0       | 5   | 0   | 5       | 0       | 0    | 10        | 15        | 5      | 0         | 40    |
| 15                            | 0      | 0       | 0       | 5   | 5   | 15      | 10      | 0    | 0         | 5         | 25     | 0         | 65    |
| 16                            | 5      | 20      | 0       | 15  | 0   | 15      | 5       | 0    | 0         | 15        | 0      | 0         | 75    |
| 17                            | 0      | 30      | 0       | 10  | 15  | 5       | 10      | 0    | 0         | 15        | 0      | 0         | 85    |
| 18                            | 10     | 10      | 0       | 10  | 10  | 15      | 5       | 0    | 0         | 5         | 0      | 5         | 70    |
| 19                            | 0      | 0       | 0       | 10  | 10  | 5       | 10      | 0    | 0         | 0         | 0      | 25        | 60    |
| 20                            | 20     | 10      | 5       | 15  | 15  | 5       | 15      | 0    | 0         | 0         | 0      | 0         | 85    |
| 21                            | 10     | 20      | 10      | 10  | 20  | 0       | 5       | 0    | 0         | 5         | 0      | 0         | 80    |
| 22                            | 20     | 15      | 10      | 5   | 10  | 0       | 5       | 0    | 0         | 10        | 0      | 0         | 75    |
| 23                            | 25     | 10      | 5       | 5   | 0   | 0       | 0       | 0    | 0         | 5         | 0      | 0         | 50    |
| 24                            | 0      | 5       | 10      | 0   | 5   | 0       | 5       | 0    | 0         | 0         | 0      | 0         | 25    |
| 25                            | 0      | 0       | 25      | 0   | 0   | 0       | 0       | 0    | 0         | 5         | 16     | 0         | 46    |
| 26                            | 0      | 0       | 15      | 0   | 0   | 10      | 0       | 0    | 0         | 0         | 14     | 0         | 39    |
| 27                            | 0      | 0       | 10      | 0   | 0   | 10      | 0       | 0    | 0         | 0         | 0      | 0         | 20    |
| 28                            | 0      | 0       | 0       | 1   | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 1     |
| 29                            | 5      | 0       | 0       | 4   | 1   | 5       | 0       | 0    | 0         | 0         | 0      | 0         | 15    |
| 30                            | 10     | 0       | 0       | 10  | 4   | 5       | 10      | 0    | 0         | 0         | 0      | 0         | 39    |
| 31                            | 0      | 0       | 1       | 15  | 10  | 0       | 10      | 0    | 0         | 0         | 0      | 0         | 36    |
| 32                            | 0      | 0       | 4       | 0   | 0   | 0       | 1       | 0    | 0         | 0         | 0      | 0         | 5     |
| 33                            | 5      | 0       | 5       | 0   | 15  | 0       | 4       | 0    | 0         | 0         | 0      | 0         | 29    |
| 34                            | 10     | 0       | 0       | 0   | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 10    |
| 35                            | 0      | 0       | 20      | 0   | 0   | 0       | 0       | 0    | 0         | 0         | 0      | 0         | 20    |
| Sum                           | 120    | 120     | 120     | 120 | 120 | 120     | 100     | 75   | 90        | 105       | 120    | 120       | 1,330 |

Notes: This table presents the number of treated countries *N* (in rows) in the subsample analysis by Legal origin, culture, and history in Section 5. Cultural proxies: Hofst is the dummy for 'collectivist' countries, Dist UK the dummy for greater dissimilarity in blood type to the UK, CL is low common language and LS low language similarity. Historical proxies: Col indicates any colonial experience, Extr is for extractive colony, Early col is colonisation before 1860, Few Settl indicates that during the colonial period less than 1% of the population were foreigners from the colonial power.